Algebraic Geometry

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1 Sheaves

1.1 Definitions

Definition 1.1.1

For a topological space X, its set of open subsets $\mathrm{open}(X)$ is partially ordered by inclusion, and thus forms a small category. It is also a cofiltered category: for every $U, V \in \mathrm{open}(X)$ they have a lower bound $U \cap V$.

Definition 1.1.2

Let X be a topological space, and $\mathcal{C}$ a category. A **$\mathcal{C}$-valued presheaf on X** is a contravariant functor $\mathcal{F}\colon \mathrm{open}(X)^{\mathrm{op}} \to \mathcal{C}$. It is customary to denote $\mathcal{F}(U \subseteq V)\colon \mathcal{F}(V) \to \mathcal{F}(U)$ by res^V_U. When $\mathcal{C}$ is a concrete category (i.e. its objects are sets), elements of $\mathcal{F}(U)$ are called **sections of $\mathcal{F}$ over U**, and for a section $s \in \mathcal{F}(V)$ we write $s|_U$ for $\mathrm{res}^V_U(s)$.

A **morphism of presheaves** is a natural transformation.

Denote the category of $\mathcal{C}$-valued presheaves on X by $\mathrm{Psh}_{\mathcal{C}}(X)$.

Example 1.1.3

Let E be a set (or a topological space, or any mathematical object which lies in the same category as X and its open subsets). We can then define the presheaf of functions on E by $\mathcal{F}(U) = \hom(U, E)$: the set of all functions $U \to E$. The restriction maps are very literally restriction: for $U \subseteq V$ and $f\colon V \to E$, we restrict f to U to obtain a function $f|_U\colon U \to E$.

Definition 1.1.4

Let $\mathcal{F}$ be a presheaf over X valued in a category $\mathcal{C}$ with filtered colimits, and $p \in X$. Then $\mathcal{U}(x) = \{U \subseteq \text{ open} \mid p \in U\}$, the set of neighborhoods of p, is a cofiltered subcategory of $\mathrm{open}(X)$, and so $\mathcal{F}$ restricts to a functor from $\mathcal{U}(x)^{\mathrm{op}}$. Define $\mathcal{F}$'s **stalk** at p to be the filtered colimit

$$\mathcal{F}_p = \mathrm{colim}_{U \in \mathcal{U}(p)} \mathcal{F}(U) \in \mathcal{C}.$$

For concrete categories, this is the set of **germs** of $\mathcal{F}$ at p:

$$\mathcal{F}_p = \{(U, s) \mid U \in \mathcal{U}(p),\ s \in \mathcal{F}(U)\},$$

modulo the relation where we identify (s, U) and (t, V) if there is a subset $p \in W \subseteq U, V$ such that $s|_W = t|_W$.

Taking colimits is functorial, so the correspondence $p \mapsto \mathcal{F}_p$ is functorial. Explicitly $\varphi\colon \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves over X valued in a concrete category, and $p \in X$ we define $\varphi_p\colon \mathcal{F}_p \to \mathcal{G}_p$ by

$$\varphi_p(U, s) = (U, \varphi_U(s)).$$

Definition 1.1.5

A presheaf $\mathcal{F}$ on X (valued in a concrete category) is a **sheaf** if it satisfies the **sheaf axiom**: for every

open subset $U \subseteq X$ and open cover $\{U_i\}$ of U and sections $s_i \in \mathcal{F}(U_i)$, if $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j then there exists a *unique* $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i.

We can split the sheaf axiom in two:

(1) the **identity axiom**: if $\{U_i\}$ is an open cover of U, and $s_1, s_2 \in \mathcal{F}(U)$ are such that $s_1|_{U_i} = s_2|_{U_i}$ for all i, then $s_1 = s_2$.

(2) the **gluability axiom**: the sheaf axiom without requiring uniqueness.

We can also state the sheaf axiom categorically. For every open cover $\{U_i\}$ of U, we can consider the following diagram:

$$\mathcal{F}(U) \xrightarrow{\text{res}} \prod_i \mathcal{F}(U_i) \underset{\text{res}_j}{\overset{\text{res}_i}{\rightrightarrows}} \prod_{i,j} \mathcal{F}(U_i \cap U_j),$$

where $\mathcal{F}(U) \xrightarrow{\text{res}} \prod_i \mathcal{F}(U_i)$ is the map $s \mapsto (s|_{U_i})_i$, res_1 and res_2 are the maps

$$\text{res}_1(s_i)_i = (s_i|_{U_i \cap U_j})_{i,j}, \qquad \text{res}_2(s_i)_i = (s_j|_{U_i \cap U_j})_{i,j}.$$

Categorically, res_1's coordinate to $\mathcal{F}(U_i \cap U_j)$ composes the ith projection with $\text{res}^{U_i}_{U_i \cap U_j}$, and res_2's coordinate composes the jth projection with $\text{res}^{U_j}_{U_i \cap U_j}$. These maps are defined for any presheaf, and we can see that $\mathcal{F}$ is a presheaf iff this is an equalizer diagram.

Indeed, the equalizer of $\text{res}_1, \text{res}_2$ are all $(s_i)_i$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, i.e. a family of sections which agree on the overlaps. And there is an $s \in \mathcal{F}(U)$ whose projections $s|_{U_i}$ are s_i when the gluing axiom is satisfied, and it is unique (so an embedding, and thus an equalizer) when the identity lemma is satisfied.

This allows us to generalize sheaves to any category $\mathcal{C}$ with arbitrary products.

Remark 1.1.6

Alternatively for an open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of U, we can define a subcategory of $\text{open}(X)$ consisting of objects $U_i, U_i \cap U_j$ with morphisms $U_i \cap U_j \subseteq U_i$. Then $\mathcal{F}$ restricts to a functor $\mathcal{F}|_{\mathcal{U}}: \mathcal{U}^{\text{op}} \to \mathcal{C}$. Then the sheaf condition says that the cone $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}}$ whose components are the restriction maps res, is a limiting cone.

The category of $\mathcal{C}$-valued sheaves on X is denoted $\text{Sh}_{\mathcal{C}}(X)$, it is a full subcategory of $\text{Psh}_{\mathcal{C}}(X)$; meaning sheaf morphisms are just presheaf morphisms between sheaves. The stalk of a sheaf at a point is just the stalk of it as a presheaf.

Example 1.1.7

If $\mathcal{F}$ is a presheaf on X, and $U \subseteq X$ is open, we define $\mathcal{F}$'s **restriction** to U to be the presheaf $\mathcal{F}|_U$ on U defined for $V \subseteq U$ by $\mathcal{F}|_U(V) = \mathcal{F}(V)$ (it is just the restriction of the functor $\mathcal{F}$ to the subcategory $\text{open}(U)$). If $\mathcal{F}$ is a sheaf, its restriction is clearly a sheaf as well.

Example 1.1.8

Let X be a topological space, $p \in X$, and S a set (or a group, module, etc.). We define the **skyscraper sheaf** p_*S by

$$p_*S(U) = \begin{cases} S & \text{if } p \in U \\ * & \text{else} \end{cases},$$

where $*$ is a single-point set; the restrictions are given by the identities or the unique morphism $S \to *$. This is indeed a sheaf: if $p \in U$ and $\{U_i\}$ is a covering of U with $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$

then for every $p \in U_i$ we have a single value $s \in S$ which is equal to all $s_i \in \mathcal{F}(U_i)$, this is the glued section. Otherwise $p \notin U$ and s_i are all the unique element in $*$, so s must be too.

Example 1.1.9

Let X be a topological space and S a set. The **constant presheaf** associated with S is the presheaf $\underline{S}_{\text{pre}}$ which maps all open sets U to S. This is a presheaf, but not a sheaf in general. For example if $X = \{p, q\}$ is the two point space with the discrete topology then take $s_p \neq s_q$; they cannot be glued to a global section on X.

If we instead define $\underline{S}(U)$ to be the set of locally constant functions $U \to S$, then this is a sheaf, called the **constant sheaf**. It is just the sheaf of continuous functions $\underline{S}(U) = C(U, S)$ with S given the discrete topology. The sheaf of continuous functions $C(\bullet, Y)$ is a sheaf in general (by the gluing lemma from topology).

Example 1.1.10

Let $\pi\colon X \to Y$ be a continuous map, and $\mathcal{F}$ a presheaf over X. Define the **pushforward** (or **direct image**) presheaf $\pi_*\mathcal{F}$ by

$$\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}V),$$

for $V \subseteq Y$, and $\operatorname{res}^V_W = \operatorname{res}^{\pi^{-1}V}_{\pi^{-1}W}$. This is clearly a presheaf; if $\mathcal{F}$ is a sheaf it is also a sheaf. Indeed, if $V \subseteq Y$ is open and $\{V_i\}_i$ is a cover with $s_i \in \pi_*\mathcal{F}(V_i) = \mathcal{F}(\pi^{-1}V_i)$. Then the glue of these is unique, because $\mathcal{F}$ is a sheaf.

The skyscraper sheaf p_*S is the pushforward of the constant sheaf $\underline{S}$ by the inclusion $\{p\} \to X$.

An important property of the direct image sheaf is that it is functorial: if $\varphi\colon \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves over X, then $\pi_*\varphi\colon \pi_*\mathcal{F} \to \pi_*\mathcal{G}$ defined by $\pi_*\varphi_V = \varphi_{\pi^{-1}V}$ is a morphism of presheaves.

Note that if $\pi\colon X \to Y$ is continuous, $\mathcal{F}$ is a sheaf over X, and $\pi(p) = q$, then there is a natural morphism of stalks

$$(\pi_*\mathcal{F})_q \to \mathcal{F}_p.$$

Indeed, the left-hand side is

$$\operatorname{colim}_{q \in V} \mathcal{F}(\pi^{-1}V) = \operatorname{colim}_{p \in \pi^{-1}V} \mathcal{F}(\pi^{-1}V).$$

Now a cone under $\mathcal{F}$ defines a unique cone under $\pi_*\mathcal{F}$ simply by restricting, and since $\operatorname{colim} \pi_*\mathcal{F}$ is the initial cone under $\pi_*\mathcal{F}$ there is a canonical map to this restricted cone, in particular for $\operatorname{colim} \mathcal{F}$.

In more concrete terms, $(\pi_*\mathcal{F})_q$ has elements (s, V) for $s \in \mathcal{F}(\pi^{-1}V)$, which we then map to $(s, \pi^{-1}V) \in \mathcal{F}_p$.

Definition 1.1.11

If $\mathcal{O}_X$ is a sheaf of rings over a topological space X (i.e. a sheaf over X valued in the category of rings), then the pair $(X, \mathcal{O}_X)$ is called a **ringed space**; $\mathcal{O}_X$ is called the **structure sheaf**. Sections of $\mathcal{O}_X$ over $U \subseteq X$ are called **functions** on U, global sections are caled **global functions**.

Definition 1.1.12

If $\mathcal{O}_X$ is a ringed space, an **$\mathcal{O}_X$-module** is a sheaf of Abelian groups $\mathcal{F}$ over X such that for each $U \subseteq X$, $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module, and all of these modules are well-behaved in that when $U \subseteq V$, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O}_X(V) \times \mathcal{F}(V) & \xrightarrow{\text{mult}} & \mathcal{F}(V) \\ \downarrow{\scriptstyle \text{res}_{V,U} \times \text{res}_{V,U}} & & \downarrow{\scriptstyle \text{res}_{V,U}} \\ \mathcal{O}_X(U) \times \mathcal{F}(U) & \xrightarrow{\text{mult}} & \mathcal{F}(U) \end{array}$$

The horizontal arrows are $\mathcal{O}_X(U)$'s action on $\mathcal{F}(U)$.

The commutative diagram means that

$$(s \cdot x)|_U = (s|_U) \cdot (x|_U),$$

for $s \in \mathcal{O}_X(V), x \in \mathcal{F}(V)$.

Note that just as modules generalize the concept of an Abelian group (Abelian groups are $\mathbb{Z}$-modules), $\mathcal{O}_X$-modules generalize the concept of a sheaf of Abelian groups. Indeed, a sheaf of Abelian groups is just an $\underline{\mathbb{Z}}$-module. Every $\underline{\mathbb{Z}}$-module is a sheaf of Abelian groups by definition, and conversely a sheaf of Abelian groups has a unique $\underline{\mathbb{Z}}$-module structure where for $U \subseteq X$ connected we have $\underline{\mathbb{Z}}(U) \cong \mathbb{Z}$ which has a unique multiplication on $\mathcal{F}(U) \in \text{Ab}$. We extend this to arbitrary $U \subseteq X$ by gluing.

Proposition 1.1.13

If $\mathcal{F}$ is an $\mathcal{O}_X$-module, then for each $p \in X$, $\mathcal{F}_p$ is an $\mathcal{O}_{X,p}$-module.

Proof

The condition that $\mathcal{F}$ be an $\mathcal{O}_X$-module means that multiplication is a natural map $\mathcal{O}_X \times \mathcal{F} \to \mathcal{F}$. Colimits are taken pointwise in functor categories (in particular for the product of functors), so this gives a natural map $\mathcal{O}_{X,p} \times \mathcal{F}_p \to \mathcal{F}_p$. The natural map is given by mapping $(s, U), (x, V)$ to $(sx, U \cap V)$, which is easily seen to give a module structure. ■

Example 1.1.14

Consider a smooth manifold X; this defines a ringed space $(X, \mathcal{O}_X)$ where $\mathcal{O}_X$ is the sheaf of smooth real-valued functions: $\mathcal{O}_X(U) = C^\infty(U)$. These are all real algebras, so we could view this as an $\mathbb{R}$-ringed space.

If $\pi: V \to X$ is a vector bundle over X, then consider the sheaf of smooth local sections of π:

$$\Gamma(U) = \{\sigma: U \to V \text{ smooth} \mid \pi \circ \sigma = \text{id}_U\}.$$

Then Γ is an $\mathcal{O}_X$-module: it is a classical result that a local section defined on U can be multiplied by a smooth real-valued function defined on U.

1.2 The category of sheaves

As discussed, presheaves form a category for which sheaves form a full subcategory. We will denote the category of $\mathcal{C}$-valued presheaves over a space X by $\mathcal{C}_X^{\mathsf{pre}}$ (instead of $\text{Psh}_{\mathcal{C}}(X)$), and we will denote the full subcategory of sheaves by $\mathcal{C}_X \subseteq \mathcal{C}_X^{\mathsf{pre}}$.

Note that if $\mathcal{F}, G \in \mathcal{C}_X$ are sheaves (or presheaves), then the hom-set of sheaf morphisms $\mathcal{F} \to \mathcal{G}$ is indeed a set, since X as a category is small (functor categories under small categories are locally small). Furthermore, if $\mathcal{C}$ is enriched over some monoidal category (in our case, usually Ab) so is $\mathcal{C}_X$. This is because it is true in general for functor categories over enriched categories.

Definition 1.2.1

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-valued sheaves defined on X. Define their **sheaf-hom** to be the sheaf whose sections are sheaf morphisms,

$$\mathbf{Hom}(\mathcal{F}, \mathcal{G})(U) = \hom_{\mathcal{C}_X}(\mathcal{F}|_U, \mathcal{G}|_U).$$

As discussed, if $\mathcal{C}$ is enriched over a monoidal category $\mathcal{V}$ then so is $\mathcal{C}_X$, and so $\mathbf{Hom}(\mathcal{F}, \mathcal{G})$ takes values in $\mathcal{V}$. In particular, if $\mathcal{F}, \mathcal{G}$ are sheaves of Abelian groups, then so is their sheaf-hom.

Proposition 1.2.2

The sheaf-hom is indeed a sheaf.

Proof

We first show it is a sheaf: if $U \subseteq V$ then a sheaf morphism $\mathcal{F}|_V \to \mathcal{G}|_V$ restricts to a sheaf morphism $\mathcal{F}|_U \to \mathcal{G}|_U$ in the obvious way: its components are the same but only taken on subsets of U. This is clearly a presheaf.

To show that it is a sheaf, let $\{U_i\}$ be an open cover for $U \subseteq X$, and $\alpha_i \colon \mathcal{F}|_{U_i} \to \mathcal{G}|_{U_i}$ be sheaf morphisms such that $\alpha_i|_{U_i \cap U_j} = \alpha_j|_{U_i \cap U_j}$ for all i, j. In order to glue them together for a sheaf morphism $\alpha \colon \mathcal{F}_U \to \mathcal{G}_U$ we need to define α_W for all $W \subseteq U$. For $s \in \mathcal{F}(W)$ consider $s_i = s|_{U_i} \in U_i \cap W$, and its image under α_i to $\mathcal{G}(U_i \cap W)$, call it t_i. Then since $U_i \cap W$ cover W and since α is a morphism as well as $\mathcal{G}$ being a sheaf, we can uniquely glue t_i together to get a $t \in \mathcal{G}(W)$; this will be the image of s.

It is not hard to see that this is the unique gluing of α_i, and is itself a sheaf morphism. ■

Note:

In the above proof we needed only for $\mathcal{G}$ to be a sheaf, $\mathcal{F}$ can be a presheaf.

Furthermore **Hom** forms a bifunctor

$$\mathbf{Hom} \colon \mathcal{C}_X^{\mathsf{op}} \times \mathcal{C}_X \to \mathrm{Set}_X,$$

or V_X instead of Set_X when $\mathcal{C}$ is enriched over V.

Note that if $\mathcal{O}_X$ is a ringed space, it is also a module over itself. Thus if $\mathcal{F}$ is an $\mathcal{O}_X$ module, we can form its **dual**:

$$\mathcal{F}^\vee = \mathbf{Hom}_{\mathrm{Mod}_{\mathcal{O}_X}}(\mathcal{F}, \mathcal{O}_X).$$

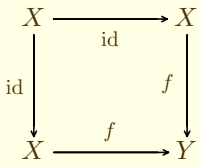
We now focus on sheaves of Abelian groups. It is a classical result that functor categories over Abelian categories are Abelian, thus $\mathrm{Ab}_X^{\mathsf{pre}}$ is an Abelian category. That is to say, the hom-sets (not hom-sheaves) have an Abelian group structure: for $\varphi, \psi \colon \mathcal{F} \to \mathcal{G}$ we can define $\varphi + \psi$ and $-\varphi$ by $(\varphi + \psi)(U) = \varphi(U) + \psi(U)$ and $(-\varphi)(U) = -\varphi(U)$). And we can form kernels and cokernels so that all monomorphisms are kernels and all epimorphisms are cokernels.

As with all limits in functor categories, these are done coordinate-wise. That is to say, if we denote by $\ker_{\mathsf{pre}}$ and $\mathrm{coker}_{\mathsf{pre}}$ the kernel and cokernel of Abelian presheaf morphisms, we have

$$(\ker_{\mathsf{pre}} \varphi)(U) = \ker(\varphi(U)), \qquad (\mathrm{coker}_{\mathsf{pre}} \varphi)(U) = \mathrm{coker}(\varphi(U)).$$

Remark 1.2.3

Let I be an indexing category and $\mathcal{C}$ be a category with arbitrary pullbacks. It is easy to see that a $\mathcal{C}$-morphism $f \colon X \to Y$ is monic iff the following square is a pullback:



Because the functor category $[I, \mathcal{C}]$ also has arbitrary pullbacks, a natural transformation $\alpha: \mathcal{F} \Rightarrow \mathcal{G}$ is monic iff a similar diagram is a pullback. Because limits in a functor category are taken coordinate-wise, this is iff $\alpha(i)$ is monic for all $i \in I$.

Thus a natural transformation is monic iff its coordinates are.

Similarly, if $\mathcal{C}$ has arbitrary pushouts, natural transformations in $[I, \mathcal{C}]$ are epic iff their coordinates are.

In lieu of this note, we see that $\mathrm{Ab}_X^{\mathsf{pre}}$ forms an Abelian category. Indeed, if $\varphi: \mathcal{F} \Rightarrow \mathcal{G}$ is a monic morphism of presheaves, then $\varphi(U): \mathcal{F}(U) \Rightarrow \mathcal{G}(U)$ is a monic morphism of Abelian groups for all $U \subseteq X$. Thus $\varphi(U) = \ker \operatorname{coker} \varphi(U)$ for all $U \subseteq X$ and thus $\varphi = \ker \operatorname{coker} \varphi$ as desired. Similar for epic morphisms.

Many properties of $\mathrm{Ab}_X^{\mathsf{pre}}$ are seen to be pointwise. For one, if we set an open set $U \subseteq X$, the functor $\mathrm{Ab}_X^{\mathsf{pre}} \to \mathrm{Ab}$ mapping $\mathcal{F} \mapsto \mathcal{F}(U)$ is an exact functor. Indeed, it is additive and preserves kernels and cokernels because these are taken pointwise.

Furthermore, a sequence of presheaves

$$\mathcal{F}_1 \to \mathcal{F}_2 \to \cdots \to \mathcal{F}_n$$

is exact iff $\mathcal{F}_1(U) \to \cdots \to \mathcal{F}_n(U)$ is for all $U \subseteq X$. Again, this is an immediate consequence of kernels and cokernels (and thus images) being taken coordinate-wise.

Everything we have said holds for presheaves valued in any Abelian category $\mathcal{A}$ (recall that Abelian categories have all finite limits and colimits). But we are more interested in the category of sheaves, in which case this is less obvious.

Note that the constant presheaf $\underline{0}_{\mathsf{pre}}$ is a sheaf, so we can denote it by $\underline{0}$ without confusion. In order to show that $\mathcal{A}_X$ is an Abelian category, we must first show that kernels and cokernels exist in it. The following proposition shows that the kernel of a sheaf morphism is the same as the kernel of a presheaf morphism, and thus we get the existence of kernels for free.

Proposition 1.2.4

If $\varphi: \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves, then $\ker_{\mathsf{pre}} \varphi$ is a sheaf. Thus $\ker \varphi = \ker_{\mathsf{pre}} \varphi$ forms a kernel in the category of sheaves.

Proof

We show this for $\mathcal{A} = \mathrm{Ab}$: if $\{U_i\}_i$ is an open cover of $U \subseteq X$ and $s_i \in \ker \varphi(U_i)$ are such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j, then by gluing they form a section $s \in \mathcal{F}(U)$. And by naturality

$$\varphi(U)(s)|_{U_i} = \varphi(U_i)(s|_{U_i}) = \varphi(U_i)(s_i) = 0.$$

The unique gluing of $0 \in \mathcal{G}(U_i)$ is $0 \in \mathcal{G}(U)$, so that $\varphi(U)(s) = 0$, meaning $s \in \ker \varphi(U)$ as desired. ■

Remark 1.2.5

A generalization of the above proposition shows that limits taken in the category of presheaves of sheaves forms a sheaf. That is, if $\Phi: I \to \mathcal{C}_X$ is a functor, then consider the inclusion $\iota: \mathcal{C}_X \hookrightarrow \mathcal{C}_X^{\mathsf{pre}}$. Then $\lim \iota \circ \Phi$ is a sheaf, so $\lim \Phi$ exists in $\mathcal{C}_X$.

Thus if $\mathcal{C}$ has arbitrary pullbacks, so too does $\mathcal{C}_X^{\mathsf{pre}}$ and thus so too does $\mathcal{C}_X$. It then follows that a morphism

of sheaves is monic iff its monic as a morphism of presheaves (considering the pullback condition remark 1.2.3) iff each component is monic (again by remark 1.2.3).

Unfortunately we now run in to our first hurdle: presheaf cokernels are not sheaves.

Example 1.2.6

Consider $\mathbb{C}$ with its usual topology, and let $\mathcal{O}_\mathbb{C}$ be the sheaf of holomorphic functions. Let $\mathcal{F}$ be the presheaf of functions admitting a holomorphic logarithm. Consider the following exact sequence of presheaves

$$0 \to \underline{\mathbb{Z}} \hookrightarrow \mathcal{O}_\mathbb{C} \xrightarrow{\exp(2\pi i)} \mathcal{F} \to 0,$$

where the second map is $f \mapsto \exp(2\pi i f)$. This is indeed an exact sequence of presheaves: the first morphism is an embedding, and the second is epic (if $f = \exp(g)$ then $g/2\pi i \in \mathcal{O}_\mathbb{C}$ is an origin for f). The composition is zero, as $\exp(2\pi i n) = 0$, and if $\exp(2\pi i f) = 0$ then $f(x) \in \mathbb{Z}$ for all $x \in \mathbb{C}$. Because $\mathbb{Z}$ is discrete, this means $f \in \underline{\mathbb{Z}}$ as desired.

This shows that $\mathcal{F}$ is the cokernel of $\underline{\mathbb{Z}} \hookrightarrow \mathcal{O}_\mathbb{C}$: but it is not a sheaf! Indeed $\mathrm{id}\colon \mathbb{C} - 0 \to \mathbb{C}$ has a local logarithm at every point in its domain, but no global logarithm. Thus $\mathcal{F}$ fails the gluing axiom.

Thus for now we must wait to demonstrate that $\mathcal{A}_X$ is an Abelian category.

1.3 Sheafification

There is a natural map

$$\mathcal{F}(U) \to \mathcal{F}_p,$$

for every $p \in U$ given by mapping $s \in \mathcal{F}(U)$ to $(U, s) \in \mathcal{F}_p$. Together these assimilate into a map

$$\mathcal{F}(U) \to \prod_{p \in U} \mathcal{F}_p.$$

Proposition 1.3.1

The natural map above is injective when $\mathcal{F}$ is a sheaf.

Proof

Suppose that $s, t \in \mathcal{F}(U)$ such that $(U, s) = (U, t)$ in $\mathcal{F}_p$. Then there exists a neighborhood $p \in W_p \subseteq U$ such that $s|_{W_p} = t|_{W_p}$. If this is true for all p, then W_p form an open cover of U and by the identity axiom, $s = t$. ∎

Definition 1.3.2

An element $(s_p)_p \in \prod_{p \in U} \mathcal{F}_p$ consists of **compatible germs** if for all $p \in U$, s_p has a representative $\widetilde{s}_p \in \mathcal{F}(U_p)$ for $p \in U_p \subseteq U$ such that the germ of $\widetilde{s}_p$ at $q \in U_p$ is s_q for all $q \in U_p$.

Equivalently, this says that there is an open cover $\{U_i\}$ of U and sections $f_i \in \mathcal{F}(U_i)$ such that if $p \in U_i$ then s_p is the germ of f_i at p. If s is a section of $\mathcal{F}$ over U, then $(s)_p$ consists of compatible germs.

Proposition 1.3.3

The natural map $\mathcal{F}(U) \to \prod_{p\in U} \mathcal{F}_p$'s image is precisely the compatible germs.

Proof

We already showed that every tuple in its image consists of compatible germs, now we show the converse. Suppose $(s_p)_p$ consists of compatible germs, so let $\widetilde{s}_p \in \mathcal{F}(U_p)$ be a representation as in the definition. Then U_p cover U, and $\widetilde{s}_p|_{U_p\cap U_q}$'s germ at q is s_q so that $\widetilde{s}_p, \widetilde{s}_q$ agree on a neighborhood of q and by symmetry in a neighborhood of p and q. ■

Proposition 1.3.4

Suppose $\mathcal{F}$ is a presheaf of sets and $\mathcal{G}$ a sheaf of sets, and $\varphi_1, \varphi_2 \colon \mathcal{F} \to \mathcal{G}$ are two morphisms. Further suppose that φ_1, φ_2 induce the same map on each stalk; show that $\varphi_1 = \varphi_2$.

Proof

Consider the following diagram for $U \subseteq X$ open,

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow[\varphi_i]{} & \mathcal{G}(U) \\ \downarrow & & \downarrow{\scriptstyle \iota} \\ \prod_{p\in U} \mathcal{F}_p & \xrightarrow{\prod \varphi_{i,p}} & \prod_{p\in U} \mathcal{G}_p \end{array}$$

for $i = 1, 2$. It is easily seen that this diagram commutes, and because $\varphi_{1,p} = \varphi_{2,p}$ for all $p \in U$ it follows that

$$\iota \circ \varphi_1 = \iota \circ \varphi_2$$

and since ι is injective by proposition 1.3.1, we get that $\varphi_1 = \varphi_2$ as desired. ■

Proposition 1.3.5

A morphism of sheaves of sets is injective iff it induces an injection on all stalks. Similarly it is an isomorphism iff it induces an isomorphism on all stalks.

Proof

If $\varphi \colon \mathcal{F} \to \mathcal{G}$ is an isomorphism, then by functorality $\varphi_p \colon \mathcal{F}_p \to \mathcal{G}_p$ is as well for all p. Similarly if it is an injection then $\varphi_p(s_p) = \varphi_p(t_p)$ then $\varphi(s)$ and $\varphi(t)$ are equal in a subset $V \subseteq U$ so that $\varphi(s|_V) = \varphi(t|_V)$ and by injectivity this means that $s|_V = t|_V$ and thus $s_p = t_p$.

Conversely consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \iota_{\mathcal{F}}} & & \downarrow{\scriptstyle \iota_{\mathcal{G}}} \\ \prod_{p\in U} \mathcal{F}_p & \xrightarrow{\sim} & \prod_{p\in U} \mathcal{G}_p \end{array}$$

where the bottom horizontal arrow is the injection/isomorphism $\prod_{p\in U} \varphi_p$. Then $\iota_{\mathcal{G}} \circ \varphi$ is injective, and so φ is.

Now to show that φ is surjective, let $s \in \mathcal{G}(U)$. Then $\iota_{\mathcal{G}}(s) \in \prod_U \mathcal{G}_p$ consists of compatible germs, and mapping it back to $\prod_U \mathcal{F}_p$ we see that it still consists of compatible germs. Thus it lies in the image of $\iota_{\mathcal{F}}$ by proposition 1.3.3, and thus has a preimage in $\mathcal{F}(U)$. Because all of the maps are injective, this is the preimage of s in $\mathcal{F}(U)$ as desired. ■

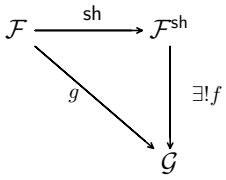
Note:
It is not true that sheaves with isomorphic stalks are isomorphic: there may not be an isomorphism between the sheaves which induces these isomorphisms.

Theorem 1.3.6 (Sheafification)

The inclusion $\mathrm{Set}_X \hookrightarrow \mathrm{Set}_X^{\mathsf{pre}}$ has a left adjoint, called **sheafification**: $\mathsf{sh}\colon \mathrm{Set}_X^{\mathsf{pre}} \to \mathrm{Set}_X$.

Proof

We need to show for any presheaf $\mathcal{F}$ there exists a sheaf $\mathcal{F}^{\mathsf{sh}}$ and morphism $\mathsf{sh}\colon \mathcal{F} \to \mathcal{F}^{\mathsf{sh}}$ such that for every sheaf $\mathcal{G}$ and morphism $g\colon \mathcal{F} \to \mathcal{G}$ there exists a unique morphism $f\colon \mathcal{F}^{\mathsf{sh}} \to \mathcal{G}$ such that the below diagram commutes:



We construct $\mathcal{F}^{\mathsf{sh}}$ explicitly. For $U \subseteq X$ open, define

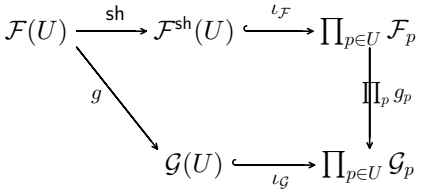
$$\mathcal{F}^{\mathsf{sh}}(U) = \left\{ (f_p \in \mathcal{F}_p)_{p\in U} \;\middle|\; \begin{array}{l} \text{for all } p \in U \text{ there exists a neighborhood } p \in V \subseteq U \text{ and } s \in \mathcal{F}(V) \\ \text{such that } s_q = f_q \text{ for all } q \in V \end{array} \right\},$$

where by s_q we mean the germ of s in $\mathcal{F}_q$. For $U \subseteq V$ there is an obvious map $\mathcal{F}^{\mathsf{sh}}(V) \to \mathcal{F}^{\mathsf{sh}}(U)$ mapping $(f_p)_{p\in V} \mapsto (f_p)_{p\in U}$; it is easily seen that $\mathcal{F}^{\mathsf{sh}}$ is a presheaf.

Note that the construction of $\mathcal{F}^{\mathsf{sh}}(U)$ is just the set of all families of compatible germs over U, which we discussed in the previous few results.

Now let $U \subseteq X$ be open and $\{U_i\}_i$ an open cover of U. Then if we have $(f_p^i)_{p\in U_i} \in \mathcal{F}^{\mathsf{sh}}(U_i)$ for all i which are compatible, this means $f_p^i = f_p^j$ for all $p \in U_i \cap U_j$. Then we must define $(f_p)_{p\in U}$ by $f_p = f_p^i$ for any $i \in U_i$ for which p is a member of. This is well-defined precisely by compatibility, and it is a member of $\mathcal{F}^{\mathsf{sh}}(U)$ because for $p \in U$ there is a $p \in U_i \subseteq U$ and then there is a $p \in V \subseteq U_i$ and $s \in \mathcal{F}(V)$ such that $s_q = f_q^i = f_q$ for all $q \in V$ as desired. Thus $\mathcal{F}^{\mathsf{sh}}$ is a sheaf.

Now we must demonstrate that $\mathcal{F}^{\mathsf{sh}}$ satisfies the universal property of a left adjoint. There is an obvious map $\mathsf{sh}\colon \mathcal{F} \to \mathcal{F}^{\mathsf{sh}}$ which maps $s \in \mathcal{F}(U)$ to $(s_p \in \mathcal{F}_p)_{p\in U} \in \mathcal{F}^{\mathsf{sh}}(U)$ (this is a section because s satisfies the condition for it to be in $\mathcal{F}^{\mathsf{sh}}(U)$). Let $g\colon \mathcal{F} \to \mathcal{G}$ be a morphism, and for $U \subseteq X$ consider the following diagram



Note that the image of $\prod_p g_p \circ \iota_{\mathcal{F}}$ consists of compatible germs in $\prod_{p\in U} \mathcal{G}_p$ and so we can fill this diagram in with a unique arrow $f\colon \mathcal{F}^{\mathsf{sh}}(U) \to \mathcal{G}(U)$. It is easy to see by uniqueness that these f assemble into a morphism $\mathcal{F}^{\mathsf{sh}} \to \mathcal{G}$. ■

Note:
In the language of category theory, this theorem says that $\mathrm{Set}_X \subseteq \mathrm{Set}_X^{\mathsf{pre}}$ is a **reflective subcategory**. Note that this proof works for other concrete categories like $\mathrm{Ab}, \mathrm{Grp}, \mathrm{Ring}$, etc. It can be shown that for sufficiently nice $\mathcal{C}$ (e.g. complete and cocomplete), sheafification exists for $\mathcal{C}_X \subseteq \mathcal{C}_X^{\mathsf{pre}}$.
This justifies the fact that inclusion $\mathcal{C}_X \subseteq \mathcal{C}_X^{\mathsf{pre}}$ preserves limits: it is a right adjoint. This explains why the presheaf kernel of a sheaf morphism is a sheaf kernel for example.

Proposition 1.3.7

The canonical morphism $\mathcal{F} \to \mathcal{F}^{\mathsf{sh}}$ induces an isomorphism on stalks.

Proof

This is easily seen by an explicit description of $\mathcal{F}^{\mathsf{sh}}$'s stalks. ■

Remark 1.3.8 (The space of sections of a presheaf)

Suppose $\mathcal{F}$ is a presheaf of sets on X. We will construct a topological space F along with a continuous map $\pi: F \to X$ as follows. As a set define F to be the disjoint unions of $\mathcal{F}$'s stalks:

$$F = \coprod_{p \in X} \mathcal{F}_p.$$

Now for $s \in \mathcal{F}(U)$ let $F_s = \{(p, s_p) \mid p \in U\} \subseteq F$; then the collection of all F_s where $s \in \mathcal{F}(U)$ for arbitrary $U \subseteq X$ forms a base for a topology on F. Indeed, for $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ we see that $F_s \cap F_t$ is the union of all F_r where $r = s|_W = t_W$ for all $W \subseteq U \cap V$ where $s|_W = t|_W$.
The map $\pi: F \to X$ is the obvious projection $(p, s_p) \mapsto p$. For $U \subseteq X$ open we have that $\pi^{-1}U$ is the union of all F_s where $s \in \mathcal{F}(U)$ so that π is continuous. Furthermore π is an open map: for $s \in \mathcal{F}(U)$ it maps F_s to U. Moreso $\pi|_{F_s}$ is a bijection, so that π is a local homeomorphism.
The space F is called the **space of sections**, or **étalé space** of $\mathcal{F}$.
We can give an alternative construction for the sheafification of $\mathcal{F}$ as the sheaf of sections of $\pi: F \to X$ (that is for $U \subseteq X$ we define $\mathcal{F}^{\mathsf{sh}}(U)$ to be the set of continuous maps $\sigma: U \to F$ where $\pi \circ \sigma = \mathrm{id}_U$). Let us denote by $\mathcal{F}^+$ this new construction.
For $U \subseteq X$ open we define $\varphi: \mathcal{F}^{\mathsf{sh}}(U) \to \mathcal{F}^+(U)$ by mapping $(f_p)_{p \in U}$ to $\sigma: U \to F$, $p \mapsto f_p$. Its inverse is given by mapping σ to $(\sigma(p))_p$.

Example 1.3.9

Let S be a set and consider the constant presheaf $\underline{S}_{\mathsf{pre}}$. Then its sheafification is the constant *sheaf*: $(\underline{S}_{\mathsf{pre}})^{\mathsf{sh}} = \underline{S}$.
We can show this with the universal property: $\mathsf{sh}: \underline{S}_{\mathsf{pre}} \to \underline{S}$ is given by inclusion (constant maps are locally constant). Then for a morphism $g: \underline{S}_{\mathsf{pre}} \to \mathcal{F}$ where $\mathcal{F}$ is a sheaf, we define $\overline{g}: \underline{S} \to \mathcal{F}$ by mapping a locally constant $f: U \to S$ to the gluing of $(\overline{g}(f|_{U_i}))_i$ (where $U_i \subseteq U$ are its connected components so that $f|_{U_i}$ is a constant).

Let $\varphi: \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves, then recall the following equivalences (see remark 1.2.5 and proposition 1.3.5):

(1) φ is monic as a morphism of sheaves,

(2) φ is monic as a morphism of presheaves,

(3) $\varphi(U)$ is monic for all $U \subseteq X$,

(4) φ_p is monic for all $p \in X$.

A similar equivalence does not hold true for epimorphisms. This boils down to the fact that inclusion $\mathcal{C}_X \subseteq \mathcal{C}_X^{\mathsf{pre}}$ does not necessarily preserve colimits, and thus not epimorphisms. So while a presheaf morphism is epic iff its components are, the same does not hold for sheaf morphisms. But it is true that epimorphisms are determined by stalks.

Proposition 1.3.10

Let $\varphi: \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then it is an epimorphism (in the category of sheaves) iff $\varphi_p: \mathcal{F}_p \to \mathcal{G}_p$ is an epimorphism for all $p \in X$.

Proof

If φ_p is epic for all p, then suppose we have two sheaf morphisms $\psi_1, \psi_2: \mathcal{G} \to \mathcal{H}$ such that $\psi_1 \circ \varphi = \psi_2 \circ \varphi$. Then taking stalks we see that for all p, $(\psi_1)_p \circ \varphi_p = (\psi_2)_p \circ \varphi_p$. Since φ_p is epic this means $(\psi_1)_p = (\psi_2)_p$ for all p and by proposition 1.3.4 this means $\psi_1 = \psi_2$ so that φ is epic.

Conversely let $p \in X$ and suppose there are maps $f, g: \mathcal{G}_p \to X$ such that $f \circ \varphi_p = g \circ \varphi_p$. We can define maps $\overline{f}, \overline{g}: \mathcal{G} \to p_* X$ by $\overline{f}_U(s) = f(s_p)$ for $p \in U$ and $\overline{f}_U = *$ otherwise. It is easy to see that $\overline{f} \circ \varphi = \overline{g} \circ \varphi$ and so $\overline{f} = \overline{g}$ which can easily be shown to mean $f = g$. ■

Note:

In remark 1.6.2 will show that there is an adjunction between taking stalks and the skyscraper sheaf: $\bullet_p \dashv p_* \bullet$. This was used in the above proof, and it is easy to see then how to construct this.

Example 1.3.11

We will show that an epimorphism of sheaves need not be epic on each of its components.

Let $\mathcal{O}_{\mathbb{C}}$ be the sheaf of holomorphic functions and $\mathcal{O}_{\mathbb{C}}^*$ be the sheaf of invertible (nowhere zero) holomorphic functions. These are sheaves of Abelian groups, and we then have maps of sheaves

$$0 \to \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \to 1.$$

We will soon show that this is an exact sequence of sheaves of Abelian groups.

In particular $\exp: \mathcal{O}_{\mathbb{C}} \to \mathcal{O}_{\mathbb{C}}^*$ is an epimorphism as we can show directly. For any $z \in \mathcal{C}$ the morphism of stalks $\exp_z$ is surjective because every invertible holomorphic function locally has a logarithm. But exp is not an epimorphism on each of its components, for example $\mathrm{id} \in \mathcal{O}_{\mathbb{C}}^*(\mathbb{C} - 0)$ has no global logarithm.

1.4 B-Sheaves

Oftentimes it is more natural to define a sheaf on X by defining its sections on a base of X, instead of on every open subset of X. Recall that a base of a topology is a collection of open subsets $\mathcal{B} \subseteq \mathrm{open}(X)$ such that every open subset of X can be written as a union of sets from $\mathcal{B}$.

In the absence of an ambient topology, we may mean a base to mean a collection $\mathcal{B}$ of subsets of X such that the intersection of any two sets from $\mathcal{B}$ can be written as a union of elements from $\mathcal{B}$. If we are given an arbitrary collection S of subsets of X, they generate a minimal topology on X; S is called a **subbase** of this topology. The topology generated by $\mathcal{B}$ consists of all finite intersections of elements from $\mathcal{B}$, and $\mathcal{B}$ is a base of this topology (in the first sense).

As $\mathcal{B}$ is a collection of subsets of X, we can view it as a poset under inclusion, and thus a subcategory of $\mathrm{open}(X)$.

Definition 1.4.1 (B-Sheaves)

Let X be a topological space with a given base $\mathcal{B}$. Define a **$\mathcal{B}$-presheaf** to be a functor $\mathcal{F}\colon \mathcal{B}^{\mathsf{op}} \to \mathrm{Set}$. This presheaf is a **$\mathcal{B}$-sheaf** if it satisfies the following two axioms:

(1) the **identity axiom**: if $B = \bigcup_i B_i$ is a basic covering and $f, g \in \mathcal{F}(B)$ such that $f|_{B_i} = g|_{B_i}$ for all i, then $f = g$.

(2) the **gluing axiom**: if $B = \bigcup_i B_i$ is a basic covering and $f_i \in \mathcal{F}(B_i)$ are given for all i such that $f_i|_{B_k} = f_j|_{B_k}$ for all $B_k \subseteq B_i \cap B_j$, then there is an $f \in \mathcal{F}(B)$ such that $f|_{B_i} = f_i$ for all i.

We can rewrite the following axioms categorically as follows. For $B = \bigcup_i B_i$, the following diagram must be an equalizer map

$$\mathcal{F}(B) \xrightarrow{\varphi} \prod_i B_i \overset{\alpha}{\underset{\beta}{\rightrightarrows}} \prod_{i,j,k} B_{i,j,k}$$

where $\{B_{i,j,k}\}_k$ ranges over all basic sets contained in $B_i \cap B_j$, and

$$\theta = (\mathrm{res}_{B,B_i})_i, \qquad \alpha = (\mathrm{res}_{B_i,B_k} \circ \mathrm{proj}_i)_{i,j,k}, \qquad \beta = (\mathrm{res}_{B_j,B_k} \circ \mathrm{proj}_j)_{i,j,k}.$$

Thus we can define $\mathcal{B}$-sheaves over any category $\mathcal{C}$ with enough limits.

Definition 1.4.2

Denote the category of $\mathcal{C}$-valued $\mathcal{B}$-presheaves by $\mathcal{C}_{\mathcal{B}}^{\mathsf{pre}}$, and the category of $\mathcal{C}$-valued $\mathcal{B}$-sheaves by $\mathcal{C}_{\mathcal{B}}$.

Theorem 1.4.3

Let $\mathcal{B}$ be a base for a topology on X, and $\mathcal{F}$ a sheaf of sets on this base. Then there is a unique sheaf of sets $\mathcal{F}^{\mathsf{ext}}$ on X extending $\mathcal{F}$.

This extends to an equivalence of categories $\mathsf{ext}\colon \mathrm{Set}_{\mathcal{B}} \xrightarrow{\sim} \mathrm{Set}_X \colon \mathrm{res}$, where $\mathrm{res}\colon \mathrm{Set}_X \to \mathrm{Set}_{\mathcal{B}}$ is the obvious restriction functor.

Proof

For a $\mathcal{B}$-sheaf we can define stalks much the same was as for a usual sheaf. For $p \in X$ define

$$\mathcal{F}_p = \mathrm{colim}_{p \in B \in \mathcal{B}} \mathcal{F}(B).$$

Then define for $U \subseteq X$,

$$\mathcal{F}^{\mathsf{ext}}(U) = \left\{ (f_p \in \mathcal{F}_p)_{p \in U} \;\middle|\; \begin{array}{l} \text{for all } p \in U \text{ there is a } B \in \mathcal{B} \text{ with } p \in B \subseteq U,\ s \in \mathcal{F}(B) \\ \text{such that } s_q = f_q \text{ for all } q \in B \end{array} \right\}.$$

There are obvious restriction maps and it is easy to see that this is a sheaf. Furthermore it is easy to see that $\mathcal{F}^{\mathsf{ext}}(B)$ is the image of $\mathcal{F}(B)$ under $\iota_{\mathcal{F}}$ which is an injection as for usual sheaves. ∎

Remark 1.4.4

Suppose we have an open cover $X = \bigcup_i U_i$ and for each i a sheaf of sets $\mathcal{F}_i$ on U_i. Furthermore suppose we have isomorphisms

$$\varphi_{ij}\colon \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j}$$

such that φ_{ii} is the identity and

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$$

for all i, j, k on $U_i \cap U_j \cap U_k$.

There then exists a unique sheaf $\mathcal{F}$ on X such that $\mathcal{F}|_{U_i}$ is isomorphic to $\mathcal{F}_i$.

This is because we can define a base $\mathcal{B}$ consisting of those open subsets contained in some U_i, i.e. $\mathcal{B} = \bigcup_i \operatorname{open}(U_i)$. Then there is a unique way to extend the $\mathcal{F}_i$s to a $\mathcal{B}$-sheaf. For $U \subseteq U_i$ we define $\mathcal{F}(U) = \mathcal{F}_i(U)$ (this is well-defined because if $U \subseteq U_i \cap U_j$ then $\mathcal{F}_i(U) \cong \mathcal{F}_j(U)$), and the restriction maps are given by the restriction maps of $\mathcal{F}_i$ and the φ_{ij}. That is, for $U \subseteq U_i$ and $V \subseteq U_j$ where $U \subseteq V$ we define $\operatorname{res}\colon \mathcal{F}_j(V) \to \mathcal{F}_i(U)$ by

$$\operatorname{res} = \mathcal{F}_j(V) \xrightarrow{\operatorname{res}_j} \mathcal{F}_j(U) \xrightarrow{\varphi_{ji}} \mathcal{F}_i(U).$$

The conditions on φ_{ij} ensure that this is a $\mathcal{B}$-sheaf, and we can extend it to a sheaf on X via the above theorem.

1.5 Abelian categories of sheaves

Recall that for a sheaf of rings $\mathcal{O}_X$, an **$\mathcal{O}_X$-module** is a sheaf of Abelian groups $\mathcal{F}$ such that each $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module in a compatible way (definition 1.1.12). A sheaf of Abelian groups is a $\underline{\mathbb{Z}}$-module, so $\mathcal{O}_X$-modules generalize the concept of sheaves of Abelian groups much like how modules generalize the concept of Abelian groups.

The goal of this section is to demonstrate that the category of $\mathcal{O}_X$-modules (and in particular sheaves of Abelian groups) forms an Abelian category.

To do this we need to show that the category of $\mathcal{O}_X$-modules has kernels and cokernels, and that monics are kernels and epics are cokernels. Firstly let us consider $\operatorname{Mod}^{\mathsf{pre}}_{\mathcal{O}_X}$, the category of presheaf $\mathcal{O}_X$-modules. The definition here is the same as definition 1.1.12 except we require only that $\mathcal{F}$ be a presheaf.

Now $\operatorname{Mod}^{\mathsf{pre}}_{\mathcal{O}_X}$ is complete and cocomplete. If $\Phi\colon I \to \operatorname{Mod}^{\mathsf{pre}}_{\mathcal{O}_X}$ is a diagram, then there is an obvious forgetful functor $\operatorname{Mod}^{\mathsf{pre}}_{\mathcal{O}_X} \to \operatorname{Ab}^{\mathsf{pre}}_X$. Composing this with Φ we can take the co/limit, and then there is an obvious $\mathcal{O}_X$-module structure on the resulting object (this is the same situation as with $\operatorname{Mod}_R \to \operatorname{Ab}$).

Then sheafification lifts to $\operatorname{Mod}^{\mathsf{pre}}_{\mathcal{O}_X} \to \operatorname{Mod}_{\mathcal{O}_X}$. This is clear by considering $\mathcal{F}^{\mathsf{sh}}(U)$ explicitly. Thus $\operatorname{Mod}_{\mathcal{O}_X}$ is a reflective subcategory of $\operatorname{Mod}^{\mathsf{pre}}_{\mathcal{O}_X}$, and since the latter is co/complete so too is the former.

In particular the kernel of a $\operatorname{Mod}_{\mathcal{O}_X}$-morphism is the kernel of it viewed as a morphism in $\operatorname{Ab}^{\mathsf{pre}}_X$, and its cokernel is the sheafification of this morphism.

Recall that the stalk of a morphism in $\operatorname{Mod}_{\mathcal{O}_X}$ is an $\mathcal{O}_{X,p}$-module. Thus we view $\bullet_p$ as a functor

$$\bullet_p\colon \operatorname{Mod}_{\mathcal{O}_X} \to \operatorname{Mod}_{\mathcal{O}_{X,p}}.$$

When $\mathcal{O}_X = \underline{\mathbb{Z}}$ then $\operatorname{Mod}_{\mathcal{O}_X}$ is the same as Ab_X, and $\mathcal{O}_{X,p}$ is the same as $\mathbb{Z}$ so that $\operatorname{Mod}_{\mathcal{O}_{X,p}} \cong \operatorname{Ab}$. Thus all of the following results hold for sheaves of Abelian groups.

Lemma 1.5.1

Taking stalks preserves kernels and cokernels: for $p \in X$ and a morphism $\varphi\colon \mathcal{F} \to \mathcal{G}$ of $\mathcal{O}_X$-modules,

$$(\ker \varphi)_p \cong \ker(\varphi_p), \qquad (\operatorname{coker} \varphi)_p \cong \operatorname{coker}(\varphi_p).$$

Proof

This is clear for cokernels, as the stalk at p amounts to taking a certain colimit, and colimits (such as cokernels) commute with colimits.

For kernels we can show this directly. We view both $(\ker \varphi)_p$ and $\ker(\varphi_p)$ canonically as subobjects of $\mathcal{F}_p$, in which case it is clear that there is an equality. ∎

Lemma 1.5.2

In $\mathrm{Mod}_{\mathcal{O}_X}$, a morphism $\varphi\colon \mathcal{F} \to \mathcal{G}$ is monic iff its stalks are monic, epic iff its stalks are epic, and an isomorphism iff its stalks are isomorphisms.

Note:

It doesn't matter if we view φ_p as existing in $\mathrm{Mod}_{\mathcal{O}_{X,p}}$ or Set; a module morphism (in the ordinary sense) is a monomorphism/epimorphism/isomorphism iff it is injective/surjective/bijective.

Note that in lieu of this, a $\mathrm{Mod}_{\mathcal{O}_X}$-morphism with any of the above properties has those properties in Ab_X and Set_X.

Proof

First we note that stalks determine a morphism: if $\varphi_p = \psi_p$ for all $p \in X$ then $\varphi = \psi$. This is because the forgetful functor $U\colon \mathrm{Mod}_{\mathcal{O}_X} \to \mathrm{Set}_X$ is faithful, and commutes with taking stalks. So $U\varphi_p = U\psi_p$, which implies $U\varphi = U\psi$ by proposition 1.3.4, and thus $\varphi = \psi$.

Then the proofs of proposition 1.3.5 and proposition 1.3.10 are valid here. ■

This allows us to prove the desired result.

Theorem 1.5.3

$\mathrm{Mod}_{\mathcal{O}_X}$ is an Abelian category.

Proof

As already shown, $\mathrm{Mod}_{\mathcal{O}_X}$ is co/complete. Furthermore it is clearly a pre-additive category. We need only show that for a monic φ and an epic ψ,

$$\varphi = \ker \operatorname{coker} \varphi, \qquad \psi = \operatorname{coker} \ker \psi.$$

But this is true only if for all $p \in X$,

$$\varphi_p = \ker \operatorname{coker} \varphi_p, \qquad \psi_p = \operatorname{coker} \ker \psi_p \tag{1}$$

(parentheses on the left-hand sides of the equations here are unnecessary, as per lemma 1.5.1). By lemma 1.5.2 φ_p, ψ_p are monic and epic respectively, so (1) holds as $\mathrm{Mod}_{\mathcal{O}_{X,p}}$ is Abelian. ■

Note:

It is actually the case that if $\mathcal{A}$ is an arbitrary (complete) Abelian category, then $\mathcal{A}_X$ is Abelian as well.

Lemma 1.5.4

Let φ be a $\mathrm{Mod}_{\mathcal{O}_X}$-morphism, then $\operatorname{im}\varphi$ is the sheafification of the image presheaf of φ. In particular $(\operatorname{im}\varphi)_p = \operatorname{im}(\varphi_p)$.

Proof

We know that $\operatorname{im}_{\mathsf{pre}}\varphi = \ker_{\mathsf{pre}}\operatorname{coker}_{\mathsf{pre}}\varphi$, while $\operatorname{im}\varphi = \ker\operatorname{coker}\varphi$. Since $\operatorname{coker} = \operatorname{coker}^{\mathsf{sh}}_{\mathsf{pre}}$ and $\ker = \ker^{\mathsf{sh}}_{\mathsf{pre}} = \ker_{\mathsf{pre}}$ so

$$\operatorname{im}\varphi = \ker^{\mathsf{sh}}_{\mathsf{pre}}\operatorname{coker}^{\mathsf{sh}}_{\mathsf{pre}}\varphi = (\ker_{\mathsf{pre}}\operatorname{coker}_{\mathsf{pre}}\varphi)^{\mathsf{sh}} = (\operatorname{im}_{\mathsf{pre}}\varphi)^{\mathsf{sh}}$$

as desired.

The second point follows from stalks commuting with kernels and cokernels. ■

Corollary 1.5.5

The following sequence of $\mathrm{Mod}_{\mathcal{O}_X}$-morphisms

$$\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$$

is exact iff

$$\mathcal{F}_p \xrightarrow{\alpha_p} \mathcal{G}_p \xrightarrow{\beta_p} \mathcal{H}_p$$

is exact for all $p \in X$.

Proof

The former is exact iff $\operatorname{im}\alpha = \ker\beta$ iff $(\operatorname{im}\alpha)_p = (\ker\beta)_p$ for all $p \in X$, which happens iff $\operatorname{im}\alpha_p = \ker\beta_p$, i.e. if the latter sequence is exact. ■

An immediate consequence of this corollary is the following.

Corollary 1.5.6

Taking stalks is an exact functor $\bullet_p\colon \mathrm{Mod}_{\mathcal{O}_X} \to \mathrm{Mod}_{\mathcal{O}_{X,p}}$.

Example 1.5.7

For $U \subseteq X$, there is a functor $\Gamma_U\colon \mathcal{C}_X \to \mathcal{C}$ mapping a sheaf $\mathcal{F} \mapsto \mathcal{F}(U)$. It is easy to see that $\Gamma_U\colon \mathrm{Mod}_{\mathcal{O}_X} \to \mathrm{Mod}_{\mathcal{O}_X(U)}$ is a left-exact functor. That is, it preserves kernels. This is clear because the kernel of a morphism of sheaves $\mathcal{F} \to \mathcal{G}$ is just the presheaf kernel, which is taken coordinate-wise.

When $U = X$ we denote this by just $\Gamma = \Gamma_X$, and it is called the **global sections functor**. Note that there is an obvious restriction functor $\mathrm{res}\colon \mathcal{C}_X \to \mathcal{C}_U$ mapping a sheaf $\mathcal{F}$ to its restriction $\mathcal{F}|_U$. Then Γ_U is equal to the composition

$$\Gamma_U = \mathcal{C}_X \xrightarrow{\mathrm{res}} \mathcal{C}_U \xrightarrow{\Gamma} \mathcal{C}.$$

And it is easy to see that res is also exact.

It is not true that Γ_U is exact though. Consider the exact sequence sequence

$$0 \to \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}^*_{\mathbb{C}} \to 1$$

of example 1.3.11. As before, $\Gamma_U(\exp)$ is not surjective as not every nowhere zero holomorphic function has a global inverse. Take for example $U = \mathbb{C} - 0$ and the function id.

Note:

Note that the maps $\mathrm{res}: \mathcal{C}_V \to \mathcal{C}_U$ for $U \subseteq V$ forms a presheaf $\mathrm{open}(X)^{\mathsf{op}} \to \mathrm{Cat}$. In fact it is a Cat-valued sheaf.

Note:
We will show soon that $\Gamma: \mathcal{C}_X \to \mathcal{C}$ is right-adjoint to the constant sheaf functor. And as we will see $\mathrm{res}: \mathcal{C}_X \to \mathcal{C}_U$ is the inverse image sheaf of the inclusion $U \hookrightarrow X$, which in general is left-adjoint to the pushforward, but in the case of an open embedding it is a right-adjoint as well. Thus $\Gamma_U = \Gamma \circ \mathrm{res}$ is a right-adjoint and thus preserves limits; in particular it is left-exact.

Example 1.5.8

As alluded to in the previous note, the pushforward functor $\pi_*: \mathcal{C}_X \to \mathcal{C}_Y$ associated to a continuous map $\pi: X \to Y$ is a right-adjoint, and thus left-exact. We can show left-exactness directly: for a morphism $\varphi: \mathcal{F} \to \mathcal{G}$ in $\mathcal{C}_X$, the components of $\pi_*\varphi$ are $(\pi_*\varphi)_V = \varphi_{f^{-1}V}$ for $V \subseteq Y$. Because kernels are taken coordinate-wise:

$$(\ker \pi_*\varphi)_V = \ker(\pi_*\varphi_V) = \ker \varphi_{f^{-1}V} = (\ker \varphi)_{f^{-1}V} = (\pi_* \ker \varphi)_V$$

so that $\ker \pi_*\varphi = \pi_* \ker \varphi$ as desired.

Definition 1.5.9

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$-modules. Then we form their **tensor product presheaf** to be the presheaf $\mathcal{F} \otimes^{\mathsf{pre}}_{\mathcal{O}_X} \mathcal{G}$ given by

$$(\mathcal{F} \otimes^{\mathsf{pre}}_{\mathcal{O}_X} \mathcal{G})(U) = \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U), \qquad \mathrm{res}_{V,U} = \mathrm{res}^{\mathcal{F}}_{V,U} \otimes \mathrm{res}^{\mathcal{G}}_{V,U}.$$

Their **tensor product** is the sheafification of the tensor product presheaf:

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} = (\mathcal{F} \otimes^{\mathsf{pre}}_{\mathcal{O}_X} \mathcal{G})^{\mathsf{sh}}.$$

Proposition 1.5.10

For a presheaf $\mathcal{F}$, $\bullet \otimes^{\mathsf{pre}} \mathcal{F}$ is left adjoint to $\mathbf{Hom}(\mathcal{F}, \bullet)$ as functors $\mathrm{Mod}^{\mathsf{pre}}_{\mathcal{O}_X} \to \mathrm{Mod}^{\mathsf{pre}}_{\mathcal{O}_X}$.
If $\mathcal{F}$ is a sheaf then $\bullet \otimes \mathcal{F}$ is left adjoint to $\mathbf{Hom}(\mathcal{F}, \bullet)$ as functors $\mathrm{Mod}_{\mathcal{O}_X} \to \mathrm{Mod}_{\mathcal{O}_X}$.
Furthermore, $\mathbf{Hom}(\bullet, \mathcal{F})$ is a self-adjoint functor. That is, there is an adjunction $\mathbf{Hom}(\bullet, \mathcal{F})^{\mathsf{op}}: \mathrm{Mod}^{\mathsf{op}}_{\mathcal{O}_X} \rightleftarrows \mathrm{Mod}_{\mathcal{O}_X}: \mathbf{Hom}(\bullet, \mathcal{F})$.

Proof

If $\varphi: \mathcal{G} \otimes^{\mathsf{pre}} \mathcal{F} \to \mathcal{H}$ is a morphism, then for all $U \subseteq X$, $\varphi_U: \mathcal{G}(U) \otimes \mathcal{F}(U) \to \mathcal{H}(U)$ corresponds to a map $\psi_U: \mathcal{G}(U) \to \mathbf{Hom}_{\mathcal{O}_X(U)}(\mathcal{F}(U), \mathcal{H}(U))$. This then defines a map $\widehat{\varphi}_U: \mathcal{G}(U) \to \mathbf{Hom}(\mathcal{F}, \mathcal{H})(U)$ which maps $x \in \mathcal{G}(U)$ to $\widehat{\varphi}_U(x): \mathcal{F}|_U \to \mathcal{H}|_U$ whose components on $V \subseteq U$ are ψ_V.

Conversely for a map $\psi: \mathcal{G} \to \mathbf{Hom}(\mathcal{F}, \mathcal{H})$, we get maps $\mathcal{G}(U) \to \mathbf{Hom}(\mathcal{F}|_U, \mathcal{H}|_U) \xrightarrow{\Gamma} \mathbf{Hom}(\mathcal{F}(U), \mathcal{H}(U)$ which then gives us a map $\widetilde{\psi}_U: \mathcal{G}(U) \otimes \mathcal{F}(U) \to \mathcal{H}(U)$. This assembles into the desired map $\widetilde{\psi}: \mathcal{G} \otimes^{\mathsf{pre}} \mathcal{F} \to \mathcal{H}$.

Thus we have adjunctions

$$\mathrm{Mod}^{\mathsf{pre}}_{\mathcal{O}_X} \underset{\mathbf{Hom}(\mathcal{F}, \bullet)}{\overset{\otimes^{\mathsf{pre}} \mathcal{F}}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} \mathrm{Mod}^{\mathsf{pre}}_{\mathcal{O}_X} \underset{\mathrm{inc}}{\overset{\mathsf{sh}}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} \mathrm{Mod}_{\mathcal{O}_X}$$

so that $\bullet \otimes \mathcal{F}$ is the left adjoint of $\mathbf{Hom}(\mathcal{F}, \bullet)$ as desired.

For the final point we note that the adjunctions above are natural in all variables, so that

$$\hom(\mathcal{H}, \mathbf{Hom}(\mathcal{G}, \mathcal{F})) \cong \hom(\mathcal{H} \otimes \mathcal{G}, \mathcal{F}) \cong \hom(\mathcal{H} \otimes \mathcal{G}, \mathcal{F}) \cong \hom(\mathcal{H}, \mathbf{Hom}(\mathcal{G}, \mathcal{F}))$$

as desired. ■

Corollary 1.5.11

For a sheaf $\mathcal{F}$, $\mathbf{Hom}(\mathcal{F}, \bullet)$ is left-exact and $\mathbf{Hom}(\bullet, \mathcal{F})$ is contravariant left-exact.

Proposition 1.5.12

If $\mathcal{F}, \mathcal{G}$ are $\mathcal{O}_X$-modules then for $p \in X$,

$$(\mathcal{F} \otimes \mathcal{G})_p = \mathcal{F}_p \otimes \mathcal{G}_p.$$

Proof

Filtered colimits in module categories commute with tensor products so

$$(\mathcal{F} \otimes \mathcal{G})_p = \operatorname{colim}_{p \in U} (\mathcal{F}(U) \otimes \mathcal{G}(U)) = (\operatorname{colim}_U \mathcal{F}(U)) \otimes (\operatorname{colim}_U \mathcal{G}(U)) = \mathcal{F}_p \otimes \mathcal{G}_p.$$ ■

1.6 Adjunctions of sheaves

In the following section, $\mathcal{C}$ is a complete and cocomplete category (or at least has all filtered colimits) for which sheafification exists.

Recall that for a continuous map $\pi: X \to Y$ there is a functor $\pi_*: \mathcal{C}_X \to \mathcal{C}_Y$ mapping a sheaf $\mathcal{F}$ to the sheaf $(\pi_* \mathcal{F})(V) = \mathcal{F}(\pi^{-1} V)$ for $V \subseteq Y$. It turns out that this functor has a left adjoint.

Theorem 1.6.1

The pushforward functor $\pi_*: \mathcal{C}_X \to \mathcal{C}_Y$ has a left adjoint, called the **inverse image functor**, $\pi^{-1}: \mathcal{C}_Y \to \mathcal{C}_X$.

Proof

First we show that $\pi_*: \mathcal{C}_X^{\mathsf{pre}} \to \mathcal{C}_Y^{\mathsf{pre}}$ has a left adjoint. For $U \subseteq X$ and $\mathcal{G} \in \mathcal{C}_Y^{\mathsf{pre}}$ define

$$\pi_{\mathsf{pre}}^{-1} \mathcal{G}(U) = \operatorname{colim}_{V \supseteq \pi(U)} \mathcal{G}(V).$$

This is clearly functorial in U and thus defines a presheaf in $\mathcal{C}_X^{\mathsf{pre}}$. Furthermore it is also clearly functorial in $\mathcal{G}$, defining a functor $\mathcal{C}_Y^{\mathsf{pre}} \to \mathcal{C}_X^{\mathsf{pre}}$.

To show the adjunction, consider we have a morphism $\varphi: \pi^{-1} \mathcal{G} \to \mathcal{F}$, which has components

$$\varphi_U: \operatorname{colim}_{V \supseteq \pi(U)} \mathcal{G}(V) \to \mathcal{F}(U).$$

Then define $\widehat{\varphi}: \mathcal{G} \to \pi_* \mathcal{F}$ whose component on $V \subseteq Y$ is

$$\widehat{\varphi}_V = \mathcal{G}(V) \to \operatorname{colim}_{W \supseteq \pi(\pi^{-1} V)} \mathcal{G}(W) \xrightarrow{\varphi_{\pi^{-1} V}} \mathcal{F}(\pi^{-1} V),$$

where the first morphism comes from the colimit cone.

Conversely given $\psi: \mathcal{G} \to \pi_* \mathcal{F}$ then for $U \subseteq X$ and $V \supseteq \pi U$ there is a morphism

$$\mathcal{G}(V) \xrightarrow{\psi_V} \mathcal{F}(\pi^{-1} V) \xrightarrow{\text{res}} \mathcal{F}(U).$$

These assemble into a cone under $(V \supseteq \pi U) \mapsto \mathcal{G}(V)$, and thus defines a map $\widetilde{\psi}_U : \operatorname{colim}_{V \supseteq \pi U} \mathcal{G}(V) \to \mathcal{F}(U)$. These assemble into a morphism $\widetilde{\psi} : \pi^{-1} \mathcal{G} \to \mathcal{F}$.

Now for sheaves, we have adjunctions

$$\mathcal{C}_Y^{\mathsf{pre}} \underset{\pi_*}{\overset{\pi_{\mathsf{pre}}^{-1}}{\rightleftarrows}} \mathcal{C}_X^{\mathsf{pre}} \underset{\text{inc}}{\overset{\mathsf{sh}}{\rightleftarrows}} \mathcal{C}_X$$

so we define $\pi^{-1} = \mathsf{sh} \circ \pi_{\mathsf{pre}}^{-1}$ and this gives us the desired left adjoint. ■

Remark 1.6.2

A point $p \in X$ may be equivalently viewed as a morphism from the one-point space $p : * \to X$ which picks out p. Sheaves on $*$ amount to objects of $\mathcal{C}$ (because the sheaf condition implies $\mathcal{F}(\emptyset)$ is the terminal object), so that the inverse image-pushforward amounts to an adjunction

$$\mathcal{C}_X \underset{p_*}{\overset{p^{-1}}{\rightleftarrows}} \mathcal{C}.$$

When viewing $A \in \mathcal{C}$ as a sheaf over $*$, we get that $p_* A$ is the sheaf on X where for $U \subseteq X$,

$$(p_* A)(U) = A(p^{-1} U) = \begin{cases} A & p \in U \\ * & \text{else} \end{cases},$$

as A is the sheaf $A(*) = A$ and $A(\emptyset) = *$. This is precisely the skyscraper sheaf (which justifies the notation $p_* A$).

On the other hand, $p^{-1} \mathcal{F} \in \mathcal{C}$ is the colimit

$$\operatorname{colim}_{U \ni p} \mathcal{F}(U) = \mathcal{F}_p$$

so that p^{-1} is the stalk functor at p.

So in this specific case, we have an adjunction $\bullet_p \dashv p_* \bullet$.

Proposition 1.6.3

The construction $\pi \mapsto \pi_*$ is functorial, and $\pi \mapsto \pi^{-1}$ is contravariantly functorial. That is, these form form functors

$$\begin{aligned} \text{Top} &\to \text{Cat}, \qquad X \mapsto \mathcal{C}_X, \quad \pi \mapsto \pi_* \\ \text{Top}^{\mathsf{op}} &\to \text{Cat}, \qquad X \mapsto \mathcal{C}_X, \quad \pi \mapsto \pi^{-1}. \end{aligned}$$

Proof

Because π^{-1} is the left adjoint of π_*, it is sufficient to show that one construction is functorial to show that the second is. We will show that if we have a composition of continuous maps $X \xrightarrow{\pi} Y \xrightarrow{\tau} Z$, then $(\tau \circ \pi)_* = \tau_* \circ \pi_*$. Indeed,

$$(\tau_* \circ \pi_* \mathcal{F})(W) = \pi_* \mathcal{F}(\tau^{-1} W) = \mathcal{F}((\tau \circ \pi)^{-1} W) = (\tau \circ \pi)_* \mathcal{F}(W).$$ ■

Note:
It should be made more clear what it is meant for $\pi \mapsto \pi^{-1}$ to be functorial. Indeed, it is not strictly true that $(\tau \circ \pi)^{-1} = \pi^{-1} \circ \tau^{-1}$; this equality is not strict and is rather a natural isomorphism. So $\pi \mapsto \pi^{-1}$ is not an ordinary functor, rather a 2-functor.

Corollary 1.6.4

Let $\pi\colon X \to Y$ be continuous, and $\mathcal{G} \in \mathcal{C}_Y$. Then for $p \in X$ there is a natural isomorphism $(\pi^{-1}\mathcal{G})_p \cong \mathcal{G}_{\pi p}$.

Proof

Note that $(\pi^{-1}\mathcal{G})_p$ is the composition $p^{-1} \circ \pi^{-1}(\mathcal{G}) = (\pi \circ p)^{-1}\mathcal{G}$, with $p\colon * \to X$. This is equal to $\pi(p)^{-1}(\mathcal{G}) = \mathcal{G}_{\pi p}$ as desired. ■

Proposition 1.6.5

Suppose $U \subseteq X$ is an open subset and consider the inclusion $\iota\colon U \hookrightarrow X$. Then the functors ι^{-1} and $\bullet|_U$ (restriction) are isomorphic.

Proof

We will show that $\bullet|_U$ is a left adjoint to ι_*. A map $\varphi\colon \mathcal{F}|_U \to \mathcal{G}$ defines $\widehat{\varphi}\colon \mathcal{F} \to \iota_*\mathcal{G}$ by

$$\widehat{\varphi}_V = \mathcal{F}(V) \xrightarrow{\text{res}} \mathcal{F}(U \cap V) \xrightarrow{\varphi_{U\cap V}} \mathcal{G}(U \cap V) = \iota_*\mathcal{G}(V).$$

Conversely $\psi\colon \mathcal{F} \to \iota_*\mathcal{G}$ defines $\widetilde{\psi}\colon \mathcal{F}|_U \to \mathcal{G}$ by $\widetilde{\psi}_V = \psi_V$ for $V \subseteq U$, as $\iota_*\mathcal{G}(V) = \mathcal{G}(V)$. ■

Proposition 1.6.6

Let $\pi\colon X \to Y$ be continuous, then $\pi^{-1}\colon \mathrm{Ab}_Y \to \mathrm{Ab}_X$ is exact.

Proof

Exactness can be checked on the level of stalks by corollary 1.5.5. Thus π^{-1} is exact iff $p^{-1} \circ \pi^{-1}$ is exact for all $p \in X$ (where p^{-1} denotes taking the stalk at p). But this is isomorphic to $\pi(p)^{-1} = \bullet_{\pi p}$ which is exact. ■

Consider the following situation: we have a commutative square of continuous maps

$$\begin{array}{ccc} W & \xrightarrow{\beta'} & X \\ {\scriptstyle \alpha'}\downarrow & & \downarrow{\scriptstyle \alpha} \\ Y & \xrightarrow{\beta} & Z \end{array}$$

For a sheaf $\mathcal{F}$ on X we can define the **push-pull map**

$$\beta^{-1} \circ \alpha_*\mathcal{F} \to \alpha'_*(\beta')^{-1}\mathcal{F}$$

as follows. Consider the unit of $(\beta')^{-1} \dashv \beta'_*$, $\eta^{\beta'}\colon \mathcal{F} \to (\beta'_*)(\beta')^{-1}\mathcal{F}$. Then apply α_* to get $\alpha_*\mathcal{F} \to \alpha_*(\beta'_*)(\beta')^{-1}\mathcal{F}$. By commutativity this is equal to $\alpha_*\mathcal{F} \to (\beta_*)(\alpha'_*)(\beta')^{-1}\mathcal{F}$, and we can apply the adjunction $\beta^{-1} \dashv \beta$ to get $\beta^{-1}\alpha_*\mathcal{F} \to \alpha'_*(\beta')^{-1}\mathcal{F}$ as desired.

Recall that given an adjunction $(\eta, \varepsilon)\colon L \dashv R$, the isomorphism $\hom(c, Rd) \xrightarrow{\sim} \hom(Lc, d)$ is given $f \mapsto \varepsilon \circ Lf$. Thus the push-pull map is given by

$$\varepsilon^{\beta}_{\alpha'_*(\beta')^{-1}} \circ (\beta^{-1} \circ \alpha_*(\eta^{\beta'})).$$

But we could define this another way. Instead we start with the counit $\varepsilon^{\alpha}\colon \alpha^{-1}\alpha_*\mathcal{F} \to \mathcal{F}$, apply $(\beta')^{-1}$: $(\beta')^{-1}\alpha^{-1}\alpha_*\mathcal{F} \to (\beta')^{-1}\mathcal{F}$, which is equivalent to $(\alpha')^{-1}\beta^{-1}\alpha_*\mathcal{F} \to (\beta')^{-1}\mathcal{F}$, then we apply the adjunction $(\alpha')^{-1} \dashv \alpha'_*$ to get the desired $\beta^{-1}\alpha_*\mathcal{F} \to \alpha'_*(\beta')^{-1}\mathcal{F}$. In this case the map becomes

$$(\alpha'_* \circ (\beta')^{-1})(\varepsilon^{\alpha}) \circ \eta^{\alpha'}.$$

These two definitions are equivalent. Indeed, we can view both of these maps as the $\mathcal{F}$-component of a certain natural transformation, given by the two paths in the following diagram (the whiskering in the diagram is left implicit):

$$\begin{array}{ccc}
\beta^{-1}\alpha_*\beta'_*(\beta')^{-1} & \xrightarrow{\sim} & \beta^{-1}\beta_*\alpha'_*(\beta')^{-1} \\
\uparrow{\scriptstyle \eta^{\beta'}} & & \downarrow{\scriptstyle \varepsilon^{\beta}} \\
\beta^{-1}\alpha_* & & \alpha'_*(\beta')^{-1} \\
\downarrow{\scriptstyle \eta^{\alpha'}} & & \uparrow{\scriptstyle \varepsilon^{\alpha}} \\
\alpha'_*(\alpha')^{-1}\beta^{-1}\alpha_* & \xrightarrow[\sim]{} & \alpha'_*(\beta')^{-1}\alpha^{-1}\alpha_*
\end{array}$$

The claim is that this diagram commutes, and it can be proven by filling out the diagram using the naturality of the units, counits, and the triangle identities.

The push-pull map is also called the **change of basis map**.

Definition 1.6.7

Let $\mathcal{F}$ be a sheaf of Abelian groups on X, and $s \in \mathcal{F}(X)$ a global section. Define its **support** to be

$$\operatorname{supp} s = \{p \in X \mid s_p \neq 0 \text{ in } \mathcal{F}_p\}.$$

Proposition 1.6.8

The support of a global section $\operatorname{supp} s$ is a closed subset of X.

Proof

Note that $p \notin \operatorname{supp} s$ iff $s_p = 0$ in $\mathcal{F}_p$ iff there is a neighborhood $p \in U$ such that $s|_U = 0$ in $\mathcal{F}(U)$. But then for all $q \in U$ one has $s_q = 0$ for this reason, so $U \subseteq (\operatorname{supp} s)^c$; so $\operatorname{supp} s$ is closed. ∎

Note:

The locus where a continuous function is nonzero is open, i.e. if f is a continuous function to an Abelian group then $\{p \in X \mid f(p) \neq 0\}$ is open. In contrast the previous proposition shows that the locus where the germ of a continuous function (i.e. the stalk of a sheaf of continuous functions) is nonzero is closed.

This is because the latter is the set of all points where s is nonzero *near* (i.e. in a neighborhood around) p, while the former is the set of all points where f is nonzero *at* p.

In the following sheaves are valued in a category with a terminal object (so not Ring for example).

Definition 1.6.9

The **support** of a sheaf $\mathcal{F} \in \mathcal{C}_X$ is the set $\operatorname{supp}\mathcal{F}$ of $p \in X$ for which $\mathcal{F}_p$ is non-terminal in $\mathcal{C}$.

If $\mathcal{F}$ is a sheaf of groups, then

$$\operatorname{supp}\mathcal{F} = \{p \in X \mid \mathcal{F}_p \neq 0\} = \bigcup_{s \in \Gamma(\mathcal{F})} \operatorname{supp} s.$$

Proposition 1.6.10

Let $Z \subseteq X$ be a closed subset, and denote by $\iota\colon Z \hookrightarrow X$ the inclusion.

(1) If $\mathcal{F}$ is a sheaf on Z then for $q \in X$,

$$(\iota_*\mathcal{F})_q = \begin{cases} \mathcal{F}_q & q \in Z \\ * & q \notin Z \end{cases}$$

where $* \in \mathcal{C}$ is the terminal object.

(2) If $\mathcal{F}$ is a sheaf on X with $\operatorname{supp}\mathcal{F} \subseteq Z$ then the unit $\mathcal{F} \to \iota_*\iota^{-1}\mathcal{F}$ is an isomorphism.

Proof

For (1) note that

$$(\iota_*\mathcal{F})_q = \operatorname{colim}_{q \in U} \mathcal{F}(U \cap Z).$$

If $q \in Z$ then this is the colimit over all neighborhoods of q in Z, so isomorphic to $\mathcal{F}_q$. Otherwise there is a neighborhood of q disjoint from Z as it is closed, and $\mathcal{F}(\emptyset) = *$.

For (2), let $p \in X$. If $p \in Z$ then by (1) and corollary 1.6.4

$$(\iota_*\iota^{-1}\mathcal{F})_p \cong (\iota^{-1}\mathcal{F})_p \cong \mathcal{F}_p.$$

This isomorphism is the stalk of the unit. And if $p \notin Z$ then $p \notin \operatorname{supp}\mathcal{F}$ so that $\mathcal{F}_p$ is terminal and so is $(\iota_*\iota^{-1}\mathcal{F})_p$ by (1).

Thus the stalks of the unit are all isomorphisms, and thus the unit itself is an isomorphism. ■

2 Schemes

2.1 Affine schemes

In this section we briefly review the definition of an affine scheme.

Definition 2.1.1

Let A be a commutative ring; its **spectrum** is the topological space whose underlying set is the set of all

prime ideals of A. We denote it by $\operatorname{Spec}(A)$, and for a prime ideal $\mathfrak{p} \leq A$ we denote the element of $\operatorname{Spec}(A)$ by $[\mathfrak{p}] \in \operatorname{Spec}(A)$.

For a subset $S \subseteq A$ we define its **vanishing set** to be

$$\mathrm{V}(S) = \{[\mathfrak{p}] \in \operatorname{Spec}(A) \mid S \subseteq \mathfrak{p}\}.$$

The collection of all such vanishing sets, $\{\mathrm{V}(S)\}_{S \subseteq A}$, forms a topology of closed sets on $\operatorname{Spec}(A)$. This topology is called the **Zariski topology**.

Notice that for a subset $S \subseteq A$, $\mathrm{V}(S) = \mathrm{V}((S))$ where $(S) \leq A$ is the ideal generated by A. Furthermore,

$$\mathrm{V}(\emptyset) = \mathrm{V}(0) = \operatorname{Spec}(A), \qquad \mathrm{V}(A) = \emptyset, \qquad \bigcap_{i \in I} \mathrm{V}(S_i) = \mathrm{V}\left(\bigcup_{i \in I} S_i\right),$$

and that for two ideals $I, J \leq A$ we have that $\mathrm{V}(I) \cup \mathrm{V}(J) = \mathrm{V}(IJ)$. It follows that the Zariski topology is indeed a topology. Notice that V is order-reversing: if $S_1 \subseteq S_2$ then $\mathrm{V}(S_2) \subseteq \mathrm{V}(S_1)$.

Recall that if $\varphi \colon A \to B$ is a ring morphism then it pulls back prime ideals: if $\mathfrak{q} \leq B$ is prime, then so is $\varphi^{-1}\mathfrak{q} \leq A$. Thus Spec is functorial:

$$\operatorname{Spec} \colon \mathrm{Ring}^{\mathrm{op}} \to \mathrm{Top}, \qquad \varphi \mapsto \varphi^*$$

where $\varphi^* \colon \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is $[\mathfrak{q}] \mapsto [\varphi^{-1}\mathfrak{q}]$. This is continuous: for $I \leq A$

$$(\varphi^*)^{-1}\,\mathrm{V}_A(I) = \left\{[\mathfrak{q}] \in \operatorname{Spec}(B) \;\middle|\; I \subseteq \varphi^{-1}\mathfrak{q}\right\} = \{[\mathfrak{q}] \in \operatorname{Spec}(B) \mid \varphi(I) \subseteq \mathfrak{q}\} = \mathrm{V}_B(\varphi(I)).$$

If $I \leq A$ is an ideal then its **radical** is the ideal

$$\sqrt{I} = \{a \in A \mid a^n \in I \text{ for some } n \geq 0\} \leq A.$$

It is easy to see that $\mathrm{V}(I) = \mathrm{V}(\sqrt{I})$: if $I \leq \mathfrak{p}$ then if $f \in \sqrt{I}$ then $f^n \in I \leq \mathfrak{p}$ for some n and since $\mathfrak{p}$ is prime, $f \in \mathfrak{p}$. Furthermore it can be seen that $\sqrt{I}$ is the intersection of all prime ideals containing I:

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p} = \bigcap \mathrm{V}(I).$$

Clearly the left-hand side is contained in the right. Conversely if $f \notin \sqrt{I}$ then $\sqrt{I}$ and $\{f^n\}_{n \geq 0}$ do not intersect, and so $(\sqrt{I})_f \leq A_f$ is a proper ideal. Thus it is contained in a prime ideal $(\sqrt{I})_f \leq \mathfrak{p}_f \leq A_f$ associated to a prime ideal $\mathfrak{p} \leq A$, so that $\sqrt{I} \leq \mathfrak{p}$. But $f \notin \mathfrak{p}$, so $f \notin \bigcap \mathrm{V}(I)$.

We also define the **nilradical** of A be the radical of the zero ideal:

$$\operatorname{nil}(A) = \sqrt{0} = \{a \in A \mid a^n = 0 \text{ for some } n \geq 0\}.$$

Thus we have that

$$\operatorname{nil}(A) = \bigcap_{[\mathfrak{p}] \in \operatorname{Spec}(A)} \mathfrak{p}.$$

We can also form a base for the Zariski topology of $\operatorname{Spec}(A)$. For $S \subseteq A$ let

$$\mathrm{D}(S) = \mathrm{V}(S)^c = \{[\mathfrak{p}] \in \operatorname{Spec}(A) \mid S \not\subseteq \mathfrak{p}\}$$

form the open sets of the Zariski topology. For $f \in A$ let

$$\mathrm{D}(f) = \mathrm{D}\{f\} = \{[\mathfrak{p}] \in \operatorname{Spec}(A) \mid f \notin \mathfrak{p}\}.$$

It is clear that

$$\mathrm{D}(S) = \bigcup_{f \in S} \mathrm{D}(f)$$

so that the $\mathrm{D}(f)$ form a base of open sets for the topology.

We can map ideals of A to closed subsets of $\mathrm{Spec}(A)$, we can also map subsets of $\mathrm{Spec}(A)$ to ideals of A. For $X \subseteq \mathrm{Spec}(A)$ we define the ideal

$$\mathrm{I}(X) = \bigcap_{[\mathfrak{p}] \in X} \mathfrak{p} \leq A.$$

This is also clearly order-reversing: if $X_1 \subseteq X_2$ then $\mathrm{I}(X_2) \subseteq \mathrm{I}(X_1)$.

Notice that for an ideal $I \leq A$,

$$\mathrm{I}(\mathrm{V}(I)) = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p} = \sqrt{I}.$$

And conversely for a subset $X \subseteq \mathrm{Spec}(A)$,

$$\mathrm{V}(\mathrm{I}(X)) = \overline{X},$$

the closure of X in the Zariski topology. Indeed: if $X \subseteq \mathrm{V}(I)$ then for $[\mathfrak{p}] \in X$ we have $I \leq \mathfrak{p}$ so that $I \leq \bigcap_{[\mathfrak{p}] \in X} \mathfrak{p} = \mathrm{I}(X)$. Thus $\mathrm{V}(\mathrm{I}(X)) \leq \mathrm{V}(I)$, so that $\mathrm{V}(\mathrm{I}(X))$ is the smallest closed subset containing X; i.e. $\overline{X}$.

Thus V and I form order-reversing bijections

$$\{\text{radical ideals of } A\} \underset{\mathrm{I}(\bullet)}{\overset{\mathrm{V}(\bullet)}{\rightleftarrows}} \{\text{closed subsets of } \mathrm{Spec}(A)\}.$$

It follows that $\mathrm{V}(I) \subseteq \mathrm{V}(J)$ iff $\sqrt{J} \leq \sqrt{I}$: applying $\mathrm{I}(\bullet)$ gives us the left-to-right direction, and $\mathrm{V}(\sqrt{I}) = \mathrm{V}(I)$ gives the right-to-left direction. In particular $\mathrm{V}(I) = \mathrm{V}(J)$ iff $\sqrt{I} = \sqrt{J}$. This carries over to D as well obviously: $\mathrm{D}(I) \subseteq \mathrm{D}(J)$ iff $\sqrt{I} \leq \sqrt{J}$. In particular $\mathrm{D}(f) \subseteq \mathrm{D}(g)$ iff $\sqrt{f} \leq \sqrt{g}$ iff $f^n \in (g)$ for some $n \geq 0$, or equivalently g is invertible in A_f (g is invertible in A_f iff $(g)_f$ is unital, iff (g) intersects the localizing set, iff $f^n \in (g)$ for some $n \geq 0$).

Because $\mathrm{V}(0) = \mathrm{Spec}(A)$, $\mathrm{V}(I) = \mathrm{Spec}(A)$ iff $\sqrt{I} = \mathrm{nil}(A)$ iff $I \subseteq \mathrm{nil}(A)$. In particular $\mathrm{D}(f) = \emptyset$ iff $f \in \mathrm{nil}(A)$. On the other hand, $\mathrm{V}(I) = \emptyset$ for an ideal I iff $I = A$, since every proper ideal is contained in a prime ideal. Thus $\mathrm{D}(f) = \mathrm{Spec}(A)$ iff f is a unit.

We now discuss some topological results regarding $\mathrm{Spec}(A)$.

Definition 2.1.2

A topological space X is said to be

(1) **quasicompact** if every open cover of X has a finite subcover. Explicitly, if $X = \bigcup_{i \in I} U_i$ for $U_i \subseteq X$, then there are $i_1, \ldots, i_n \in I$ such that $X = \bigcup_{j=1}^{n} U_{i_j}$.

(2) **connected** if X cannot be written as the disjoint union of two nonempty open sets. Equivalently, X has no clopen subsets (subsets which are simultaneously open and closed) other than X and $\emptyset$. Otherwise X is **disconnected**.

(3) **irreducible** if X cannot be written as the union of two proper closed subsets. That is if $X = Y \cup Z$ for Y, Z closed then either $X = Y$ or $X = Z$. Equivalently the intersection of any two nonempty open subsets is nonempty.

Note:

The term "quasicompact" may be odd to you, as this is sometimes taken to be the definition of a "compact" space. For geometers, oftentimes a compact space is a space which is both quasicompact and Hausdorff.

Remark 2.1.3

A space X is quasicompact iff every collection of closed sets with the finite intersection property has nonempty intersection. That is, if $\{Z_i\}_{i \in I}$ is a collection of closed subsets of X such that $\bigcap_{i \in I_0} Z_i$ is nonempty for all finite $I_0 \subseteq I$, then $\bigcap_{i \in I} Z_i \neq \emptyset$.

Remark 2.1.4

It is immediate that a space X is irreducible iff all nonempty open subsets of X are dense.

It is true that $\operatorname{Spec}(A)$ is always quasicompact. It is sufficient to consider basic closed sets. If $\{\mathrm{V}(f_i)\}_{i\in I}$ has the finite intersection property, then

$$\bigcap_{i\in I}\mathrm{V}(f_i)=\mathrm{V}(\{f_i\}_{i\in I})$$

and if this is empty, then $(f_i \mid i\in I)=A$. So there are $i_1,\dots,i_n\in I$ such that $1\in(f_{i_1},\dots,f_{i_n})$ so that

$$\emptyset=\mathrm{V}(f_{i_1},\dots,f_{i_n})=\bigcap_{j=1}^{n}\mathrm{V}(f_{i_j}),$$

contradicting the finite intersection property.

Definition 2.1.5

An **idempotent** in a ring A is an element $f\in A$ such that $f^2=f$. A **nontrivial idempotent** is an idempotent which is not 0 or 1. A ring A is **connected** if it has no nontrivial idempotents; otherwise it is **disconnected**.

Proposition 2.1.6

Let A be a ring. Then the following conditions are equivalent:

(1) A is disconnected;

(2) A is isomorphic to the product of two non-trivial rings;

(3) $\operatorname{Spec}(A)$ is disconnected.

In particular $\operatorname{Spec}(A\times B)\cong\operatorname{Spec}(A)\amalg\operatorname{Spec}(B)$.

Proof

Suppose (1), then if $f\in A$ is a nontrivial idempotent let $g=1-f$; this is also a nontrivial idempotent. The ideal $A^f=(f)$ has a ring structure whose addition and multiplication is inherited from A, its identity is f. We define $\varphi\colon A\to A^f\times A^g$ by $a\mapsto(fa,ga)$. This is injective because $fa+ga=a$, and it is surjective because $fg=0$ so $\varphi(fa+gb)=(fa,gb)$. Thus $A\cong A^f\times A^g$, and these are nontrivial because $f,g\neq 0$; so (1) impies (2).

Conversely if $A\cong B\times C$ then $(1,0),(0,1)\in A$ are nontrivial idempotents so A is disconnected. Thus (2) implies (1).

Now again suppose (2), so consider the ring $A\times B$. Then notice that $\operatorname{Spec}(A\times B)=\mathrm{D}(1,0)\sqcup\mathrm{D}(0,1)$. Indeed:

$$\mathrm{D}(1,0)\cup\mathrm{D}(0,1)=\mathrm{D}((1,0),(0,1))=\mathrm{D}(1)=\operatorname{Spec}(A\times B),\qquad \mathrm{D}(1,0)\cap\mathrm{D}(0,1)=\mathrm{D}((1,0)\cdot(0,1))=\mathrm{D}(0)=\emptyset.$$

Thus (2) implies (3).

Notice that the projection maps $A\times B\to A,B$ induce $\operatorname{Spec}(A)\amalg\operatorname{Spec}(B)\to\operatorname{Spec}(A\times B)$. It is easy to see that this maps $\operatorname{Spec}(A)$ to $\mathrm{D}(1,0)$ and $\operatorname{Spec}(B)$ to $\mathrm{D}(1,0)$, showing that this is an isomorphism.

Now assume (3), so $\operatorname{Spec}(A)=\mathrm{V}(I)\sqcup\mathrm{V}(J)$ where I,J are nonzero (so $\mathrm{V}(I),\mathrm{V}(J)$ are proper subsets). This means that $\mathrm{V}(I+J)=\emptyset$ so that $I+J=A$ and $\mathrm{V}(IJ)=\operatorname{Spec}(A)$ so that $IJ=0$. Thus by the Chinese remainder theorem, $A\to A/I\times A/J$ is an isomorphism, proving (2). ∎

Proposition 2.1.7

$\operatorname{Spec}(A)$ is irreducible iff $\operatorname{nil}(A)$ is prime.

Proof

If $\operatorname{nil}(A)$ is not prime then there are $f, g \notin \operatorname{nil}(A)$ such that $fg \in \operatorname{nil}(A)$. So $\mathrm{V}(f), \mathrm{V}(g) \neq \operatorname{Spec}(A)$ and $\operatorname{Spec}(A) = \mathrm{V}(fg) = \mathrm{V}(f) \cup \mathrm{V}(g)$; thus $\operatorname{Spec}(A)$ is reducible as desired.

Conversely if $\operatorname{Spec}(A)$ is reducible then suppose $\operatorname{Spec}(A) = \mathrm{V}(I) \cup \mathrm{V}(J)$ for proper subsets, so I, J are not subsets of $\operatorname{nil}(A)$. But then $\mathrm{V}(IJ) = \operatorname{Spec}(A)$ so $IJ \leq \operatorname{nil}(A)$, so $\operatorname{nil}(A)$ is not prime. ■

Corollary 2.1.8

If A is a domain then $\operatorname{Spec}(A)$ is irreducible.

Note:

While it is true that $\operatorname{Spec} \prod_{i=1}^{n} A_i \cong \coprod_{i=1}^{n} \operatorname{Spec} A_i$ for finite products, this does not hold for infinite products. There is a canonical map $\coprod_{i \in I} \operatorname{Spec} A_i \to \operatorname{Spec} \prod_{i \in I} A_i$; and it is easy to see that this map is an embedding. But if all the A_i are nonzero and I is infinite, the left hand side is not compact (because $\{\operatorname{Spec} A_i\}_{i \in I}$ does not have a finite subcover) and thus it cannot be an isomorphism.

Definition 2.1.9

Let X be a topological space. A **closed point** of X is a point $x \in X$ such that $\overline{\{x\}} = \{x\}$.

In a Hausdorff space, all points are closed. For $\operatorname{Spec} A$, we know that $\overline{[\mathfrak{p}]} = \mathrm{V}(\mathrm{I}([\mathfrak{p}])) = \mathrm{V}(\mathfrak{p})$. Thus $[\mathfrak{p}]$ is a closed point iff $\mathfrak{p}$ is not contained in any prime ideal other than itself; equivalently it is a maximal ideal. That is: $[\mathfrak{p}] \in \operatorname{Spec} A$ is a closed point iff $\mathfrak{p} \leq A$ is maximal.

Definition 2.1.10

A subset of a topological space $Z \subseteq X$ is an **irreducible component** if it is irreducible and maximally so: it is not properly contained in an irreducible set.

Proposition 2.1.11

A subset $Z \subseteq X$ is irreducible iff $\overline{Z} \subseteq X$ is.

Proof

If Z is irreducible and $\overline{Z} = Y \cup W$ are closed, then $Z \subseteq Y \cup W$ so that $Z \subseteq Y$ or $Z \subseteq W$. Then $\overline{Z} = Y$ or W because they are closed in $\overline{Z}$ and thus in X. Conversely if $\overline{Z}$ is irreducible then if $Z = Y \cup W$, then $\overline{Z} = \overline{Y} \cup \overline{W}$ so $\overline{Z} = \overline{Y}$ or $\overline{W}$. Suppose the former, so $Z = \overline{Y} \cap Z = Y$ since $Y \subseteq Z$ is closed. ■

Proposition 2.1.12

Every point $p \in X$ is contained in a (not necessarily unique) irreducible component, and all irreducible components of X are closed.

Proof

The first claim is due to Zorn. Clearly $\{p\}$ is irreducible, so we can form the nonempty poset of irreducible sets containing p. For a chain of such irreducible sets $\{Z_i\}_{i\in I}$, we have that $Z = \bigcup_{i\in I} Z_i$ is irreducible. Indeed if $Z = Y \cup W$ are closed then suppose $Y \subsetneq Z$ then there is an $f \in Z_i$ such that $f \notin Z$. Because Z_i is irreducible we must have $Z_i \subseteq W$. Then for all $j \geq i$ we have that $Z_j \subseteq W$ for the same reason; then $Z = \bigcup_{j\geq i} Z_j = W$ so that Z is irreducible as desired.

The second claim follows from the previous proposition: if Z is an irreducible component then $\overline{Z}$ is irreducible as well and since $Z \subseteq \overline{Z}$ by maximality $Z = \overline{Z}$. ■

Definition 2.1.13

Let $x, y \in X$ be points in a topological space such that $x \in \overline{\{y\}}$. Then x is called a **specialization** of y, and y a **generalization** of x.

A point $p \in X$ is a **generic point** if $\overline{\{p\}} = X$; i.e. if it is a generalization of all the points of X.

In $\operatorname{Spec} A$ because $\overline{[\mathfrak{p}]} = \mathrm{V}(\mathfrak{p})$, $[\mathfrak{p}]$ is a generalization of $[\mathfrak{q}]$ iff $[\mathfrak{q}] \in \mathrm{V}(\mathfrak{p})$ iff $\mathfrak{p} \leq \mathfrak{q}$. And $[\mathfrak{p}]$ is a generic point iff $\mathrm{V}(\mathfrak{p}) = \operatorname{Spec}(A)$ iff $\mathfrak{p} = \operatorname{nil} A$, so that $\operatorname{Spec} A$ has a generic point iff $\operatorname{nil} A$ is prime iff $\operatorname{Spec} A$ is irreducible.

Definition 2.1.14

A poset Σ has the **descending chain condition** (**DCC**) if any of the following two equivalent conditions are satisfied:

(1) every subset $\Sigma' \subseteq \Sigma$ has a minimal element;

(2) every chain of elements $p_1 \geq p_2 \geq p_3 \geq \cdots$ eventually stabilizes: $p_n = p_{n+1} = p_{n+2} = \cdots$ for some n onward.

Dually Σ has the **ascending chain condition** (**ACC**) if Σ^{op} has the DCC.

The equivalence of the conditions is clear: (1) implies (2) by defining $\Sigma' = \{p_n\}_n$ and taking the minimum. Conversely if $\Sigma' \subseteq \Sigma$ has no minimal element, then we can form an infinite decreasing chain.

Definition 2.1.15

A topological space X is **Noetherian** if its poset of closed subsets has the DCC (equivalently its poset of open subsets has the ACC).

There is a notion of **Noetherian induction**: if Σ has DCC and we have some property P (which one can view as a function $\Sigma \to \{0,1\}$) such that $P(x)$ holds for all minimal elements of Σ, and for $y \in \Sigma$ non-minimal if $P(x)$ holds for all $x < y$ then $P(y)$. Then $P(x)$ holds for all $x \in \Sigma$. This is easy to prove: take Σ' to be the set of all $x \in \Sigma$ for which $P(x)$ does not hold.

Using Noetherian induction one can prove the following.

Proposition 2.1.16

If X is a Noetherian topological space, then every nonempty closed subset $Z \subseteq X$ can be uniquely expressed as a finite union $Z = Z_1 \cup \cdots \cup Z_n$ where the Z_i are all irreducible closed subsets, none containing any other.

Remark 2.1.17

It is clear that if A is a Noetherian ring (its poset of ideals has ACC) then $\operatorname{Spec} A$ is a Noetherian space. The converse is not true: if $\operatorname{Spec} A$ is a Noetherian space then the poset of A's *radical* ideals has ACC.

2.2 The structure sheaf an affine scheme

Definition 2.2.1

Let A be a ring, then we define a sheaf of rings $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ as follows. For a basic open set $\mathrm{D}(f)$, define $\mathcal{O}_{\operatorname{Spec} A}(\mathrm{D}(f))$ to be the localization of A at all functions which do not vanish outside of $\mathrm{V}(f)$, i.e. all $g \in A$ such that $\mathrm{V}(g) \subseteq \mathrm{V}(f)$.

For $f \in A$ let $S = \{g \in A \mid \mathrm{V}(g) \subseteq \mathrm{V}(f)\} = \{g \in A \mid \mathrm{D}(f) \subseteq \mathrm{D}(g)\}$. Recall that $\mathrm{V}(g) \subseteq \mathrm{V}(f)$ iff $f^n \in (g)$ iff $g \in A_f$ is invertible. Thus S is the saturation of $\{f^n\}_{n\geq 0}$, so that $S^{-1}A = \mathcal{O}_{\operatorname{Spec} A}(\mathrm{D}(f))$ and A_f are naturally isomorphic.

If $\mathrm{D}(f') \subseteq \mathrm{D}(f)$ then there is an obvious map $\mathcal{O}_{\operatorname{Spec} A}(\mathrm{D}\, f) \to \mathcal{O}_{\operatorname{Spec} A}(\mathrm{D}\, f')$ as the right ring is a localization of the left. Taking these maps as the restriction maps, this makes $\mathcal{O}_{\operatorname{Spec} A}$ into a presheaf of rings (or A-algebras). To show that $\mathcal{O}_{\operatorname{Spec} A}$ is a sheaf, we will make use of the following lemmas.

Lemma 2.2.2

Let A be a ring and $f_1, \ldots, f_n$ generate the unit ideal. Then the following sequence of A-module maps is exact

$$0 \to A \xrightarrow{\varphi} \prod_i A_{f_i} \xrightarrow{\psi} \prod_{i,j} A_{f_i f_j},$$

where $\varphi(a) = (a)_i$ and $\psi(a_i) = (a_i - a_j)_{i,j}$. (We write $a \in A_f$ for the localization $a = a/1$.)

Proof

First we show that φ is injective: if $a = 0$ in all A_{f_i} then there are $m_1, \ldots, m_n$ such that $f_i^{m_i} a = 0$ for all $i = 1, \ldots, n$. But because $f_i \in \sqrt{(f_j^{m_j})_j}$ for all i and $(f_1, \ldots, f_n) = A$, we see that $(f_1^{m_1}, \ldots, f_n^{m_n}) = A$ as well. Thus there are $g_1, \ldots, g_n \in A$ such that $g_1 f_1^{m_1} + \cdots + g_n f_n^{m_n} = 1$; multiplying by a shows that $a = 0$.
Next we show that $\psi \circ \varphi = 0$. Indeed:

$$\psi \circ \varphi(a) = \psi(a)_i = (a - a)_{i,j} = 0,$$

so that $\operatorname{im} \varphi \subseteq \ker \psi$. Now suppose that $(a_i/f_i^{m_i})_i \in \ker \psi$, then by multiplying both the numerator and denominator by f_is, we can assume that all the m_is are equal to some m. Thus

$$0 = \psi(a_i/f_i^m)_i = (a_i/f_i^m - a_j/f_j^m)_{i,j}$$

so that there is an M where $(f_i f_j)^M (f_j^m a_i - f_i^m a_j) = 0$ for all i, j.
As before, the f_i^{M+m} generate the unit ideal, so there are $b_i \in A$ such that $\sum_i b_i f_i^{M+m} = 1$. Then define $a = \sum_i b_i f_i^M a_i$ so that

$$f_j^{M+m} a = \sum_i b_i (f_i f_j)^M f_j^m a_i = \sum_i b_i (f_i f_j)^M f_i^m a_j = f_j^M \sum_i b_i f_i^{M+m} a_j = f_j^M a_j.$$

Thus in A_{f_j}, $a = a_j/f_j^m$, so that $\varphi(a) = (a_i/f_i^m)_i$ as desired. ■

We will say that a presheaf on X **has the sheaf property at** U for an open $U \subseteq X$ if the sheaf axiom holds for all open covers of U. So a sheaf is a presheaf which has the sheaf property at every open subset of X. Furthermore we can say that a presheaf has the sheaf property at U **relative to** $\{U_i\}_i$ for a specified open cover $\{U_i\}_i$ of U if the sheaf condition holds specifically for $\{U_i\}_i$.

Lemma 2.2.3

If X is a quasicompact space, then a presheaf $\mathcal{F}$ has the sheaf property at X iff it has the sheaf property at X relative to all finite covers of X.

Proof

One direction is clear. Conversely let $\mathcal{F}$ be a presheaf with the sheaf property at X relative to all finite covers of X, and let $\mathcal{U} = \{U_i\}_{i\in I}$ be an open cover of X. Then there is a finite subset $I_0 \subseteq I$ such that $\mathcal{U}_0 = \{U_i\}_{i\in I_0}$ covers X. To show that $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}}$ is a limiting cone (see remark 1.1.6), we note that a cone over $\mathcal{F}|_U$ induces a cone over $\mathcal{F}|_{\mathcal{U}_0}$. Because $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}_0}$ is limiting, any morphism of cones $(C \Rightarrow \mathcal{F}|_{\mathcal{U}}) \to (\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}})$ must be unique.

To show existence, it is sufficient to show that a morphism of cones $(C \Rightarrow \mathcal{F}|_{\mathcal{U}_0}) \to (\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}_0})$ is also a morphism of cones over $\mathcal{F}|_U$. If not, then there is a finite diagram which doesn't commute: there is a $V \in \mathcal{U}$ such that $C \to \mathcal{F}(V)$ is not equal to $C \to \mathcal{F}(U) \to \mathcal{F}(V)$. Adding V to $\mathcal{U}_0$ yields a finite cover of U, and since $\mathcal{F}$ has the sheaf property relative to *all* finite covers of X, this is a contradiction. ■

Theorem 2.2.4

$\mathcal{O}_{\operatorname{Spec} A}$ is a sheaf on the distinguished base, thus determines a unique sheaf on the topological space Spec A.

Proof

We must show that if $\mathrm{D}(f) = \bigcup_{i\in I} \mathrm{D}(f_i)$ then the following sequence is exact

$$0 \to A_f \xrightarrow{\varphi} \prod_i A_{f_i} \xrightarrow{\psi} \prod_{i,j} A_{f_i f_j}$$

where φ, ψ are defined as above (recall that $\mathrm{D}(f) \cap \mathrm{D}(g) = \mathrm{D}(fg)$). For a finite I, this follows from the lemma before the previous with A_f in place of A. For general I this follows from the above lemma, along with the fact that $\mathrm{D}(f) \cong \operatorname{Spec}(A_f)$ is quasicompact. ■

Note:

It is *not* the case that $\mathcal{O}_{\operatorname{Spec} A}(U)$ is the localization of A at all $f \in A$ not vanishing outside of U.

For a ring A, we can embed Mod_A fully faithfully into $\mathrm{Mod}_{\mathcal{O}_{\operatorname{Spec} A}}$. For an A-module M, let $\widetilde{M} \in \mathrm{Mod}_{\mathcal{O}_{\operatorname{Spec} A}}$ be the presheaf where $\widetilde{M}(\mathrm{D}(f))$ is the localization of M at all $g \in A$ which don't vanish outside of $\mathrm{V}(f)$. For the same reason as before, $\widetilde{M}(\mathrm{D}(f)) \cong M_f$. The restriction maps are similar to before, and the proof that $\widetilde{M}$ is a B-sheaf, and thus determines a sheaf, is very similar to above.

In particular, $\widetilde{A}$ is just Spec A.

Lemma 2.2.5

Let A be a ring and $S \subseteq A$ a multiplicative set. We can view S as a preorder whose objects are the $f \in S$, and there is an arrow $f \to g$ iff $\mathrm{D}(f) \subseteq \mathrm{D}(g)$. Then for an A-module M, $f \mapsto M_f$ forms a functor $S^{\mathrm{op}} \to \mathrm{Mod}_{S^{-1}A}$, and

$$S^{-1}M \cong \operatorname{colim}_{f \in S^{\mathrm{op}}} M_f.$$

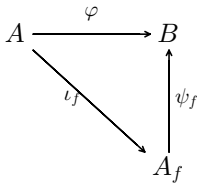
Proof

It is sufficient to prove this for $M = A$, as S is a cofiltered category ($fg \to f, g$) so the colimit is filtered. Because tensor products commute with filtered colimits,

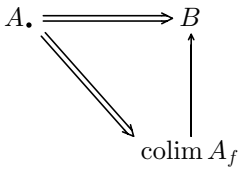
$$\operatorname{colim} M_f \cong \operatorname{colim}(A_f \otimes M) \cong (\operatorname{colim} A_f) \otimes M \cong S^{-1}A \otimes M \cong S^{-1}M.$$

The following proof works just as well for M, but is clearer with A.

Let $\varphi\colon A \to B$ be a ring map such that $\varphi(S) \subseteq B^\times$. Then $\varphi(f) \in B^\times$ so that this induces unique maps $\psi_f\colon A_f \to B$ for all $f \in S$ such that the following diagrams commute:



By uniqueness this forms a cone $(f \mapsto A_f) \Rightarrow B$, and thus a unique map $\operatorname{colim} A_f \to B$. So we have a commutative diagram



Because $1 \in S$ and $A = A_1$, this gives rise to the desired map. This also shows uniqueness of the map: a map $\operatorname{colim} A_f \to B$ commuting with φ gives rise to maps $A_f \to B$ commuting with $A \to A_f$ (again because $A = A_1$), and by the universal property of A_f, they are unique. ■

Corollary 2.2.6

Let $M \in \mathrm{Mod}_A$ be an A-module and $\widetilde{M} \in \mathrm{Mod}_{\mathcal{O}_{\operatorname{Spec} A}}$ be the induced $\mathcal{O}_{\operatorname{Spec} A}$-module. Then for $[\mathfrak{p}] \in \operatorname{Spec} A$ there is a natural isomorphism between the stalk of $\widetilde{M}$ at $[\mathfrak{p}]$ and the localization $M_{\mathfrak{p}}$:

$$\widetilde{M}_{[\mathfrak{p}]} \cong M_{\mathfrak{p}}.$$

In particular

$$(\operatorname{Spec} A)_{[\mathfrak{p}]} \cong A_{\mathfrak{p}}.$$

Proof

The lefthand side is

$$\widetilde{M}_{[\mathfrak{p}]} = \operatorname{colim}_{[\mathfrak{p}] \in \mathrm{D}(f)} \widetilde{M}(\mathrm{D}\, f) = \operatorname{colim}_{f \notin \mathfrak{p}} M_f,$$

which is $(A - \mathfrak{p})^{-1}M = M_{\mathfrak{p}}$ by the previous lemma. ■

Proposition 2.2.7

Let M be an A-module. Then the natural map

$$M \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$$

is injective.

Proof

This map corresponds to the map

$$M = \widetilde{M}(\operatorname{Spec} A) \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} \widetilde{M}_p \cong \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$$

which is an injection by proposition 1.3.1. ■

Note:
Alternatively we could prove this directly: if $m \in M$ is equal to zero in $M_{\mathfrak{p}}$ for all $\mathfrak{p} \in \operatorname{Spec} A$. Define the ideal

$$I = \operatorname{ann}_A(m) = \{a \in A \mid am = 0\} \leq A.$$

If m were nonzero, then I is a proper ideal and thus contained in a prime ideal $\mathfrak{p}$. For $a \notin \mathfrak{p}$ then $a \notin I$ so $am \neq 0$, and thus $m \neq 0$ in $M_{\mathfrak{p}}$ a contradiction.

Note that $M \mapsto \widetilde{M}$ is functorial. Indeed, an A-module map $\varphi \colon M \to N$ defines maps

$$\varphi_f \colon \widetilde{M}(\mathrm{D}\, f) = M_f \to N_f = \widetilde{N}(\mathrm{D}\, f)$$

which assemble into a morphism $\widetilde{\varphi} \colon \widetilde{M} \to \widetilde{N}$.

Note that this functor is fully faithful. It is faithful because $\widetilde{\varphi}_{\mathrm{D}\,1} = \varphi$. And it is full because if $\psi \colon \widetilde{M} \to \widetilde{N}$ is a morphism, then let $\varphi = \psi_{\mathrm{D}\,1}$, and then $\psi = \widetilde{\varphi}$. Indeed, by naturality we must have commutative squares for all $f \in A$:

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & N \\ \downarrow & & \downarrow \\ M_f & \xrightarrow{\psi_{\mathrm{D}\,f}} & N_f \end{array}$$

By the universal property of localization we must have $\psi_{\mathrm{D}\,f} = \varphi_f$ as desired. Thus $M \mapsto \widetilde{M}$ is indeed a fully faithful embedding $\operatorname{Mod}_A \to \operatorname{Mod}_{\operatorname{Spec} A}$.

Definition 2.2.8

Let M be an A-module and $m \in M$. Define its **support** to be

$$\operatorname{supp} m = \{[\mathfrak{p}] \in \operatorname{Spec} A \mid m_{\mathfrak{p}} \neq 0\},$$

which is the support of m viewed as a global section of $\widetilde{M}$. We similarly define

$$\operatorname{supp} M = \operatorname{supp} \widetilde{M} = \{[\mathfrak{p}] \in \operatorname{Spec} A \mid M_{\mathfrak{p}} \neq 0\}.$$

2.3 Schemes

Note that when $\pi\colon X \to Y$ is a homeomorphism, there is an isomorphism of categories $\pi\colon \mathrm{open}(Y) \cong \mathrm{open}(X)$ mapping $V \mapsto \pi^{-1}V$. This lifts to an isomorphism of categories $\pi_*\colon \mathcal{C}_X^{\mathsf{pre}} \cong \mathcal{C}_Y^{\mathsf{pre}}$ mapping a presheaf $\mathcal{F} \in \mathcal{C}_X^{\mathsf{pre}}$ to precomposition with π: $\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}V)$. This is the usual pushforward, and it then induces an isomorphism $\mathcal{C}_X \cong \mathcal{C}_Y$.

Because π^{-1} is its left adjoint, which is unique up to isomorphism, this means that π^{-1}, π_* form an adjoint equivalence. In particular, the adjunction preserves categorical properties of morphisms, e.g. a morphism $\mathcal{F} \to \pi_*\mathcal{G}$ is an isomorphism iff $\pi^{-1}\mathcal{F} \to \mathcal{G}$ is.

Definition 2.3.1

Let $\mathcal{C}$ be a category (with enough limits and colimits). A **$\mathcal{C}$-space** is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a $\mathcal{C}$-valued sheaf on X. A morphism of $\mathcal{C}$-spaces $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a continuous map $\pi\colon X \to Y$ along with a map of sheaves $\varphi\colon \mathcal{O}_Y \to \pi_*\mathcal{O}_X$. Composition is given in the obvious manner: if $X \xrightarrow{\pi} Y \xrightarrow{\tau} Z$ are continuous, and $\varphi\colon \mathcal{O}_Y \to \pi_*\mathcal{O}_X$ and $\psi\colon \mathcal{O}_Z \to \tau_*\mathcal{O}_Y$ are morphisms, then define

$$(\tau, \psi) \circ (\pi, \varphi) = \mathcal{O}_Z \xrightarrow{\psi} \tau_*\mathcal{O}_Y \xrightarrow{\tau_*\varphi} (\tau \circ \pi)_*\mathcal{O}_X.$$

We denote the category of $\mathcal{C}$-spaces by $\mathrm{Space}_{\mathcal{C}}$.

Note that composition is indeed associative. If we have maps

$$(X, \mathcal{O}_X) \xrightarrow{(\pi,\varphi)} (Y, \mathcal{O}_Y) \xrightarrow{(\tau,\psi)} (Z, \mathcal{O}_Z) \xrightarrow{(\sigma,\chi)} (W, \mathcal{O}_W)$$

then we have compositions

$$\big((\sigma, \chi) \circ (\tau, \psi)\big) \circ (\pi, \varphi) = (\sigma \circ \tau, \sigma_*\psi \circ \chi) \circ (\pi, \varphi) = (\sigma \circ \tau \circ \pi, (\sigma \circ \tau)_*\varphi \circ \sigma_*\psi \circ \chi$$
$$(\sigma, \chi) \circ \big((\tau, \psi) \circ (\pi, \varphi)\big) = (\sigma, \chi) \circ (\tau \circ \pi, \tau_*\varphi \circ \psi) = (\sigma \circ \tau \circ \pi, \sigma_*(\tau_*\varphi \circ \psi) \circ \chi$$

which are equal.

Note that by the adjunction $\pi^{-1} \dashv \pi_*$, to give a morphism $\mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is to give a morphism $\pi^{-1}\mathcal{O}_Y \to \mathcal{O}_X$. In which case the composition of $\varphi\colon \pi^{-1}\mathcal{O}_Y \to \mathcal{O}_X$ and $\psi\colon \tau^{-1}\mathcal{O}_Z \to \mathcal{O}_Y$ is

$$(\tau, \psi) \circ (\pi, \varphi)\colon (\tau \circ \pi)^{-1}\mathcal{O}_Z \xrightarrow{\pi^{-1}\psi} \pi^{-1}\mathcal{O}_Y \xrightarrow{\varphi} \mathcal{O}_X.$$

Definition 2.3.2 (Schemes)

A **ringed space** is a Ring-space; we write RSp for $\mathrm{Space}_{\mathrm{Ring}}$.

An **affine scheme** is a ringed space which is isomorphic to $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ for some ring A. A **scheme** is a ringed space $(X, \mathcal{O}_X)$ which is locally affine: for every $p \in X$ there is an open neighborhood $p \in U$ such that $(U, \mathcal{O}_X|_U)$ is affine. The topology of a scheme is called the **Zariski topology**.

Schemes are ringed spaces, but we will see that not all morphisms of ringed spaces should be considered morphisms of schemes. Though we will see that an isomorphism of schemes as ringed spaces gives an isomorphism of rings (when we define what morphisms of schemes are).

We can define the **global sections functor**

$$\Gamma\colon \mathrm{Space}_{\mathcal{C}}^{\mathsf{op}} \to \mathcal{C}, \qquad \Gamma(X, \mathcal{O}_X) = \mathcal{O}_X(X).$$

A morphism $(\pi, \varphi)\colon (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ determines a map $\varphi\colon \mathcal{O}_Y \to \pi_*\mathcal{O}_X$, which has component $\varphi_Y\colon \mathcal{O}_Y(Y) \to \mathcal{O}_X(\pi^{-1}Y) = \mathcal{O}_X(X)$. Thus we define $\Gamma(\pi, \varphi) = \varphi_Y$, turning Γ into a functor.

If X is a topological space and $U \subseteq X$ an open set, we may write $\Gamma(U, \mathcal{O}_X)$ instead of $\Gamma(U, \mathcal{O}_X|_U)$. In general we may write $(U, \mathcal{O}_X)$ instead of $(U, \mathcal{O}_X|_U)$.

Note:
Note that the global sections functor on sheaves is covariant, while the global sections functor on $\mathcal{C}$-spaces is contravariant.

Lemma 2.3.3
There is a natural isomorphism

$$(\mathrm{D}(f), \mathcal{O}_{\operatorname{Spec} A}) \cong (\operatorname{Spec} A_f, \mathcal{O}_{\operatorname{Spec} A_f}).$$

Proof

Consider the canonical map $\iota\colon A \to A_f$, which induces the continuous injective $\iota\colon \operatorname{Spec}(A_f) \to \operatorname{Spec}(A)$. Its image consists of all prime ideals not intersecting $\{f^k\}_{k\geq 0}$, equivalently not containing f, so $\mathrm{D}(f)$. It is also an open map: $\mathrm{D}_{A_f}(g/f^n)$ contains all prime ideals of A_f not containing g/f^n; since f^n is a unit, this is just all prime ideals not containing g. Thus $\iota\,\mathrm{D}_{A_f}(g/f^n) = \mathrm{D}(g) \cap \mathrm{D}(f) = \mathrm{D}(fg)$ which is open. Thus ι forms a homeomorphism $\operatorname{Spec}(A_f) \xrightarrow{\sim} \mathrm{D}(f)$.

Now we need to show that there is an isomorphism $\mathcal{O}_{\operatorname{Spec} A}|_{\mathrm{D}\, f} \cong \iota_* \mathcal{O}_{\operatorname{Spec} A_f}$. Note that for $\mathrm{D}(g) \subseteq \mathrm{D}(f)$ we have

$$\mathcal{O}_{\operatorname{Spec} A}(\mathrm{D}\, g) = A_g, \qquad \iota_* \mathcal{O}_{\operatorname{Spec} A_f}(\mathrm{D}\, g) = \mathcal{O}_{\operatorname{Spec} A_f}(\iota^{-1}\, \mathrm{D}\, g) = \mathcal{O}_{\operatorname{Spec} A_f}(\mathrm{D}\, g) = (A_f)_g \cong A_g.$$

So there is indeed an isomorphism, as desired. ■

Corollary 2.3.4
If $(X, \mathcal{O}_X)$ is a scheme and $U \subseteq X$ is open, then $(U, \mathcal{O}_X|_U)$ is a scheme.

Proof

For an affine scheme this is due to the previous lemma: because the $\mathrm{D}(f)$ form a base, for any $p \in U$ there is an $f \in A$ such that $p \in \mathrm{D}(f) \subseteq U$. Then $(\mathrm{D}(f), \mathcal{O}_X|_{\mathrm{D}\, f}) \subseteq (U, \mathcal{O}_X|_U)$ is an open affine subscheme.

In general let $p \in U$, then take $p \in V \subseteq X$ an affine neighborhood. Then $p \in U \cap V \subseteq U$ is an open subset of an affine scheme and thus there is an affine neighborhood $p \in W \subseteq U \cap V$. ■

Definition 2.3.5
If $(X, \mathcal{O}_X)$ is a scheme and $U \subseteq X$ open, then $(U, \mathcal{O}_X|_U)$ is called an **open subscheme** of X. If U is also an affine scheme, then U is an **affine open subscheme** or an **affine open subset**.

It is clear that for a scheme X, its affine open subsets form a base for the topology.

Definition 2.3.6

Let $\mathcal{C}$ be a category, a subcategory $\mathcal{C}_0 \subseteq \mathcal{C}$ is said to be **replete** if it is closed under isomorphisms: if $c \in \mathcal{C}_0$ and $f\colon c \xrightarrow{\sim} d$ is an isomorphism then both d and f are in $\mathcal{C}_0$.

Let $\mathcal{C}_0 \subseteq \mathcal{C}$ be a replete category. A $(\mathcal{C}, \mathcal{C}_0)$-space is a $\mathcal{C}$-space $(X, \mathcal{O}_X)$ such that for every $p \in X$, the stalk $\mathcal{O}_{X,p}$ is in $\mathcal{C}_0$. A morphism of $(\mathcal{C}, \mathcal{C}_0)$-spaces is a $\mathcal{C}$-space morphism $(\pi, \varphi)\colon (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ such that for

every $p \in X$ the morphism of stalks $\varphi_p \colon (\pi^{-1}\mathcal{O}_Y)_p \cong \mathcal{O}_{Y,\pi(p)} \to \mathcal{O}_{X,p}$ is in $\mathcal{C}_0$.

Note:
We can write $\varphi_p \colon \mathcal{O}_{Y,\pi p} \to \mathcal{O}_{X,p}$ more explicitly. For $\pi(p) \in V \subseteq Y$ open we have a map

$$\varphi_V \colon \mathcal{O}_Y(V) \to \mathcal{O}_X(\pi^{-1}V) \to \operatorname{colim}_{p \in U} \mathcal{O}_X(U) = \mathcal{O}_{X,p}.$$

These then assemble into a cone $\mathcal{O}_Y(\bullet) \Rightarrow \mathcal{O}_{X,p}$ which then limits to $\mathcal{O}_{Y,\pi(p)} \to \mathcal{O}_{X,p}$.

Definition 2.3.7
Define LocRing to be the category of local rings $(A, \mathfrak{m})$. A morphism $\varphi \colon (A, \mathfrak{m}) \to (B, \mathfrak{n})$ is a ring morphism $A \to B$ such that $\varphi(\mathfrak{m}) \subseteq \mathfrak{n}$.
A **locally ringed space** is a $(\mathrm{Ring}, \mathrm{LocRing})$-space. We denote the category of locally ringed spaces by LocRSp.

Note:
Importantly, a locally ringed space is not a LocRing-space. In fact, generally a locally ringed space's sections will not be local rings.

A scheme which is isomorphic to the spectrum of a ring is called an **affine scheme**. We denote the full subcategory of affine schemes by AffSch.

Schemes are locally ringed spaces: if X is a scheme and $p \in X$ then p has an affine neighborhood. Since stalks can be taken locally, it is sufficient to show that $\operatorname{Spec} A$ is a locally ringed space for any ring A. And this is clear as the stalk of $\operatorname{Spec} A$ at $[\mathfrak{p}] \in \operatorname{Spec} A$ is isomorphic to $A_{\mathfrak{p}}$ which is local.

Definition 2.3.8
The **category of schemes** is the full subcategory of LocRing whose objects are schemes; it is denoted by Sch. The full subcategory of Sch consisting of affine schemes is denoted AffSch.

Note that if X, Y are locally ringed space which are isomorphic as ringed spaces, then they are isomorphic as locally ringed spaces. Indeed in general if X, Y are $(\mathcal{C}, \mathcal{C}_0)$-spaces which are isomorphic as $\mathcal{C}$-spaces, they are isomorphic as $(\mathcal{C}, \mathcal{C}_0)$-spaces. This is because $\mathcal{C}_0$ is replete (contains all isomorphisms).

Note that Spec forms a contravariant functor $\mathrm{Ring}^{\mathrm{op}} \to \mathrm{AffSch}$ which maps a ring A to is spectrum. Then for a ring morphism $\varphi \colon A \to B$ we have a continuous map $\varphi^* \colon \operatorname{Spec} B \to \operatorname{Spec} A$ mapping $\mathfrak{q} \leq B$ to $\varphi^{-1}\mathfrak{q}$. And $\varphi^\sharp \colon \operatorname{Spec} A \to \varphi^*_* \operatorname{Spec} B$ which for $f \in A$ maps $\varphi^\sharp_f \colon A_f \to \operatorname{Spec} B((\varphi^*)^{-1} \operatorname{D}_B(f)) = B_{\varphi f}$. This is just the natural map induced from the universal $A \xrightarrow{\varphi} B \to B_{\varphi f}$.

This is clearly seen to be functorial.

Definition 2.3.9
If $(X, \mathcal{O}_X)$ is a locally ringed space, then for $p \in X$ denote the maximal ideal of the stalk $\mathcal{O}_{X,p}$ by $\mathfrak{m}_{X,p}$ or $\mathfrak{m}_p$. The **residue field** is $\kappa(p) = \mathcal{O}_{X,p}/\mathfrak{m}_p$. For a function f on an open subset $U \subseteq X$ has a value at every $p \in U$ denote by $f(p)$: it is the image of the germ of f at p under the quotient $\mathcal{O}_{X,p} \to \kappa(p)$. We say that f **vanishes** at $p \in X$ if $f(p) = 0$.

Proposition 2.3.10

If f is a function on a locally ringed space X, then the subset of where f vanishes is closed. Conversely if f vanishes nowhere then it is invertible.

Proof

Note that X does not vanish at p iff $f_p \notin \mathfrak{m}_p$ iff f_p is invertible in $\mathcal{O}_{X,p}$. For a ringed space X the subset of points where f is invertible in the stalk is open. Indeed, if f is invertible in $\mathcal{O}_{X,p}$ then there is a $p \in U$ and a $g \in \mathcal{O}_X(U)$ such that $fg = 1$ in $\mathcal{O}_{X,p}$ so that $fg = 1$ in a neighborhood around p; so f is invertible in that neighborhood.

Thus the set of points where f is not invertible, for X locally ringed $f(p) = 0$, is closed.

For the second point, at every $p \in X$ there is an open $g_p \in U_p$ which is the inverse of f in U_p. Because inverses are unique, the g_p agree on their overlaps, so we can glue them together to get a global inverse for f. ■

For an affine scheme $\operatorname{Spec} A$ and a point $[\mathfrak{p}] \in \operatorname{Spec} A$, the stalk at $[\mathfrak{p}]$ is isomorphic to $A_\mathfrak{p}$. This is local with maximal ideal $\mathfrak{p}A_p$, so that $\kappa(\mathfrak{p}) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. Localization commutes with quotients, so this is isomorphic to the localization of $A/\mathfrak{p}$ at the image of $\mathfrak{p}$, which is the zero ideal. Thus

$$\kappa(\mathfrak{p}) = \frac{A_\mathfrak{p}}{\mathfrak{p}A_\mathfrak{p}} = \operatorname{Frac}(A/\mathfrak{p}).$$

Example 2.3.11

Consider the scheme $\mathbb{A}^2_k = \operatorname{Spec} k[x, y]$, where k is a field of characteristic not 2. Then $(x^2+y^2)/x(y^2-x^5)$ is a function away from the y-axis and $y^2 - x^5$ (i.e. it exists in $\mathrm{D}(y, y^2 - x^5)$). Its value at $(2, 4)$, meaning $(x-2, y-4)$ is

$$\frac{x^2+y^2}{x(y^2-x^5)} \equiv \frac{2^2+4^2}{2(4^2-2^5)}.$$

Its value at (y), the generic point of the x-axis, is

$$\frac{x^2+y^2}{x(y^2-x^5)} \equiv \frac{x^2}{-x^6} = -\frac{1}{x^4}.$$

Remark 2.3.12

We can define manifolds of arbitrary flavor similar to schemes. Instead of our base space being $\operatorname{Spec} A$, we can choose different base spaces. For instance a topological manifold is a ringed space $(X, \mathcal{O}_X)$ which is locally isomorphic to an open ball in $\mathbb{R}^n$ with the sheaf of real-valued continuous functions on it. Similarly a C^k-manifold is a ringed space which is locally isomorphic to an open in $\mathbb{R}^n$ with the sheaf of real-valued C^k functions on it. And a complex manifold is a ringed space locally isomorphic to an open ball in $\mathbb{C}^n$ with the sheaf of holomorphic complex-valued functions on it.

In each of these examples we must also require that X be second-countable and Hausdorff.

Theorem 2.3.13

There is a contravariant equivalence of categories $\operatorname{Spec}\colon \mathrm{Ring}^{\mathrm{op}} \to \mathrm{AffSch}\colon \Gamma$.

Proof

Because

$$\Gamma \operatorname{Spec} A = \operatorname{Spec} A(\mathrm{D}\, 1) = A_1 = A$$

naturally, it is sufficient to show that Spec is fully faithful (as it is by definition essentially surjective). First we note that for a ring morphism $\varphi\colon A \to B$, we can obtain φ from $\operatorname{Spec}\varphi\colon \operatorname{Spec} B \to \operatorname{Spec} A$ as $\varphi = \Gamma \operatorname{Spec}\varphi$. Indeed $\operatorname{Spec}\varphi$ acts on global sections by the canonical map $A = A_1 \to B_{\varphi 1} = B$, which is φ. Thus Spec is faithful.

Similarly if $\varphi\colon \operatorname{Spec} B \to \operatorname{Spec} A$ is a morphism of schemes, then we claim $\varphi = \operatorname{Spec}\Gamma\varphi$. Indeed, let $\varphi_1 = \Gamma\varphi\colon A \to B$ and $\psi = \operatorname{Spec}\varphi_1$; further let φ denote the underlying topological map of φ and $\varphi^\sharp\colon \mathcal{O}_{\operatorname{Spec} A} \to \varphi_*\mathcal{O}_{\operatorname{Spec} B}$ be the sheaf morphism. By definition for $\mathfrak{q} \in \operatorname{Spec} B$, $\varphi^\sharp_{\mathfrak{q}}\colon \mathcal{O}_{\operatorname{Spec} A,\varphi\mathfrak{q}} \to \mathcal{O}_{\operatorname{Spec} B,\mathfrak{q}}$ is local; meaning

$$(\varphi^\sharp_{\mathfrak{q}})^{-1}(\mathfrak{m}_{\mathfrak{q}}) = \mathfrak{m}_{\varphi q}.$$

Recalling $\mathcal{O}_{\operatorname{Spec} A,\mathfrak{p}} \cong A_{\mathfrak{p}}$, we have the following commutative square

$$\begin{array}{ccc} A & \xrightarrow{\varphi_1} & B \\ \downarrow & & \downarrow \\ A_{\varphi\mathfrak{q}} & \xrightarrow{\varphi^\sharp_{\mathfrak{q}}} & B_{\mathfrak{q}} \end{array}$$

The image of $a \in A$ in the diagonal is then $\varphi^\sharp_{\mathfrak{q}}(a/1) = \varphi_1(a)/1$. Thus $\varphi^\sharp_{\mathfrak{q}}(a/1) \in \mathfrak{m}_{\mathfrak{q}}$ iff $\varphi_1(a) \in \mathfrak{q}$. The former occurs only if $a/1 \in \mathfrak{m}_{\varphi\mathfrak{q}}$ iff $a \in \varphi\mathfrak{q}$. So $a \in \varphi\mathfrak{q}$ iff $\varphi_1(a) \in \mathfrak{q}$, so that $\varphi_1^{-1}\mathfrak{q} = \varphi\mathfrak{q}$; meaning $\varphi_1^* = \varphi$ as continuous maps.

Now we need to show that $\varphi_1^\sharp = \varphi^\sharp$ as sheaf maps. We do this on the distinguished open subsets. For $f \in A$ we have that

$$\varphi^{-1}\,\mathrm{D}_A(f) = (\varphi_1^*)^{-1}\,\mathrm{D}_A(f) = \left\{\mathfrak{q} \leq B \;\middle|\; f \notin \varphi_1^{-1}\mathfrak{q}\right\} = \mathrm{D}_B(\varphi_1 f).$$

And we want to show that

$$\varphi^\sharp_f\colon \mathcal{O}_{\operatorname{Spec} A}(\mathrm{D}\, f) = A_f \to \mathcal{O}_{\operatorname{Spec} B}(\varphi^{-1}\,\mathrm{D}\, f) = B_{\varphi_1 f}$$

is equal to induced map $A_f \to B_{\varphi_1 f}$ given by $A \xrightarrow{\varphi_1} B \to B_{\varphi_1 f}$. But this immediate from the naturality of φ: the following diagram commutes

$$\begin{array}{ccc} A & \xrightarrow{\varphi_1} & B \\ \downarrow & & \downarrow \\ A_f & \xrightarrow{\varphi^\sharp_f} & B_{\varphi_1 f} \end{array}$$

Thus we get the desired result. ∎

Example 2.3.14

Let $A = k[x, y]$, so $\operatorname{Spec} A = \mathbb{A}^2_k$; define $U = \mathbb{A}^2 - \{(0,0)\} = \mathbb{A}^2 - \{(x,y)\}$. Note that $U = \mathrm{D}(x) \cup \mathrm{D}(y)$, which forms an affine cover so that U is a scheme. We claim that U is not affine; and thus not a distinguished open set.

Indeed, inclusion $U \hookrightarrow \operatorname{Spec} A$ induces a map $\Gamma U \to A$. We claim it is an isomorphism: this is sufficient as the inclusion is not an isomorphism, and if U were affine as Γ is an equivalence this would imply it were.

Indeed, functions on U are functions on $\mathrm{D}(x)$ and $\mathrm{D}(y)$ which agree on their intersection: $\mathrm{D}(x) \cap \mathrm{D}(y) = \mathrm{D}(xy)$. I.e. rational maps in A_x and A_y which are in A_{xy}. These are just the polynomials so $U = k[x, y] = A$.

We can extend the result of theorem 2.3.13. The proof is similar, and we rewrite it.

Lemma 2.3.15

The global sections functor $\Gamma\colon \mathrm{LocRing}^{\mathrm{op}} \to \mathrm{Ring}$ is fully faithful on morphisms from a locally ringed space to an affine scheme. That is for a locally ringed space X and a ring A,

$$\mathrm{LocRing}(X, \operatorname{Spec} A) \xrightarrow{\Gamma} \mathrm{Ring}(A, \Gamma X)$$

is bijective.

Proof

Suppose we have two morphisms $\varphi, \psi\colon X \to \operatorname{Spec} A$ where X is a locally ringed space, such that $\Gamma\varphi = \Gamma\psi$. Let us write $\varphi^\sharp$ for the sheaf morphism part of φ, and similar for ψ; we use φ for the continuous part. Furthermore write φ_1 for $\Gamma\varphi$, similar for ψ. Let $p \in X$: we want to show that $\varphi(p) = \psi(p)$.

Let $\mathfrak{q}_1 = \varphi(p)$ and $\mathfrak{q}_2 = \psi(p)$, and consider the following commutative square:

$$\begin{array}{ccc} A & \xrightarrow[\varphi_1]{} & \Gamma X \\ \downarrow & & \downarrow{\scriptstyle \mathrm{loc}} \\ A_{\mathfrak{q}_1} & \xrightarrow{\varphi_p} & \mathcal{O}_{X,p} \end{array}$$

Because φ_p is a local morphism we have that for $a \in A$, $a \in \mathfrak{q}_1$ iff $\varphi_p(a/1) \in \mathfrak{m}_{X,p}$. This is iff $\mathrm{loc} \circ \varphi_1(a) \in \mathfrak{m}_{X,p}$. Because $\varphi_1 = \psi_1$, this means that $a \in \mathfrak{q}_1$ iff $a \in \mathfrak{q}_2$ so that $\varphi(p) = \psi(p)$ as desired.

Now we want to show that $\varphi^\sharp = \psi^\sharp$; we can do this on distinguished open sets. For $f \in A$ we have the commutative square

$$\begin{array}{ccc} A & \xrightarrow[\varphi_1]{} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow \\ A_f & \xrightarrow{\varphi_f} & \mathcal{O}_X(\varphi^{-1}\,\mathrm{D}(f)) \end{array}$$

Note that by localization φ_f is the unique morphism making this square commute. Since $\varphi_1 = \psi_1$ this implies $\varphi_f = \psi_f$. Thus Γ is faithful.

For fullness, suppose $\sigma\colon A \to \Gamma X$ is a ring morphism. For $p \in X$ let $\sigma_p\colon A \xrightarrow{\sigma} \Gamma X \to \mathcal{O}_{X,p}$ be the composite and define $\varphi(p) = \sigma_p^{-1}(\mathfrak{m}_{X,p})$. The preimage of a maximal ideal is prime, so φ forms a map $X \to \operatorname{Spec} A$. It is continuous: for $f \in A$:

$$\varphi^{-1}\,\mathrm{D}(f) = \{p \in X \mid \sigma_p(f) \notin \mathfrak{m}_{X,p}\} = \{p \in X \mid \sigma(f)(p) \neq 0\}\,,$$

where $\sigma(f)(p)$ is the value of the global section $\sigma(f)$ at p. By proposition 2.3.10 this is an open set.

Now we define $\varphi^\sharp\colon \operatorname{Spec} A \to \varphi_* X$ again on the distinguished open sets. For $f \in A$, $\sigma(f)$ vanishes nowhere in $\varphi^{-1}\,\mathrm{D}(f)$, so it is invertible in $\mathcal{O}_X(\varphi^{-1}\,\mathrm{D}\,f)$ again by proposition 2.3.10. Thus there is a unique map $\varphi_f\colon A_f \to \mathcal{O}_X(\varphi^{-1}\,\mathrm{D}\,f)$ making the following square commute

$$\begin{array}{ccc} A & \xrightarrow[\sigma]{} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow \\ A_f & \xrightarrow{\varphi_f} & \mathcal{O}_X(\varphi^{-1}\,\mathrm{D}\,f) \end{array}$$

This forms a morphism of sheaves, and $(\varphi, \varphi^\sharp)$ is a morphism of locally ringed spaces. For $p \in X$, let $\mathfrak{q} = \varphi(p) = \sigma_p^{-1}(\mathfrak{m}_{X,p})$. We want to show that $\varphi_p(\mathfrak{q}A_{\mathfrak{q}}) \subseteq \mathfrak{m}_{X,p}$. By considering the following diagram

$$\begin{array}{ccc} A & \xrightarrow[\sigma]{} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow{\scriptstyle \mathrm{loc}} \\ A_{\mathfrak{q}} & \xrightarrow{\varphi_p} & \mathcal{O}_{X,p} \end{array}$$

it is sufficient to show that $\mathrm{loc} \circ \sigma(\mathfrak{q}) \subseteq \mathfrak{m}_{X,p}$. By definition this is just $\sigma_p(\mathfrak{q}) \subseteq \mathfrak{m}_{X,p}$, which is clear by definition of $\mathfrak{q}$. ■

Theorem 2.3.16

The functors $\mathrm{Spec}\colon \mathrm{Ring}^{\mathrm{op}} \to \mathrm{LocRing}$ and $\Gamma\colon \mathrm{LocRing}^{\mathrm{op}} \to \mathrm{Ring}$ are mutually right adjoint. That is, there is a natural isomorphism

$$\mathrm{LocRing}(X, \mathrm{Spec}\, A) \cong \mathrm{Ring}(A, \Gamma X), \qquad A \in \mathrm{Ring},\ X \in \mathrm{LocRing}.$$

We can restrict this adjunction to Sch instead of LocRing.

Proof

By the previous theorem, $\mathrm{LocRing}(X, \mathrm{Spec}\, A) \xrightarrow{\Gamma} \mathrm{Ring}(A, \Gamma X)$ is a bijection, thus a natural isomorphism. ■

An immediate corollary is the following.

Corollary 2.3.17

The global sections and Spec functors contravariantly preserve colimits.

Note:
By *contravariantly* preserving (co)limits, we mean it maps limiting cones to colimiting cones or colimiting cones to limiting cones.

2.4 Projective schemes

Lemma 2.4.1

Suppose we have a collection of schemes $\{X_i\}_{i \in I}$, open subschemes $X_{ij} \subseteq X_i$ for $i, j \in I$ with $X_{ii} = X_i$, and isomorphisms $f_{ij}\colon X_{ij} \cong X_{ji}$. Further suppose that these isomorphisms satisfy the **cocycle condition**:

$$f_{ik}|_{X_{ij} \cap X_{ik}} = f_{jk}|_{X_{ji} \cap X_{jk}} \circ f_{ij}|_{X_{ij} \cap X_{ik}}.$$

In this condition we leave implicit that $f_{ij}(X_{ij} \cap X_{ik}) \subseteq X_{jk}$.
Then there exists a unique scheme X which has open subschemes isomorphic to X_i whose intersections are isomorphic to X_{ij}, which respects the gluing data in the obvious sense.

Proof

TODO: This

Example 2.4.2 (Projective space)

Let A be a ring, we define **projective n-space** $\mathbb{P}^n_A$ by gluing together $n+1$ copies of $\mathbb{A}^n_A = \operatorname{Spec} A[x_1, \dots, x_n]$ in a particular manner. Let us denote these copies by $U_0, \dots, U_n$ where we write U_i as

$$U_i = \operatorname{Spec} A[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1).$$

Then we let $U_{ij} = \mathrm{D}(x_{j/i})$, and we identify U_{ij} and U_{ji} via the isomorphism

$$A[x_{0/i}, \dots, x_{n/i}, 1/x_{j/i}]/(x_{i/i} - 1) \cong A[x_{0/j}, \dots, x_{n/j}, 1/x_{i/j}]/(x_{j/j} - 1)$$

which maps $x_{k/i}$ to $x_{k/j}/x_{i/j}$. Because Spec is an equivalence, this forms an isomorphism of schemes.

It is easy to see that these isomorphisms satisfy the cocycle conditions, so we can glue them together to get the scheme $\mathbb{P}^n_A$.

We can write $\mathbb{P}^n_A$ as a colimit of the U_i and U_{ij}s, and because Γ contravariantly preserves colimits (corollary 2.3.17), we can write $\Gamma(\mathbb{P}^n_A)$ as a limit of $A[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1)$ and $A[x_{0/i}, \dots, x_{n/i}, 1/x_{j/i}]/(x_{i/i} - 1)$. It is easy to see that this colimit is isomorphic to A, so that $\Gamma(\mathbb{P}^n_A) \cong A$. But for $n > 0$, $\mathbb{P}^n_A$ is not isomorphic to $\operatorname{Spec} A$, so $\mathbb{P}^n_A$ is not affine.

We can generalize the previous example.

Definition 2.4.3

If M is a commutative monoid, a **M-graded ring** is a ring $S_\bullet = \bigoplus_{m \in M} S_m$ (where S_m are Abelian groups, whose elements are called **homogeneous of degree** m) such that multiplication in $S_\bullet$ respects the grading: it forms maps $S_m \times S_n \to S_{n+m}$ for $n, m \in M$.

Clearly then S_0 is a subring and $S_\bullet$ is an S_0-algebra. An ideal I of $S_\bullet$ is **homogeneous** if it is generated by homogeneous elements.

If $S_\bullet$ is M-graded, then there are obvious projections $S_\bullet \to S_m$ for all $m \in M$; the image of $a \in S_\bullet$ under this projection is called its **degree m part** and is denoted a_m. An ideal $I \leq S_\bullet$ is homogeneous iff it is closed under these projections: if $a \in I$ then $a_m \in I$. Clearly if this holds then I is homogeneous: it is generated by the images of itself under all the projections. Conversely if I is homogeneous, say generated by I_{hom}, then it is easily seen that any combination of elements in I_{hom} have degree m parts which are also combinations of I_{hom}.

Thus a homogeneous ideal can be decomposed into homogeneous pieces $I = \bigoplus_{m \in M} I_m$ where $I_m \leq S_m$ is a subgroup (and $I_0 \leq S_0$ is an ideal). Thus $S_\bullet/I \cong \bigoplus_{m \in M} S_m/I_m$ has a natural M-grading.

Clearly the sum and product of homogeneous ideals are homogeneous. The intersection of homogeneous ideals is homogeneous as well: if $a \in I \cap J$ then $a_m \in I \cap J$ for all $m \in M$ so by the above consideration $I \cap J$ is homogeneous. And if I is homogeneous, so is $\sqrt{I}$: if $a^n \in I$ then

$$(a^n)_m = \sum_{k_1 + \cdots + k_n = m} a_{k_1} \cdots a_{k_n}$$

is in I as all the a_{k_i} are.

If M is a commutative monoid, we associate to it the Abelian group $\widetilde{M}$ which is the quotient of the monoid $M \times M$ by the equivalence $(m, n) \sim (m + k, n + k)$ for all $k \in M$. We can embed $M \to \widetilde{M}$ by $m \mapsto (m, 0)$, and we note that this is the identity of an adjunction CommMon $\rightleftarrows$ Ab.

If $T \subseteq S_\bullet$ is a multiplicative subset containing only homogeneous elements, then $T^{-1}S_\bullet$ has an obvious grading over $\widetilde{M}$ where a_m/t_n has degree $m - n$. It is easy to check that this a well-defined grading.

Note:
In order to avoid excessive subscripts, we will also use the notation $S_\bullet[T^{-1}]$ for $T^{-1}S_\bullet$. In particular, $(S_\bullet)_f$ will instead be denoted $S_\bullet[1/f]$.

Now we consider only the case where $M = \mathbb{Z}$ or $\mathbb{Z}_{\geq 0}$ (every $\mathbb{Z}_{\geq 0}$-grading can be turned into a $\mathbb{Z}$ grading by setting $S_n = 0$ for $n < 0$). We define the **degree** of $a \in S_\bullet$ to be

$$\deg(a) = \max\{n \in \mathbb{Z} \mid a_n \neq 0\}.$$

In order for an homogeneous ideal $I < S_\bullet$ to be prime, it is sufficient to check the usual condition on homogeneous elements only: that is, if $ab \in I$ then $a, b \in I$ for homogeneous a, b. Suppose that $a, b \notin I$, then define $a_I = \sum_{m, a_m \in I} a_m$ and $a^I = \sum_{m, a_m \notin I} a_m$ so that $a = a_I + a^I$, thus

$$ab = a_I b_I + a_I b^I + a^I b_I + a^I b^I.$$

As $a_I \in I$ we have that $a^I \notin I$, and we need to show that $a^I b^I \notin I$. That is, it is sufficient to consider the case where $a_m \notin I$ for all $m \in M$ (or is zero). Let $n = \deg(a)$ and $m = \deg(b)$ so that $(ab)_{n+m} = a_n b_m$. Because $a_n, b_m \notin I$ we have that $(ab)_{n+m} \notin I$ (by primality on homogeneous elements) and thus $ab \notin I$ (because I is homogeneous).

Note:
For the remainder of these notes, $S_\bullet$ will be a $\mathbb{Z}_{\geq 0}$-graded ring unless otherwise specified.

If $S_0 = A$, then $S_\bullet$ is said to be **graded over** A. For example $A[x_0, \ldots, x_n]$ is graded over A (where the homogeneous elements are homogeneous polynomials), or $A[x_0, \ldots, x_n]/I$ where I is an homogeneous ideal and $I \cap A = 0$. For a graded ring $S_\bullet$, $S_+ = \bigoplus_{n>0} S_n$ is an ideal called the **irrelevant ideal**. If S_+ is a finitely generated ideal, then $S_\bullet$ is said to be **finitely generated over** A.

Note that $S_\bullet$ is finitely generated over A iff $S_\bullet$ is a finitely generated graded A-algebra (i.e. generated by finitely many homogeneous elements). If $S_\bullet$ is finitely generated over A, so $S_+ = (f_1, \ldots, f_n)$ homogeneous, then by considering degrees we see that S_1 is A-spanned by the f_i. Then we can see that S_2 is A-spanned by the $f_i, f_i f_j$s, and so on by induction. That is, S_k is spanned by the products of f_is, so that the f_is generate $S_\bullet$ as an A-algebra. Conversely if $S_\bullet$ is finitely generated as a graded A-algebra by $f_1, \ldots, f_n$, then these must also generate S_+.

Furthermore $S_\bullet$ is Noetherian iff A is, and $S_\bullet$ is finitely generated over A. Indeed if $S_\bullet$ is Noetherian, then S_+ is finitely generated as it is an ideal; by the previous paragraph this means that $S_\bullet$ is finitely generated. And any ideal of A can be extended to an homogeneous ideal of $S_\bullet$ (with the same generators), which is finitely generated. It is clear that the degree zero generators of this ideal generate the ideal from A. Alternatively we could note that the extension of A's ideal is unique, as it adds no additional elements from A, and consider the definition of a Noetherian ring. Conversely if $S_\bullet$ is finitely generated and A is Noetherian, then $S_\bullet$ is a Noetherian A-algebra and thus a Noetherian ring.

Now we can see why graded rings are important. The idea is that for a graded ring $S_\bullet$ we will define a scheme $\operatorname{Proj} S_\bullet$, motivated by $\mathbb{P}^n_A$. That is, $\operatorname{Proj} A[x_0, \ldots, x_n]$ will give us $\mathbb{P}^n_A$.

Proposition 2.4.4

Let $f \in S_+$ be homogeneous. Then there is a bijection between the prime ideals of $(S_\bullet)[1/f]_0$ and the homogeneous prime ideals of $(S_\bullet)[1/f]$.

Proof

We generalize this result as follows: for any $\mathbb{Z}$-graded ring A with an invertible homogeneous element $f \in A$ of positive degree, there is a bijection between the prime ideals of A_0 and the homogeneous prime ideals of A.

Consider the ring map $A_0 \to A$, which then maps homogeneous prime ideals of A to prime ideals of A_0 via the inverse image. Note that this maps a prime ideal $P = \bigoplus_i Q_i \leq A$ to Q_0. Conversely if $P_0 \leq A_0$ is a prime ideal, then define $P \leq A$ as $P = \bigoplus_i Q_i$ where $a \in Q_i$ iff $a^{\deg f}/f^i \in P_0$. Here we have that $Q_0 = P_0$. It remains to show that P is an ideal, and prime.

Note that $a \in Q_i$ iff $a^2 \in Q_{2i}$: $a^{\deg f}/f^i \in P_0$ iff $a^{2\deg f}/f^{2i} \in P_0$ as P_0 is prime, iff $a^2 \in Q_{2i}$. And if $a_1, a_2 \in Q_i$ then $(a_1 + a_2)^2 \in Q_{2i}$; indeed by the binomial theorem (letting $d = \deg f$)

$$\frac{(a_1 + a_2)^{2d}}{f^{2i}} = \frac{\sum_k \binom{2d}{k} a_1^k a_2^{2d-k}}{f^{2i}},$$

and for all k, one of k or $2d - k$ is at least d, so each summand is in P_0. Thus $a_1 + a_2 \in Q_i$, meaning P is closed under addition.

And if $a \in Q_i$ and $b \in S_\bullet$ has degree k, then $ab \in Q_{i+k}$ because

$$\frac{(ab)^d}{f^{i+k}} = \frac{a^d}{f^i} \cdot \frac{b^d}{f^k} \in P_0.$$

Thus P is closed under multiplication so it is an ideal.

It is sufficient to check primality on homogeneous elements, so if $a, b \in S_\bullet$ are homogeneous of degrees i, j respectively, then if $ab \in Q_{i+j}$ that means $(ab)^d/f^{i+j} \in P_0$. By primality $a^d/f^i, b^d/f^j \in P_0$ so $a \in Q_i$ and $b \in Q_j$ as desired. Thus $P \leq S_\bullet$ is a prime ideal.

Now if $P \leq S_\bullet$ is a prime ideal with $P = \bigoplus_i P_i$, we want to show that $a \in P_i$ iff $a^d/f^i \in P_0$. Indeed if $a^d/f^i \in P_0$ then by primality $a \in P_i$. Conversely if $a \in P_i$ then $a^d/f^i \in P$ is homogeneous of degree 0, so it is in P_0 as desired. So the two maps are mutual inverses, as desired. ■

Now we can construct $\operatorname{Proj} S_\bullet$ as the gluing of $\operatorname{Spec} S_\bullet[1/f]_0$ along the $U_{fg} = \mathrm{D}(g^{\deg f}/f^{\deg g})$, where f ranges over the homogeneous elements of positive degree. Note that for $S_\bullet = A[x_0, \ldots, x_n]$ and $f = x_i$ then $S_\bullet[1/f]_0 \cong A[x_{0/i}, \ldots, x_{n/i}]/(x_{i/i} - 1)$ by $x_{k/i} \mapsto x_k/x_i$. And $U_{ij} = \mathrm{D}(x_j/x_i)$ corresponds to $\mathrm{D}(x_{j/i})$, so this gives us the original definition of $\mathbb{P}^n_A$ (we will see that we need only glue along generators).

We note that U_{fg} is isomorphic to $\operatorname{Spec} S_\bullet[1/fg]_0$. This isomorphism is induced from the map $S_\bullet[1/f] \to S_\bullet[1/fg]$. Thus there is an isomorphism $U_{fg} \cong U_{gf}$, and it is routine to check that these isomorphisms satisfy the cocycle condition. Thus we have defined $\operatorname{Proj} S_\bullet$; we will continue this section by providing more descriptions of $\operatorname{Proj} S_\bullet$.

By the above proposition, the prime ideals of $S_\bullet[1/f]_0$ correspond to the homogeneous prime ideals of $S_\bullet[1/f]$. We can construct a bijection between the points of $\operatorname{Proj} S_\bullet$ and the homogeneous prime ideals of $S_\bullet$. For a homogeneous prime ideal $\mathfrak{p} \leq S_\bullet$, then for all $f \notin \mathfrak{p}$ we have the homogeneous prime ideal $\mathfrak{p}_f \in S_\bullet[1/f]$. All of these ideals are identified: if $f, g \notin \mathfrak{p}$ then $fg \notin \mathfrak{p}$ so that $\mathfrak{p}_f$ and $\mathfrak{p}_g$ are identified in $\operatorname{Spec} S_\bullet[1/fg]_0$. And all of homogeneous prime ideals of $S_\bullet[1/f]$ are of this form, so mapping $\mathfrak{p} \mapsto \mathfrak{p}_f$ is surjective. Because $\mathfrak{p} \mapsto \mathfrak{p}_f$ is an injection (for fixed f, $f \notin \mathfrak{p}$), we see that distinct ideals are not identified.

Note that we must consider only $\mathfrak{p}$ which don't contain some homogeneous $f \in S_+$; that is S_+ cannot be a subset of $\mathfrak{p}$. This is because we consider only f with positive degree. These homogeneous prime ideals which do not contain S_+ are called **relevant**; otherwise they are **irrelevant**.

Thus the points of $\operatorname{Proj} S_\bullet$ correspond to the relevant homogeneous prime ideals of $S_\bullet$.

Definition 2.4.5

Let $T \subseteq S_\bullet$ be a collection of homogeneous elements. Define its **vanishing set** to be

$$\mathrm{V}(T) = \{\mathfrak{p} \leq S_\bullet \text{ homogeneous prime} \mid T \subseteq \mathfrak{p}, \text{ and } \mathfrak{p} \text{ does not contain the irrelevant ideal}\}.$$

And if f is a homogeneous element of *positive degree*, define $\mathrm{D}_+(f) = \operatorname{Proj} S_\bullet - \mathrm{V}(f)$; its **projective distinguished open set**.

Note that $\mathrm{D}_+(f)$ can be written as

$$\mathrm{D}_+(f) = \{\mathfrak{p} \leq S_\bullet \text{ homogeneous prime} \mid f \notin \mathfrak{p}\}.$$

It is immediat that this does not contain any irrelevant ideals as $f \in S_+$. Furthermore this corresponds to $\operatorname{Spec} S_\bullet[1/f]_0$. Indeed, the homogeneous prime ideals of $S_\bullet[1/f]$ correspond to the homogeneous prime ideals of $S_\bullet$ not containing f.

The projective vanishing sets satisfy the same properties as those in the affine case. Thus they form a topology of closed sets on $\operatorname{Proj} S_\bullet$, and $\mathrm{D}_+(f)$ form a basis.

Note that any homogeneous ideal $I \leq S_\bullet$ is contained in a homogeneous prime ideal. Indeed, for an ideal I define its **homogenization** to be

$$I_{\mathsf{hom}} = \bigoplus_n (I \cap S_n) = (a \in I \mid a \text{ is homogeneous}) .$$

Because primality can be checked on homogeneous elements, if $\mathfrak{p}$ is prime so is $\mathfrak{p}_{\mathsf{hom}}$. Thus if $I \leq \mathfrak{p}$ is prime, then $I \leq \mathfrak{p}_{\mathsf{hom}}$ is homogeneous prime. Note that because $\mathfrak{p}_{\mathsf{hom}} \leq \mathfrak{p}$, we have that

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p} \text{ homogeneous}} \mathfrak{p}.$$

Here the intersection is almost over $\mathrm{V}(I)$, but it may be the case that the $\mathfrak{p}$s are irrelevant. We will remedy this now.

Note that the following are equivalent for any homogeneous ideal $I \subseteq S_+$.

(1) $\mathrm{V}(I) = \emptyset$,

(2) if $I = (f_i)_i$ are homogeneous generators, then $\bigcup_i \mathrm{D}_+(f_i) = \operatorname{Proj} S_\bullet$,

(3) $\sqrt{I} \supseteq S_+$.

(1) and (2) are equivalent because

$$\mathrm{V}(f_i)_i = \bigcap_i \mathrm{V}(f_i),$$

which is then empty iff its complement is $\operatorname{Proj} S_\bullet$, which is precisely (2). Note that $\mathrm{V}(I) = \emptyset$ iff all of the homogeneous prime ideals containing I also contain S_+. This is iff their intersection contains S_+. Because this intersection is $\sqrt{I}$, this gives us (3).

Call an ideal I satisfying any of the above equivalent conditions **irrelevant**. Clearly by (3) this agrees with our definition of an irrelevant prime ideal.

For a subset $Z \subseteq \operatorname{Proj} S_\bullet$, define the relevant homogeneous ideal $\mathrm{I}(Z) \leq S_\bullet$ by

$$\mathrm{I}(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}.$$

The intersection of homogeneous ideals is homogeneous, so $\mathrm{I}(Z)$ is homogeneous. Then we have that

$$\mathrm{V}\,\mathrm{I}(Z) = \overline{Z}$$

whose proof is the same as in the affine case.

We also have that for $I \leq S_+$ relevant homogeneous,

$$\mathrm{I}\,\mathrm{V}(I) = \sqrt{I}.$$

Indeed because $I \leq S_+$ we can ignore degree zero, in which case all irrelevant prime ideals all look like S_+, and thus contribute nothing to the intersection, as there exists a relevant prime ideal. Thus the intersection of all homogeneous prime ideals containing I looks the same as the intersection of all relevant prime ideals containing I.

Thus as in the affine case, this clearly demonstrates that $\mathrm{V}(I) \subseteq \mathrm{V}(J)$ iff $\sqrt{J} \leq \sqrt{I}$.

Now we want to define the structure sheaf on $\operatorname{Proj} S_\bullet$. Recall that there is a correspondence $\varphi\colon \mathrm{D}_+(f) \cong \operatorname{Spec} S_\bullet[1/f]_0$; we claim that this correspondence is a homeomorphism. Indeed, the map $\mathfrak{p} \mapsto (\mathfrak{p}_f)_0$ is continuous: for $g/f^k \in S_\bullet[1/f]_0$ we have that

$$\varphi^{-1}\,\mathrm{D}(g/f^k) = \left\{ \mathfrak{p} \in \mathrm{D}_+(f) \;\middle|\; g/f^k \notin (\mathfrak{p}_f)_0 \right\}.$$

This is equal to $\{\mathfrak{p} \mid g \notin \mathfrak{p}\}$; because $f \notin \mathfrak{p}$, we have that $g/f^k \in \mathfrak{p}_f$ iff $gf^t \in \mathfrak{p}$ for some t iff $g \in \mathfrak{p}$. Thus

$$\varphi^{-1} \mathrm{D}(g/f^k) = \mathrm{D}_+(g)$$

so that φ is continuous. And it is open, since for g homogeneous

$$\varphi \mathrm{D}_+(fg) = \{(\mathfrak{p}_f)_0 \mid fg \notin \mathfrak{p}\} = \mathrm{D}\left(\frac{g^{\deg f}}{f^{\deg g}}\right).$$

The last equality is because $g^d/f^k \notin \mathfrak{p}_f$ iff $g^d \notin \mathfrak{p}$ iff $g \notin \mathfrak{p}$ when $f \notin \mathfrak{p}$.
Now $\mathrm{D}_+(fg)$ corresponds to $\mathrm{D}(g^{\deg f}/f^{\deg g})$ in $\operatorname{Spec} S_\bullet[1/f]_0$ and to $\mathrm{D}(f^{\deg g}/g^{\deg f})$ in $\operatorname{Spec} S_\bullet[1/g]_0$. We want to define the structure sheaf on $D_+(f) \subseteq \operatorname{Proj} S_\bullet$ to be the structure sheaf of $\operatorname{Spec} S_\bullet[1/f]_0$. It is routine to check that this agrees with triple intersections (the cocycle condition).
Note that if $S_+ = (f_i)_i$ then we can consider only the $\mathrm{D}_+(f_i)$. Indeed, this comes down to the fact that if $X = \bigcup_i U_i$ then any sheaf $(X, \mathcal{O}_X)$ can be written as the gluing of the $(U_i, \mathcal{O}_{U_i})$.
So in terms of $A[x_0, \dots, x_n]$, we have that the x_i generate the irrelevant ideal, so we can focus on the structure sheaf restricted to

$$\mathrm{D}_+(x_i) \cong \operatorname{Spec} A[x_0, \dots, x_n, 1/x_i]_0 \cong \operatorname{Spec} A[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1).$$

Thus defining $\mathbb{P}^n_A = \operatorname{Spec} A[x_0, \dots, x_n]$ agrees with our previous definition.

2.5 Properties of schemes

2.5.1 Topological properties

Proposition 2.5.1

If X is a topological space with a finite open cover $\{X_i\}_{i=1}^n$ such that each X_i is irreducible and $\bigcap_i X_i \neq \emptyset$, then X is irreducible.

Proof

Let $Y = \bigcap_i X_i$; as a finite intersection of open sets it is open. For $U, V \subseteq X$ nonempty open, we must have that $U \cap X_i \neq \emptyset$ for some i. Since X_i is irreducible and $Y \subseteq X_i$ is open, we must have that $U \cap Y$ is nonempty. Similarly $V \cap Y$ is nonempty, and these are both nonempty open subsets of any X_i, so they have nonempty intersection. Thus $U \cap V \neq \emptyset$ as desired. ■

Corollary 2.5.2

If A is a domain, then $\mathbb{P}^n_A$ is irreducible.

Proof

We know that $\mathrm{D}_+(x_i) \cong \operatorname{Spec} A[x_0, \dots, x_n]$ is irreducible by corollary 2.1.8. And the intersection of all these $\mathrm{D}_+(x_i)$ is nonempty (it contains the zero ideal). ■

Recall that for an affine scheme, there is a bijection between points and irreducible closed subsets, given by $p \mapsto \overline{\{p\}}$. Recall this is because closed irreducible subsets are all of the form $\mathrm{V}(\mathfrak{p})$ for a prime ideal $\mathfrak{p}$. This is true for schemes in general. Indeed, if $p \mapsto \{p\}$ is a bijection between points and closed irreducible subsets of $X_i \subseteq X$ for an open cover $X = \bigcup_i X_i$, then it is one for X as well.
Let $Z \subseteq X$ be closed irreducible, so there is an i such that $Z \cap X_i \subseteq X_i$ is nonempty. Thus there is $p \in X_i$ such that $Z \cap X_i = \mathrm{cl}_{X_i}(p)$. Because $Z \cap X_i \subseteq Z$ is open, it is dense, so that

$$\mathrm{cl}_X(p) = \mathrm{cl}_X(\mathrm{cl}_{X_i}(p)) = \mathrm{cl}_X(Z \cap X_i) = Z$$

as desired.
Now if $\mathrm{cl}(p) = \mathrm{cl}(q) = Z$ then $p = q$. Indeed, because $\{p\}, \{q\}$ are dense in Z, they intersect all of its nonempty open subsets. So choose i such that $Z \cap X_i \neq \emptyset$, then $p, q \in X_i$ and $\mathrm{cl}_{X_i}(p) = \mathrm{cl}_{X_i}(q) = Z \cap X_i$ and then by assumption $p = q$.

Corollary 2.5.3

If X is a scheme, $p \mapsto \{p\}$ is a bijection between points of X and its closed irreducible subsets.

Proof

X has an open cover of affine schemes, which have this property. By the above discussion, X has this property. ■

Note:
A space with this property is called a **Sober space**.

Proposition 2.5.4

If $X = \bigcup_i X_i$ is a finite covering of X by Noetherian subspaces, then X is Noetherian.

Proof

Let $Z_0 \supset Z_1 \supset \cdots$ be a descending chain of closed subsets of X. Then intersecting with X_i gives a stabilizing chain, say it stabilizes at n_i. Then the original chain stabilizes at $N = \max n_i$ (because the original chain is the union of the intersection with the X_is). ■

Corollary 2.5.5

If X is a scheme with a finite cover $X = \bigcup_{i=1}^{n} \operatorname{Spec} A_i$ where each A_i is Noetherian, then X is a Noetherian space.

Corollary 2.5.6

If A is Noetherian, then $\mathbb{P}^n_A$ is.

Proof

We know that $A[x_0, \ldots, x_n]$ is Noetherian by Hilbert's basis theorem, and $\mathbb{P}^n_A$ has a finite covering of the spectrum of this algebra. ■

Note that if X has a finite covering by quasicompact open subsets, then X is quasicompact. This is pretty immediate by taking an open cover and then considering it over each open set. Thus the following is immediate (as affine schemes are quasicompact).

Corollary 2.5.7

A scheme is quasicompact iff it has a finite open affine covering.

Proposition 2.5.8

Let X be a quasicompact scheme, then every closed subset of X has a closed point.

Proof

For affine schemes this is true, because every ideal $I \leq A$ is contained in a maximal ideal, which is a closed point in $\mathrm{V}(I)$.

Now let $X = \bigcup_{i=1}^{n} U_i$ form an affine cover, and let $p_1 \in Z \cap U_1$ be a closed point. If p_1 is closed in X we are done, otherwise $\overline{p_1}$ contains points other than p_1, which necessarily cannot lie in U_1. So $\overline{p_1}$ intersects some other U_i, call it U_2, and then again it contains some closed point (of U_2) $p_2 \in \overline{p_1} \cap U_2$. If p_2 is closed in X we are done, otherwise $\overline{p_2} \subseteq \overline{p_1}$, and by corollary 2.5.3 this is a strict inclusion. Thus it intersects some other U_3, and we can inductively choose a closed point $p_{i+1} \in \overline{p_i} \cap U_i$.

This process must halt eventually, as there are only n affine sets, so eventually we must find a closed point. ■

Remark 2.5.9

This proposition can help us prove properties of schemes. Specifically suppose we have some property P of schemes which is **open**: if a point p has P, then there is a neighborhood of p, all of whose points have P. Then in order to check if all points of a quasicompact scheme have P, it is sufficient to check that the closed points have P.

Note that $p \in \overline{q}$ iff $N(p) \subseteq N(q)$ (where $N(p)$ is the collection of p's neighborhoods). Indeed if $p \notin \overline{q}$ then $X - \overline{q}$ is a neighborhood of p which does not contain q. Conversely if $p \in U$ and $q \notin U$, then $\overline{q} \subseteq X - U$ and so $p \notin \overline{q}$.

So if P is open and all closed points of X have P, then let $q \in X$. Choose a closed point $p \in \overline{q}$ so that there is a neighborhood $p \in U$ such that all points of U have P. Because $N(p) \subseteq N(q)$, we have that $q \in U$ and thus q has P.

Definition 2.5.10

A space X is **quasiseparated** if the intersection of any two quasicompact open subsets is quasicompact.

A scheme X is quasiseparated iff the intersection of any two open affine subschemes is a finite union of affine open subsets. If X is quasiseparated this is clear, as affine subschemes are quasicompact, and open subsets are themselves schemes (so to be quasicompact they must have a finite affine covering). Conversely if $U, V \subseteq X$ are quasicompact open, then they each have a finite affine covering, and their intersection then has a finite affine covering, so is quasicompact.

It is sufficient to only consider a basis of affine subschemes, and as such it is easy to see that affine schemes are quasiseparated: $\mathrm{D}(f) \cap \mathrm{D}(g) = \mathrm{D}(fg)$.

Thus if X is covered by affine subschemes, such that the intersection of any two of these affine subschemes is quasicompact. Indeed, if $X = \bigcup_i U_i$ and $U_i \cap U_j = \bigcup_k V_{ijk}$ is a finite union of affine subschemes, then let $D_{i\ell}$ form an affine basis for U_i. Then $D_{i\ell} \cap D_{j\ell'} \subseteq U_i \cap U_j = \bigcup_k V_{ijk}$, and so $D_{i\ell} \cap D_{j\ell'} \cap V_{ijk}$ is quasicompact as V_{ijk} is affine and thus quasiseparated. Thus $D_{i\ell} \cap D_{j\ell'} \cap V_{ijk}$ can be written as a finite union of affine schemes, and taking their unions, so too can $D_{i\ell} \cap D_{j\ell'}$. Because the $D_{i\ell}$ form an affine basis of X, we have the desired result.

It is clear that a scheme X is quasicompact and quasiseparated iff it can be covered by a finite union of affine subschemes which have quasicompact intersection. Quasicompact and quasiseparated is often abbreviated as **qcqs**.

Example 2.5.11

A **projective A-scheme** is a scheme of the form $\operatorname{proj} S_\bullet$ where $S_0 = A$ and $S_\bullet$ is finitely generated over A. A projective A-scheme is qcqs.

Indeed, let $S_\bullet$ be generated by the homogeneous elements $(x_0, \ldots, x_n)$ over A. Then

$$\operatorname{proj} S_\bullet = \bigcup_{i=0}^{n} \mathrm{D}_+(x_i)$$

is a finite covering of affine subschemes (recall $\mathrm{D}(x_i) = \operatorname{Spec} S_\bullet[1/x_i]_0$). Furthermore $\mathrm{D}_+(x_i) \cap \mathrm{D}_+(x_j) = \mathrm{D}_+(x_i x_j)$ is affine as well, showing that $\operatorname{proj} S_\bullet$ is qcqs as desired.

Example 2.5.12

Let $X = \operatorname{Spec} k[x_1, x_2, \ldots]$ and $U = X - [\mathfrak{m}]$ where $\mathfrak{m}$ is the maximal ideal $(x_1, x_2, \ldots)$. Then let Y be the scheme obtained by gluing two copies of X along their common copy of U. Now both copies of X in Y are quasicompact (as they are affine), but their intersection is U which is not quasicompact.

Indeed we can cover U as follows

$$U = \mathrm{D}(\mathfrak{m}) = \bigcup_{i=1}^{\infty} \mathrm{D}(x_i).$$

And any finite union of $\mathrm{D}(x_i)$s is not equal to $\mathrm{D}(\mathfrak{m})$ (say because $x_{n+1} \notin \sqrt{(x_1, \ldots, x_n)}$).

2.5.2 Reducedness and integrality

Recall a ring A is **reduced** if $\operatorname{nil}(A) = 0$, i.e. it has no nonzero nilpotent elements.

Definition 2.5.13

A scheme is **reduced** if its stalks are all reduced.

Proposition 2.5.14

If f, g are two functions on a reduced scheme which agree at all points, then $f = g$.

Proof

It is sufficient to show that if f is zero at all points, then $f = 0$. That is if $f_p \in \mathfrak{m}_p$ for all $p \in X$, then $f = 0$.

It is sufficient to consider this locally, so we first show this for $X = \operatorname{Spec} A$. We have that $f_\mathfrak{p} \in \mathfrak{p}A_\mathfrak{p}$, meaning $f \in \mathfrak{p}$ for all prime $\mathfrak{p} \in \operatorname{Spec} A$. Thus $f \in \operatorname{nil}(A)$, and because ring morphisms preserve nilpotents ($\varphi(f)^n = \varphi(f^n) = 0$), we have that $f_\mathfrak{p} \in \operatorname{nil}(A_\mathfrak{p}) = 0$. Thus $f_\mathfrak{p} = 0$ in $A_\mathfrak{p}$ for all $\mathfrak{p}$. Because $A \to \prod_\mathfrak{p} A_\mathfrak{p}$ is injective, this means that $f = 0$.

Now if $X = \bigcup_i \operatorname{Spec} A_i$, then by above we have $f = 0$ in each A_i. By gluing this means $f = 0$ over X. ∎

Proposition 2.5.15

A ring A is reduced iff the affine scheme $\operatorname{Spec} A$ is. And a scheme X is reduced iff $\mathcal{O}_X(U)$ is reduced for all open $U \subseteq X$. It is sufficient to check this for an affine open cover $X = \bigcup_i U_i$.

Proof

The first claim amounts to saying A is reduced iff $A_\mathfrak{p}$ is for all prime $\mathfrak{p}$. By the injection $A \to \prod_\mathfrak{p} A_\mathfrak{p}$, if the $A_\mathfrak{p}$ are all reduced then so is A. And if A is reduced, recall that radicals commute with localization, so

$$\operatorname{nil}(A_\mathfrak{p}) = \sqrt{0_\mathfrak{p}} = \sqrt{0}_p = \operatorname{nil}(A)_\mathfrak{p} = 0_\mathfrak{p}.$$

Thus $A_\mathfrak{p}$ is reduced as desired.

Because being reduced is local, if X is reduced then every open $U \subseteq X$ is reduced. By the injection $\mathcal{O}_X(U) \to \prod_{p \in U} \mathcal{O}_{X,p}$, we have that $\mathcal{O}_X(U)$ is reduced.

Conversely if there is an affine open cover $X = \bigcup_i U_i$ where $\mathcal{O}_X(U_i)$ are reduced, then the U_i are all reduced as they are isomorphic to $\operatorname{Spec} \mathcal{O}_X(U_i)$, and this is reduced by the first paragraph. Stalks in X can be taken local to a U_i, so we have the desired result. ■

Example 2.5.16

$\mathbb{A}^n_A$ is reduced for any domain A, as $A[x_1, \ldots, x_n]$ is a domain and thus reduced. Similarly $\mathbb{P}^n_A$ is reduced for any domain A, because it has an affine covering by $A[x_0, \ldots, x_n]$ which are reduced.

Remark 2.5.17

Suppose X is a quasicompact scheme and f vanishes at all points of X. Then there is an n such that $f^n = 0$ in X.

Indeed suppose $X = \bigcup_i \operatorname{Spec} A_i$, following the proof of proposition 2.5.14, we see that $f \in \operatorname{nil}(A_i)$ for all i. So there are n_i such that $f^{n_i} = 0$ in A_i. Taking N to be the maximum n_i, we see that $f^n = 0$ in all A_i, so by gluing $f = 0$.

Definition 2.5.18

A nonempty scheme X is **integral** if $\mathcal{O}_X(U)$ is an integral domain for all open $U \subseteq X$.

Proposition 2.5.19

A scheme is integral iff it is irreducible and reduced.

Proof

If X is integral, then let $U, V \subseteq X$ be nonempty open. If $U \cap V = \emptyset$ then $\mathcal{O}_X(U \sqcup V) = \mathcal{O}_X(U) \times \mathcal{O}_X(V)$ is not a domain. Thus $U \cap V$ is nonempty so X is irreducible. And X is reduced because $\mathcal{O}_X(U)$ is a domain and thus reduced for all U, and by proposition 2.5.15.

Conversely suppose X is irreducible and reduced. Let $U \subseteq X$ be nonempty open, and suppose $f, g \neq 0$ in $\mathcal{O}_X(U)$. Let $U_f = \{p \in U \mid f(p) \neq 0\}$, so that U_f is nonempty by proposition 2.5.14. Because X is irreducible $V = U_f \cap U_g \neq \emptyset$. For $p \in V$ we have that $f(p), g(p) \neq 0$ and thus $(fg)(p) \neq 0$. This means that $fg \neq 0$ in V and thus is nonzero in U as well. ■

Proposition 2.5.20

Spec A is integral iff A is an integral domain.

Proof

Immediately by definition if Spec A is integral, then A is a domain.

Conversely if A is an integral domain, it is reduced. Furthermore nil $A = 0$ is a prime ideal, so A is irreducible by proposition 2.1.7. By the previous proposition this means Spec A is integral. ■

If X is an irreducible scheme, it is closed and so by proposition 2.5.3 it has a unique generic point $\eta \in X$. Now further suppose that X is integral, and let Spec $A \subseteq X$ be an affine neighborhood. Because η is dense in X, it is in Spec A and it must furthermore be a generic point there. Thus $\eta = \text{nil}(A) = 0$ in Spec A, and thus because localization is local

$$\mathcal{O}_{X,\eta} = A_\eta = A_0 = K(A),$$

the fraction field of A.

We denote $\mathcal{O}_{X,\eta}$ by $K(X)$ and call it the **function field** of X; its elements are **rational functions**.

Note that the canonical map $\mathcal{O}_X(U) \to K(X) = \mathcal{O}_{X,\eta}$ is injective. Indeed, if $f_\eta = 0$ then $\eta \in \{f_p = 0\}$ which is closed, and since η is generic, f is zero at all points of X. By proposition 2.5.14 this means $f = 0$.

Now if $U \subseteq V$ are nonempty, then inclusion forms a morphism of schemes $U \hookrightarrow V$. Thus localizing at η forms a map $\mathcal{O}_{V,\eta} \to \mathcal{O}_{U,\eta}$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}_X(V) & \xrightarrow{\text{res}_{V,U}} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{V,\eta} & \longrightarrow & \mathcal{O}_{U,\eta} \\ \| & & \| \\ K(X) & \longrightarrow & K(X) \end{array}$$

Because $K(X)$ is a field, the bottom horizontal arrow is injective. Since the vertical arrows are injective, this means $\text{res}_{V,U}$ is injective as well. Thus we have shown the following.

Proposition 2.5.21

Let X be an integral scheme. Then the restriction maps are all injections, as well as the canonical maps $\mathcal{O}_X(U) \to K(X)$.

2.5.3 Affine-locality

Proposition 2.5.22

Suppose Spec A, Spec B are affine neighborhoods of a scheme X. Then Spec $A \cap$ Spec B is the union of all the open sets which are simultaneously distinguished open subschemes of Spec A and Spec B.

Proof

We need to show that if $p \in \text{Spec } A \cap \text{Spec } B$, then there are $f \in A, g \in B$ such $p \in \text{D}_A(f) = \text{D}_B(g)$.

Let $f \in A$ be such that Spec A_f is a distinguished neighborhood of p in Spec $A \cap$ Spec B. And let Spec B_g be a distinguished neighborhood of Spec B in Spec A_f containing p. Then $g \in \Gamma(\text{Spec } B, \mathcal{O}_X)$ restricts to an

element $g' \in \Gamma(\operatorname{Spec} A_f, \mathcal{O}_X) = A_f$. Now

$$\operatorname{Spec} B_g = \operatorname{Spec} A_f - \{\mathfrak{p} \mid g' \in \mathfrak{p}\} = \operatorname{Spec}(A_f)_{g'}.$$

We can write $g' = g''/f^n$ for $g'' \in A$ and then $\operatorname{Spec}(A_f)_{g'} = \operatorname{Spec} A_{fg''}$. Thus

$$\operatorname{Spec} B_g = \operatorname{Spec} A_{fg''}$$

is a neighborhood of p which is distinguished both in $\operatorname{Spec} A$ and $\operatorname{Spec} B$. ■

Lemma 2.5.23 (Affine communication lemma)

Let P be some property satisfied by some affine open subsets of a scheme X, such that for any affine open subset $\operatorname{Spec} A \hookrightarrow X$,

(1) if $\operatorname{Spec} A \hookrightarrow X$ has property P, then for any $f \in A$, $\operatorname{Spec} A_f \hookrightarrow X$ has P as well;

(2) if $(f_1, \ldots, f_n) = A$, and $\operatorname{Spec} A_{f_i} \hookrightarrow X$ has P for all i, then so does $\operatorname{Spec} A \hookrightarrow X$.

Now suppose that $X = \bigcup_i \operatorname{Spec} A_i$ and $\operatorname{Spec} A_i$ all have property P. Then *every* affine open subset of X has P.

Proof

Let $\operatorname{Spec} A \hookrightarrow X$ be an affine open subscheme. Then we can cover $\operatorname{Spec} A$ with a finite number of distinguished open subsets $\operatorname{Spec} A_{g_i}$, which are each distinguished in some $\operatorname{Spec} A_i$. This is possible by the previous proposition. As these are all distinguished in $\operatorname{Spec} A_i$ which as P, by (1) each $\operatorname{Spec} A_{g_i}$ has P. Then by (2) so too does $\operatorname{Spec} A$. ■

Note:

An **affine local** property is a property which can be checked on any affine cover. Thus a property which satisfies the above two conditions is affine local. Note that if $U \subseteq X$ is an open subscheme, then U inherits any affine local property X has (because affine subschemes of U are affine in X).

Furthermore note that any stalk local property (a scheme has P iff every stalk has property Q) is affine local in the sense that the scheme has property P iff an affine cover has property R (the property R being that the affine scheme has stalks satisfying Q).

Definition 2.5.24

A scheme X is **locally Noetherian** if it can be covered by affine open sets $\operatorname{Spec} A$ where A is Noetherian. And it is a **Noetherian scheme** if it is also quasicompact, or equivalently can be covered by finitely many affine open sets $\operatorname{Spec} A$ where A is Noetherian.

By corollary 2.5.5 a Noetherian scheme has an underlying Noetherian topological space. The converse is in general not true; it is possible for $\operatorname{Spec} A$ to be a Noetherian space without A itself being Noetherian.

A locally Noetherian scheme X is quasiseparated: take a cover by Noetherian affine schemes, then intersection of any two is then Noetherian (any subspace of a Noetherian space is Noetherian) and in particular quasicompact. Thus a Noetherian scheme is qcqs.

A Noetherian space has a finite number of irreducible components by proposition 2.1.16. Because every connected component is the union of irreducible components, a Noetherian space also has finitely many connected components each of which is a finite union of irreducible components.

Lemma 2.5.25

For a scheme X, the minimal prime ideals of the stalk $\mathcal{O}_{X,p}$ are in bijection with the irreducible components of X containing p.

Proof

We know this is true for affine schemes: irreducible subsets of $\operatorname{Spec} A$ are the vanishing sets of prime ideals, and they are maximal precisely when the prime ideal is minimal.

Now if $U \subseteq X$ is open, then there is a bijection between irreducible components of U containing p and irreducible components of X containing p. Indeed if $p \in Z \subseteq U$ is an irreducible component, then $\overline{Z} = \operatorname{cl}_X(Z)$ is irreducible. And if $p \in Y \subseteq X$ is an irreducible component then $Y \cap U \subseteq U$ is irreducible (open subsets of irreducible spaces are irreducible).

Note that $Y = \overline{Y \cap U} \cup (Y - U)$ are open; since Y is irreducible one must be Y. Since $p \notin Y - U$, we have that $\overline{Y \cap U} = Y$. And $\overline{Z} \cap U = \operatorname{cl}_U(Z) = Z$ because Z is closed; so these are mutual inverses between closed irreducible subsets of U and of X containing p. These also preserve order, so they preserve maximal elements, i.e. irreducible components.

Thus for $p \in \operatorname{Spec} A \subseteq X$, there is a bijection between irreducible components of X containing p and irreducible components of $\operatorname{Spec} A$ containing p. The latter is in bijection with minimal prime ideals of $\mathcal{O}_{X,p}$. ■

Proposition 2.5.26

A nonempty Noetherian scheme X is integral iff X is connected and all of its stalks are integral.

Proof

In general an integral scheme X is irreducible and thus connected. Furthermore its stalks are all domains, as the direct colimit of domains is a domain: if $fg = 0$ then $f|_U g|_U = 0$ in some neighborhood U, but then $\mathcal{O}_X(U)$ is a domain so $f|_U$ or $g|_U$ is zero in $\mathcal{O}_X(U)$, so f_p or g_p is zero in $\mathcal{O}_{X,p}$.

Conversely suppose that X is connected and all of its stalks are integral. Then X is reduced, so by proposition 2.5.19 we just need to show that X is irreducible. Now suppose that X has two distinct irreducible components Z, Z'. If $p \in Z \cap Z'$ then $\mathcal{O}_{X,p}$ is not a domain, as by the previous lemma $\mathcal{O}_{X,p}$ has multiple minimal prime ideals. A domain has a unique minimal prime ideal, 0.

Thus X is the disjoint union of its irreducible components. There are finitely many of these, and as they are all closed, they must then be open. But a connected space has no non-trivial clopen subsets, so X has a unique irreducible component and is therefore irreducible. ■

Definition 2.5.27

Let A be a ring. An **A-scheme** is a scheme X whose sections are all A-algebras and its restriction maps are A-algebra morphisms. That is, it is a $(\mathrm{Alg}_A, \mathrm{LocAlg}_A)$-space (where LocAlg_A obviously means local A-algebras).

If X is an A-scheme which can be covered by affine open subschemes $\operatorname{Spec} B_i$ where each B_i is a finitely generated A-algebra, then X is **locally of finite type** over A, or a **locally finite type A-scheme**. If X is furthermore quasicompact, then X is of **finite type** over A, or a **finite type A-scheme**.

Recall that if A is Noetherian and B is a finitely generated A-algebra, then B is Noetherian as well (by Hilbert's basis theorem). Thus if X is a locally finite type A-scheme, then it is locally Noetherian. Similarly a finite type A-scheme is Noetherian.

For example $\mathbb{P}^n_A$ is a finite type A-scheme.

Definition 2.5.28

An affine scheme which is reduced and of finite type over a field k is said to be an **affine variety** over k. A reduced (quasi)projective k-scheme is a **(quasi)projective variety** over k.

Proposition 2.5.29

Suppose A is a ring and $(f_1, \dots, f_n) = A$.

(1) if A is Noetherian, then so is A_{f_i}. Conversely if each A_{f_i} is Noetherian, so is A.

(2) suppose B is a ring and A is a B-algebra. Then A is a finitely generated B-algebra iff each A_{f_i} is.

Proof

For (1) we first note that the localization of a Noetherian ring is Noetherian. Recall that ideals of $S^{-1}A$ are all of the form $S^{-1}I$ for an ideal $I \leq A$. Now if we have a collection Σ of ideals of $S^{-1}A$, we can define $\Sigma' = \left\{ I \leq A \;\middle|\; S^{-1}I \in \Sigma \right\}$. Because A is Noetherian, this has a maximum element I, and then $S^{-1}I \in \Sigma$ is a maximum element.

Conversely if $I_1 \subseteq I_2 \subseteq \cdots$ is an increasing chain of ideals in A, then let $I_{i,j} = (I_j)_{f_i}$. Thus $I_{i,1} \subseteq I_{i,2} \subseteq \cdots$ is increasing in A_{f_i} and thus must stabilize. Now suppose that the $I_{i,j}$ all stabilize for $j \geq N$. Define $M = I_{j+1}/I_j$ for $j \geq N$, then M embeds in

$$M \hookrightarrow \prod_i M_{f_i} \cong \prod_i I_{i,j+1}/I_{i,j} \cong 0$$

so that $I_j = I_{j+1}$ as desired.

For (2) note that A_{f_i} is a finitely generated A algebra (by $1/f_i$), so if A is finitely generated then so is A_{f_i}. Conversely if $r_{ij}/f_i^{k_j}$ generate A_{f_i} for $r_{ij} \in A$, and $1 = \sum_i c_i f_i$, then $\{f_i, c_i, r_{ij}\}$ generate A. ■

Combining (1) above with Affine communication lemma we get the following immediate corollary.

Corollary 2.5.30

A scheme X is locally Noetherian iff for every affine subscheme $\operatorname{Spec} A \subseteq X$, A is Noetherian.

In particular, an affine scheme $\operatorname{Spec} A$ is a Noetherian scheme iff A is Noetherian.

Remark 2.5.31

Recall that $\operatorname{Spec} A$ and $\operatorname{Spec} A/\operatorname{nil}(A)$ are homeomorphic as topological spaces.

Let $A = k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$. Then $\operatorname{nil}(A) = (x_1, x_2, \dots)$ and $A/\operatorname{nil}(A) \cong k$ so that $\operatorname{Spec} A \cong \operatorname{Spec} k$ is a one-point space and thus Noetherian. But $\operatorname{nil}(A)$ is not a finitely generated ideal, so that A is not a Noetherian ring.

Thus $\operatorname{Spec} A$ being a Noetherian space does not mean it is a Noetherian ring. In particular a scheme which has an underlying Noetherian space is not necessarily a Noetherian scheme.

2.5.4 Normality and factorality

Recall that if B is an A-algebra, then an element $b \in B$ is **integral** over A if it is the root of some *monic* polynomial over A. It is a standard result that the set of all integral elements of B forms an A-subalgebra. We say that A is **integrally closed** in B if this subalgebra is A (more precisely, A's image in B). That is, the only integral elements of B are those which are already in A.

An integral domain A is **normal** (or **integrally closed**) if it is integrally closed in its field of fractions $A \subseteq K(A)$.

Definition 2.5.32

A scheme X is **normal** if its stalks are all normal. That is, $\mathcal{O}_{X,p}$ is an integral domain and integrally closed in its fraction field.

Thus normal schemes are reduced (but not necessarily integral, as that is not a stalk-local property).

It is not hard to see that A is normal iff $A_\mathfrak{p}$ is for all prime $\mathfrak{p}$. Indeed $K(A_\mathfrak{p}) \cong K(A)$, and the intersection of the $A_\mathfrak{p}$s in $K(A)$ is equal to A. In fact the intersection of the $A_\mathfrak{m}$s in $K(A)$ is equal to A. Indeed, if $a/b \in K(A)$ is in $A_\mathfrak{m}$ then $b \notin \mathfrak{m}$, thus

$$\frac{a}{b} \in \bigcap_\mathfrak{m} A_\mathfrak{m} \iff b \notin \bigcup_\mathfrak{m} \mathfrak{m} \iff b \in A^\times,$$

in which case $a/b \in A$. So if $f \in K(A)$ is integral over A, it is integral over each $A_\mathfrak{p}$, thus in each $A_\mathfrak{p}$ and so in A. Conversely the localization of a normal domain is normal.

Because the stalks of $\operatorname{Spec} A$ are $A_\mathfrak{p}$, $\operatorname{Spec} A$ is a normal scheme iff A is normal.

Remark 2.5.33

Recall that stalk-local properties are affine-local. Thus the following are equivalent:

(1) X is normal,

(2) for every affine subscheme $\operatorname{Spec} A \subseteq X$, A is normal,

(3) there exists an affine cover $X = \bigcup_i \operatorname{Spec} A_i$ such that A_i is normal for all i.

Proposition 2.5.34

A scheme X is normal and Noetherian iff it is the finite disjoint union of integral Noetherian normal schemes.

Proof

Suppose X is a normal Noetherian scheme. Then let $X = \bigsqcup_{i=1}^n X_i$ be its connected components. These are closed subsets of X, thus quasicompact, and because X is locally Noetherian so too are the X_i. Thus each X_i is a connected Noetherian scheme. Furthermore the stalks of X_i are stalks of X, which are normal. Thus by proposition 2.5.26, each X_i is integral.

Thus $X = \bigsqcup_{i=1}^n X_i$ is a finite disjoint union of integral Noetherian normal schemes, as desired.

Conversely if $X = \bigsqcup_{i=1}^n X_i$ is a finite disjoint union of integral Noetherian normal schemes, then X is locally Noetherian by covering each X_i with a Noetherian affine cover. Because the X_i are all Noetherian, so too is X; thus X is a Noetherian scheme. The stalks of X are stalks of X_i which are all normal, so X is a normal Noetherian scheme. ■

Recall that an integral domain A is a **unique factorization domain** (**UFD**) if every nonzero element $a \in A$ can be written uniquely as a product of irreducible elements $a = p_1 \cdots p_n$. Recall that an irreducible element $p \in A$ is an element that cannot be written as a nontrivial product: if $p = ab$, then one of a, b must be a unit.

Definition 2.5.35

A scheme X is **factorial** if all of its stalks are UFDs.

A UFD is also normal. The proof follows essentially the same as the classical one for $\mathbb{Z}$. If $p/q \in K(A)$ is integral over A, then suppose it is the root of $x^n + a_{n-1}x^{n-1} + \cdots + a_0$. We may assume, as A is a UFD, that

p, q are coprime. Plugging in $x = p/q$ and multiplying by q^n gives us

$$p^n + a_{n-1}p^{n-1}q + \cdots + a_0 q^n = 0$$

which implies that q divides p, which means that q is a unit (as p, q are coprime). Thus $p/q \in A$ as desired. Thus factorial schemes are also normal.

Lemma 2.5.36

If A is a UFD then so too are all of its localizations $S^{-1}A$.

Proof

Firstly, $S^{-1}A$ is an integral domain.

If $p \in A$ is irreducible, then it is irreducible in $S^{-1}A$. Indeed, if $p = (ab)/s$ then $ps = ab$ and so p divides ab. Since p is irreducible and A is a UFD, it is prime, so suppose it then divides a. Then it divides a/s, so p is prime and thus irreducible in $S^{-1}A$.

Conversely if $p/s \in S^{-1}A$ is irreducible, then because $1/s$ is a unit, $p \in S^{-1}A$ is irreducible. But since A embeds in $S^{-1}A$, this means that $p \in A$ is irreducible. Thus, up to multiplication by a unit, the irreducible elements of $S^{-1}A$ are those of A.

Then any $a \in A$ has a unique factorization in $S^{-1}A$ by irreducibles, and thus $a/s \in S^{-1}A$ does as well. ■

In particular if A is a UFD, $\operatorname{Spec} A$ is factorial.

Note:

The converse is not true, though. Even if $\operatorname{Spec} A$ is factorial, A need not be a UFD.

Remark 2.5.37

UFDs can arise from some common constructions. Firstly, every PID is a UFD.

Indeed if A is a PID and $a \in A$ is nonzero, then if a is not irreducible we have $a = a_{1,1}a_{1,2}$ for non-units. If these are not irreducible, we can continue with $a = a_{2,1}a_{2,2}a_{2,3}a_{2,4}$. And so on, we get an ascending chain of ideals

$$(a) \subsetneq (a_{1,1}, a_{1,2}) \subsetneq \cdots \subsetneq (a_{i,1}, \ldots, a_{i,2^i}) \subsetneq \cdots.$$

But A is Noetherian, a contradiction.

And in a PID, irreducible elements are prime. Indeed, if $p \in A$ is irreducible then $(p) \leq A$ is maximal. Since if $(p) \subseteq (a)$ then $p = ab$ so either a is a unit or b is; so $(a) = A$ or $(p) = (a)$. Thus (p) is prime which means that if p divides ab, p divides either a or b, meaning p is prime.

Thus if $p_1 \cdots p_n = q_1 \cdots q_m$, then p_1 divides one of the q_is, which means they are equal (up to unit), and so on. Thus A is a UFD.

Thus if k is a field, $k[x]$ is a PID and thus a UFD. In more generality it is true that if A is a UFD then $A[x]$ is a UFD.

Thus for example $\mathbb{A}^n_k, \mathbb{P}^n_k$ are factorial, as $\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \ldots, x_n]$ where the polynomial algebra is a UFD. And $\mathbb{P}^n_k$ can be covered by $\mathbb{A}^{n+1}_k$s. For the same reason, we can replace k with any UFD A.

2.6 Quasicoherent sheaves

2.6.1 The category of quasicoherent sheaves

Recall that for a ringed space $\mathcal{O}_X$ we have an Abelian category $\operatorname{Mod}_{\mathcal{O}_X}$ of $\mathcal{O}_X$-modules. Unfortunately this category is often unwieldy, especially when X is a scheme as we would like our modules to also locally look nice. Recall the embedding $\operatorname{Mod}_A \hookrightarrow \operatorname{Mod}_{\operatorname{Spec} A}$ which maps $M \mapsto \widetilde{M}$, whose sections on the distinguished open sets are

$$\widetilde{M}(\mathrm{D}\, f) = M_f.$$

Definition 2.6.1

If X is a scheme, then an $\mathcal{O}_X$-module $\mathcal{F}$ is called a **quasicoherent sheaf** if for every affine $\operatorname{Spec} A \subseteq X$, $\mathcal{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ for some A-module. The full subcategory of $\operatorname{Mod}_{\mathcal{O}_X}$ consisting of quasicoherent sheaves is denoted QCoh_X.

Theorem 2.6.2

To check that an $\mathcal{O}_X$-module $\mathcal{F}$ is quasicoherent, it is sufficient to check that $\mathcal{F}$ restricts to an A-module on any affine cover of X.

Proof

We will show that the property $\mathcal{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ satisfies the conditions of Affine communication lemma. The first condition is easier, as if $\mathcal{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$, then $\mathcal{F}|_{\operatorname{Spec} A_f} \cong \widetilde{M}_f$.

Now suppose $A = (f_1, \dots, f_n)$, $\mathcal{F}$ is an $\mathcal{O}_{\operatorname{Spec} A}$-module such that $\mathcal{F}|_{\mathrm{D}\, f_i} \cong \widetilde{M}_i$ where M_i are A_{f_i}-modules. We want to show that there is an A-module M such that $\mathcal{F} \cong \widetilde{M}$.

Now consider the isomorphisms

$$\begin{array}{ccc} & \mathcal{F}|_{\mathrm{D}\, f_i \cap \mathrm{D}\, f_j} & \\ \swarrow & & \searrow \\ \widetilde{M}_i|_{\mathrm{D}\, f_i \cap \mathrm{D}\, f_j} & & \widetilde{M}_j|_{\mathrm{D}\, f_i \cap \mathrm{D}\, f_j} \end{array}$$

By taking sections over $\mathrm{D}\, f_i \cap \mathrm{D}\, f_j$, this yields an isomorphism $\varphi_{ij} \colon (M_i)_{f_j} \to (M_j)_{f_i}$ of $A_{f_i f_j}$-modules. Furthermore these φ_{ij} satisfy the cocycle conditions, as is easily verified. Let $M_{ij} = (M_i)_{f_j}$ which we identify with $M_{ji} = (M_j)_{f_i}$ via φ_{ij}.

Now define M to be such that the following sequence is exact (i.e. $M = \ker \gamma$):

$$0 \to M \to \prod_{i=1}^{n} M_i \xrightarrow{\gamma} \prod_{i<j} M_{ij}, \qquad \gamma(m_i)_i = (m_i - m_j)_{i<j}.$$

Now we claim that $\widetilde{M}_{\mathrm{D}\, f_i} \cong \widetilde{M}_i$, i.e. $M_{f_i} \cong M_i$. Furthermore we claim that these isomorphisms fit into a commutative triangle

$$\begin{array}{ccc} & M_{f_i f_j} & \\ \swarrow & & \searrow \\ M_{ij} & \xrightarrow{\varphi_{ij}} & M_{ji} \end{array}$$

Let us assume that $i = 1$, then because localization is exact, we have that $M_{f_1} = \ker \gamma_{f_1}$. On the other hand the following sequence

$$0 \to M_1 \xrightarrow{\beta} \prod_{i=1}^{n} (M_i)_{f_1} \xrightarrow{\gamma_{f_1}} \prod_{i<j} (M_{ij})_{f_1}, \qquad \beta(m) = (m_i)_i$$

is exact. This is because we can rewrite it as

$$0 \to M_1 \to \prod_{i=1}^{n} (M_1)_{f_i} \to \prod_{i<j} (M_i)_{f_i f_j},$$

which is exact. This gives us an isomorphism $M_1 \cong M_{f_1}$ as desired.

Now it is straightforward to prove that these isomorphisms commute with φ_{ij}. Thus $\widetilde{M}$ is isomorphic to the gluing of the $\widetilde{M}_i$, but this gluing is $\mathcal{F}$, as desired. ■

Example 2.6.3

Let $X = \operatorname{Spec} k[t]$, and let $\mathcal{F}$ be the skyscraper sheaf supported at $[(t)]$ with group $k(t)$ and the usual $k[t]$-module structure. Because $k(t)$ is the field of fractions of $k[t]$, it has a canonical $k[t]_f$-module structure for all $f \in k[t]$. Thus $\mathcal{F}$ is an $\mathcal{O}_X$-module.

But it is not quasicoherent: if it were then $\mathcal{F}$ would be isomorphic to $\Gamma(\widetilde{X, \mathcal{F}}) = \widetilde{k(t)}$. But this is not the case as $\mathcal{F}(\mathrm{D}\, t) = 0$ (since $(t) \notin \mathrm{D}\, t$), while $k(t)_t \cong k(t) \neq 0$.

In general if X is an integral scheme with function field $K(X)$, then the skyscraper sheaf $p_* K(X)$ is a non-quasicoherent $\mathcal{O}_X$-scheme unless $p = \eta$ is generic. We can restrict this to an affine subscheme containing p. Then the above argument works as well, as there is an $f \in A$ such that $p \notin \mathrm{D}\, f$ and so $p_* K(X)(\mathrm{D}\, f) = 0$ while $\widetilde{K(X)}(\mathrm{D}\, f) = K(X)_f \cong K(X) \neq 0$. Meanwhile if $p = \eta$ is generic then $\eta_* K(X)(\mathrm{D}\, f) = K(X)$ for all $f \in A$ because η lies in all of the open subsets of X.

Recall that an A-module M is **torsion-free** if $am = 0$ implies that either a is a zerodivisor or $m = 0$. If A is an integral domain this is equivalent to $a = 0$ or $m = 0$. An element $m \in M$ for which there exists a non-zerodivisor $a \in A$ with $am = 0$ are called **torsion elements**. If M is an A-module, define M_{tor} to be the set of all torsion elements in M. This is a submodule of M: if $am, bn = 0$ for non-zerodivisors $a, b \in A$ then $ab \in A$ is also a non-zerodivisor and $(ab)(m + n) = 0$ so that $m + n \in M_{\mathsf{tor}}$. Clearly M_{tor} is also closed under scalar multiplication.

Note:

In the case of A being an integral domain, the definition of the torsion submodule becomes

$$M_{\mathsf{tor}} = \{m \in M \mid am = 0 \text{ for some nonzero } a \in A\}.$$

Sometimes this is taken as the general definition of the torsion subset, and this is not in general a submodule of M as it is not necessarily closed under addition.

For example let $A = \mathbb{Z}/\mathbb{Z}_6$. Then $2 \cdot 3 = 0$ so they are considered torsion (in this new definition). But $2 + 3 = 5 = -1$, which is not torsion.

We will basically only care about torsion when A is an integral domain, so we don't need to worry.

A module is **torsion** if all of its elements are torsion, i.e. $M = M_{\mathsf{tor}}$. Over an integral domain this is equivalent to $M \otimes_A K(A) = 0$. Indeed if M is torsion, let $m \in M$ then there is a nonzero $a \in A$ such that $am = 0$. Then for all nonzero $b \in A$,

$$\frac{m}{b} = \frac{am}{ab} = 0$$

so that $M = 0$ (recall that $M \otimes_A K(A) \cong M_{(0)}$). Conversely if $m \in M$ is torsion-free then $m/1 \in M \otimes_A K(A)$ is nonzero, as it is zero only when there is a nonzero $a \in A$ for which $am = 0$.

Definition 2.6.4

If X is a scheme, an $\mathcal{O}_X$-module is **torsion-free** if $\mathcal{F}_p$ is a torsion-free $\mathcal{O}_{X,p}$-module for all $p \in X$.

If X is reduced, then a quasicoherent sheaf $\mathcal{F}$ is **torsion** if its stalk at the generic point of every irreducible component is zero.

If X is integral then this is the same as saying $\mathcal{F}_p$ is a torsion $\mathcal{O}_{X,p}$-module for all $p \in X$. Indeed if this is true then $\mathcal{F}_\eta$ is a torsion $\mathcal{O}_{X,\eta} = K(X)$-module, and since $K(X)$ is a field its only torsion modules is the zero module. Conversely $\mathcal{O}_{X,\eta}$ is the field of fractions of $\mathcal{O}_{X,p}$ for all $p \in X$, and so $\mathcal{F}_\eta$ is the localization of $\mathcal{F}_p$, i.e. $\mathcal{F}_\eta = \mathcal{F}_p \otimes K(X)$. If this is zero then $\mathcal{F}_p$ is torsion.

Definition 2.6.5

If X is a scheme, then its **distinguished affine base** is the subcategory of $\operatorname{open}(X)$ whose objects are the

distinguished open subsets of its open affine subsets. Then for any $\operatorname{Spec} A \subseteq X$ we add only the morphisms $\mathrm{D}_A(f) \subseteq \mathrm{D}_A(g)$.
In other words this category contains all of X's affine open subsets, and for an inclusion $\operatorname{Spec} B \subseteq \operatorname{Spec} B'$, we add it as a morphism only if $\operatorname{Spec} B = \operatorname{Spec} A_f$ and $\operatorname{Spec} B' = \operatorname{Spec} A_g$ for some $\operatorname{Spec} A \subseteq X$.
Let us denote this category of $\mathrm{open}_{\mathsf{aff}}(X)$.

Note that $\mathrm{open}_{\mathsf{aff}}(X)^{\mathsf{op}}$ is a cofinal subposet of $\mathrm{open}(X)^{\mathsf{op}}$. If for $p \in X$ we denote by $\mathrm{open}_p(X) \subseteq \mathrm{open}(X)$ the set of neighborhoods of p, then $\mathrm{open}_{p,\mathsf{aff}}(X)^{\mathsf{op}}$ is cofinal in $\mathrm{open}(X)^{\mathsf{op}}$.

Remark 2.6.6

Note here that $\mathrm{open}_{\mathsf{aff}}(X)$ is not strictly a subposet of $\mathrm{open}(X)$, as it does not include all of its arrows. Nevertheless, we can still formulate what it means for it to be cofinal.

If $\mathcal{C}, \mathcal{D}$ are two categories, a functor $F\colon \mathcal{C} \to \mathcal{D}$ is **final** if for all $d \in D$, the comma category d/F is nonempty and **connected**: all objects can be connected by a zig-zag of morphisms. It can be shown that this is equivalent to precomposition with F not changing colimits (i.e. the canonical $\operatorname{colim} G \circ F \to \operatorname{colim} G$ is an isomorphism). For posets the term used is **cofinal** as opposed to final, for historic reasons (the dual of a final functor is **initial**).

If $\mathcal{C}, \mathcal{D}$ are posets then d/F is the subcategory of $\mathcal{C}$ consisting of elements $c \in C$ such that $d \leq Fc$. If $\mathcal{C}$ is filtered then if it is non-empty it is immediately connected. And being nonempty just means that $F(\mathcal{C}) \subseteq \mathcal{D}$ is cofinal in the usual sense (every $d \in \mathcal{D}$ is bound by some $c \in \mathcal{C}$).

In our case $\mathrm{open}_{\mathsf{aff}}(X)$ and $\mathrm{open}_{p,\mathsf{aff}}(X)$ are both cofiltered. If $p \in \operatorname{Spec} A \cap \operatorname{Spec} B$ then by proposition 2.5.22, p is contained in a common affine neighborhood of $\operatorname{Spec} A, \operatorname{Spec} B$. And clearly $\mathrm{open}_{p,\mathsf{aff}}(X) \subseteq \mathrm{open}_p(X)$ is cofinal, as the affine neighborhoods of X form a base.

Because cofinal subposets don't affect colimits, for any sheaf $\mathcal{F}$ on X, its stalks can be computed relative to the distinguished affine base:

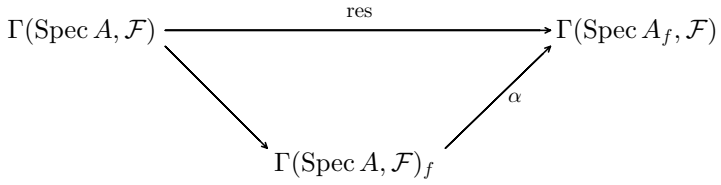
$$\mathcal{F}_p := \operatorname{colim} \mathcal{F}|_{\mathrm{open}_p(X)^{\mathsf{op}}} \cong \operatorname{colim} \mathcal{F}|_{\mathrm{open}_{p,\mathsf{aff}}(X)^{\mathsf{op}}}.$$

Any sheaf on X defines a sheaf on the distinguished affine base. But conversely any sheaf on $\mathrm{open}_{\mathsf{aff}}(X)^{\mathsf{op}}$ defines one on X. The construction is by taking the sheaf $\mathcal{F}^b$ and defining the sheaf $\mathcal{F}$ on an open $U \subseteq X$ by taking the collection of compatible germs:

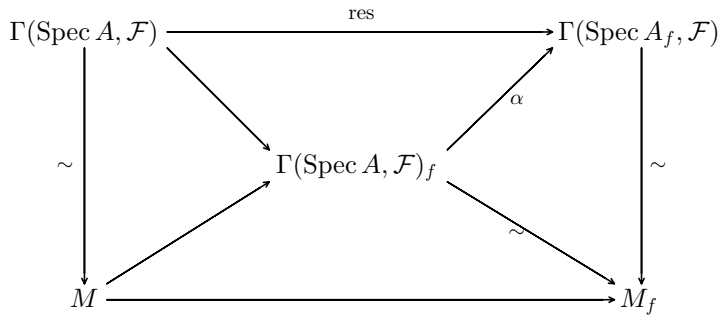
$$\mathcal{F}(U) = \left\{ (f_p \in \mathcal{F}^b_p)_{p \in U} \;\middle|\; \begin{array}{l} \text{for all } p \in U \text{ there is a } U_p \in \mathrm{open}_{p,\mathsf{aff}}(X) \text{ and a } s \in \mathcal{F}^b(U_p) \\ \text{such that } s_q = f_q \text{ for all } q \in U_p \end{array} \right\}.$$

This defines an equivalence between sheaves on X and those defined on the distinguished affine base.

Suppose $\mathcal{F}$ is an $\mathcal{O}_X$-module, then for any distinguished $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$ we have the restriction map $\mathrm{res}\colon \Gamma(\operatorname{Spec} A, \mathcal{F}) \to \Gamma(\operatorname{Spec} A_f, \mathcal{F})$. Because the domain is an A-module and the codomain an A_f-module, by the universal property of localization this factors through $\Gamma(\operatorname{Spec} A, \mathcal{F})_f$:



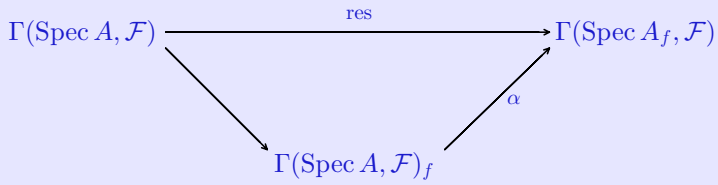
Now $\mathcal{F}$ is quasicoherent iff this α is an isomorphism. Indeed if $\mathcal{F}$ is quasicoherent, say $\mathcal{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$, then this diagram fits into a larger diagram



Showing that α is an isomorphism. On the other hand if α is an isomorphism then it defines an isomorphism $\Gamma(\operatorname{Spec} A_f, \mathcal{F}) \to M_f$ making the rightmost triangle commute. Thus the whole diagram above commutes, showing we have an isomorphism $\mathcal{F} \to \widetilde{M}$. We summarize this in the following lemma.

Lemma 2.6.7

An $\mathcal{O}_X$-module $\mathcal{F}$ is quasicoherent iff for every distinguished $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A \subseteq X$ the canonical map α below is an isomorphism:



Definition 2.6.8

Let $(X, \mathcal{O}_X)$ be a ringed space. Then we say that $f \in \mathcal{O}_X(U)$ is **locally nilpotent** if for every point $p \in U$ there is a neighborhood $p \in V \subseteq U$ such that $f|_V$ is nilpotent in $\mathcal{O}_X(V)$.

We define X's **sheaf of nilpotents** to be the sheaf $\operatorname{nil}(X)$ whose sections are given by

$$\mathcal{O}_{\operatorname{nil} X}(U) = \{f \in \mathcal{O}_X(U) \mid f \text{ is locally nilpotent}\}.$$

Note that $f \in \mathcal{O}_X(U)$ is locally nilpotent iff for every $p \in U$, f_p is in $\operatorname{nil}(\mathcal{O}_{X,p})$. This is because $f \in \operatorname{nil}(\mathcal{O}_{X,p})$ iff there is a neighborhood $p \in V$ such that $f|_V \in \operatorname{nil}(\mathcal{O}_X(V))$. This clearly shows that $\operatorname{nil}(X)$ is an $\mathcal{O}_X$-module, as $\operatorname{nil}(\mathcal{O}_{X,p})$ is an ideal for all p.

We can define the **presheaf of nilpotents** to be $\mathcal{O}_{\operatorname{nil}_{\mathsf{pre}} X}(U) = \operatorname{nil} \mathcal{O}_X(U)$. This is in general not a sheaf, but we can see that $\operatorname{nil} X$ is the sheafification of $\operatorname{nil}_{\mathsf{pre}} X$. We do this by showing it satisfies the universal property of sheafification. There is an inclusion map $\operatorname{nil}_{\mathsf{pre}} X \hookrightarrow \operatorname{nil} X$, which we will show is the sheafification map.

Suppose $\mathcal{F}$ is a sheaf and $\operatorname{nil}_{\mathsf{pre}} X \to \mathcal{F}$ is a presheaf map. Then there is a unique extension to $\operatorname{nil} X \to \mathcal{F}$: for $f \in \operatorname{nil} X(U)$ there is a covering $U = \bigcup_i U_i$ such that $f|_{U_i}$ is nilpotent. Thus it has an image in $\mathcal{F}(U_i)$ via $\operatorname{nil}_{\mathsf{pre}} X \to \mathcal{F}$. These images agree on the overlaps, so we can glue them together to get f's image. This will give the same result for any covering of U, as if $U = \bigcup_j V_j$ is another covering then gluing the images of $f|_{U_i \cap V_j}$ will give the images of $f|_{U_i}$ and $f|_{V_j}$.

It is easily seen that this is a map of sheaves, and uniqueness is clear.

Note:

$\operatorname{nil}(X)$ is an example of an **ideal sheaf** (of X). This is a subobject of X in $\operatorname{Mod}_{\mathcal{O}_X}$. Because $\operatorname{Mod}_{\mathcal{O}_X}$ is complete and its limits are taken pointwise, by an argument identical to that in remark 1.2.3 a morphism of $\mathcal{O}_X$-modules is monic iff each component is. Thus an ideal sheaf of X is just a subsheaf I whose sections

are all ideals of X's sections: $\mathcal{O}_I(U) \leq \mathcal{O}_X(U)$.

Recall that the tensor product of $\mathcal{O}_X$-modules is defined as the sheafification of the presheaf whose sections are $\mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$. We will show that when $\mathcal{F}, \mathcal{G}$ are quasicoherent sheaves on a scheme X, there is no need to sheafify. That is, the presheaf tensor is already a sheaf.

For an affine $\operatorname{Spec} A \subseteq X$ suppose $\Gamma(\operatorname{Spec} A, \mathcal{F}) = M$ and $\Gamma(\operatorname{Spec} A, \mathcal{G}) = N$. Then define $\Gamma(\operatorname{Spec} A, \mathcal{F} \otimes \mathcal{G}) = M \otimes_A N$. And if $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$ is distinguished then the restriction map $\Gamma(\operatorname{Spec} A, \mathcal{F} \otimes \mathcal{G}) \to \Gamma(\operatorname{Spec} A_f, \mathcal{F} \otimes \mathcal{G})$ corresponds to the localization map $M \otimes_A N \to M_f \otimes_{A_f} N_f \cong (M \otimes_A N)_f$. Thus $(\mathcal{F} \otimes \mathcal{G})|_{\operatorname{Spec} A} \cong \widetilde{M \otimes_A N}$, so that $\mathcal{F} \otimes \mathcal{G}$ is a sheaf on all the open affine subsets of X. In particular $\mathcal{F} \otimes \mathcal{G}$ forms a sheaf on the distinguished affine base of X, and thus is itself a sheaf on X.

Note:
Here we used the fact that $M_f \otimes_{A_f} N_f \cong (M \otimes_A N)_f$. In general we have

$$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \cong M \otimes_A S^{-1}A \otimes_{S^{-1}A} S^{-1}N \cong M \otimes_A N \otimes_A S^{-1}A \cong S^{-1}(M \otimes_A N).$$

Definition 2.6.9
If X is a scheme and $f \in \mathcal{O}_X(X)$ is a global section, let $X_f \subseteq X$ be the subset on which f doesn't vanish. This is an open subset and as such an open subscheme of X.

Note that for $f \in A$, $(\operatorname{Spec} A)_f$ and $\operatorname{Spec} A_f = \mathrm{D}\, f$ are equal. Indeed, f doesn't vanish at $\mathfrak{p}$ iff $f_{\mathfrak{p}} \notin \mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$, i.e. iff $f \notin \mathfrak{p}$.

Lemma 2.6.10 (The Qcqs lemma)
Suppose X is a quasicompact quasiseparated scheme, $\mathcal{F}$ a quasicoherent sheaf on X, and $f \in X$. Then the restriction map

$$\operatorname{res}_{X_f \subseteq X} \colon \Gamma(X, \mathcal{F}) \to \Gamma(X_f, \mathcal{F})$$

is precisely the localization by f map.

Proof

Because X is qcqs it is covered by finitely many affine subschemes, the pairwise intersections of which are covered by finitely many affine subschemes. So suppose $X = \bigcup_i U_i$ is a finite affine covering, and $U_i \cap U_j = \bigcup_k U_{ijk}$ is a finite affine covering for all i, j.

The sheaf condition on $\mathcal{F}$ necessitates that the following sequence is exact (the arrows are the standard arrows):

$$0 \to \Gamma(X, \mathcal{F}) \to \bigoplus_i \Gamma(U_i, \mathcal{F}) \to \bigoplus_{i,j,k} \Gamma(U_{ijk}, \mathcal{F}).$$

This is because the direct sums are finite and thus are equivalently products. Localization by f is exact and commutes with direct sums, so the following sequence is also exact:

$$0 \to \Gamma(X, \mathcal{F})_f \to \bigoplus_i \Gamma(U_i, \mathcal{F})_f \to \bigoplus_{i,j,k} \Gamma(U_{ijk}, \mathcal{F})_f.$$

Now for an affine $\operatorname{Spec} A \subseteq X$, $\mathcal{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ for some A-module M. Then

$$\Gamma((\operatorname{Spec} A)_f, \mathcal{F}) = \Gamma(\operatorname{Spec} A_f, \mathcal{F}) = M_f.$$

Thus we have an exact sequence

$$0 \to \Gamma(X, \mathcal{F})_f \to \bigoplus_i \Gamma((U_i)_f, \mathcal{F}) \to \bigoplus_{i,j,k} \Gamma((U_{ijk})_f, \mathcal{F})$$

and the last arrow is the usual one induced from restrictions.

But the kernel of this map is $\Gamma(X_f, \mathcal{F})$ as per the sheaf axiom. Thus we get the desired result. ■

Our goal now is to show that $\mathrm{QCoh}_X \subseteq \mathrm{Mod}_{\mathcal{O}_X}$ is an Abelian category. In general if $\mathcal{B} \subseteq \mathcal{A}$ is a full subcategory of an Abelian category $\mathcal{A}$, then for $\mathcal{B}$ to be an Abelian category itself it is sufficient that

(1) $\mathcal{B}$ contain $\mathcal{A}$'s zero element,

(2) $\mathcal{B}$ be closed under finite sums,

(3) $\mathcal{B}$ be closed under kernels and cokernels.

Indeed, if $\mathcal{B}$ is closed under colimits taken in $\mathcal{A}$, then those are $\mathcal{B}$'s colimits as well. Thus if f is monic in $\mathcal{B}$, then $\ker f = 0$ as the kernel of a monic is zero, and thus $f = \ker \operatorname{coker} f$ in $\mathcal{A}$ and thus in $\mathcal{B}$. Similar for epimorphisms.

Clearly the zero $\mathcal{O}_X$-module is quasicoherent. And (2) follows from the fact that $\widetilde{M} \oplus \widetilde{N} \cong \widetilde{M \oplus N}$, so the sum of quasicoherent sheaves is quasicoherent. For (3) if $\alpha\colon \mathcal{F} \to \mathcal{G}$ is a morphism of quasicoherent sheaves, then let $U \subseteq X$ be affine, and denote by $\beta\colon M \to N$ the induced morphism of A-modules. We then *define* $\ker \alpha(U) = \ker \beta$ and $\operatorname{coker} \alpha(U) = \operatorname{coker} \beta$. Because localization is exact, it commutes with kernels and cokernels, we see that on U, $\ker \alpha \cong \widetilde{\ker \beta}$ and $\operatorname{coker} \alpha \cong \widetilde{\operatorname{coker} \beta}$. This definition forms a sheaf on the distinguished affine base, so $\ker \alpha, \operatorname{coker} \alpha$ are quasicoherent sheaves.

Recall that exactness can be checked stalk-locally. Thus it can be checked on an affine cover, which shows that

$$0 \to \ker \alpha \to \mathcal{F} \xrightarrow{\alpha} \mathcal{G} \to \operatorname{coker} \alpha \to 0$$

is exact, as it resorts to checking on affine subschemes, which just gives

$$0 \to \ker \beta \to M \xrightarrow{\beta} N \to \operatorname{coker} \beta \to 0.$$

So we have proven the following.

Theorem 2.6.11

QCoh_X is an Abelian subcategory of $\mathrm{Mod}_{\mathcal{O}_X}$.

For the last part of this section, we will show an alternative method of defining quasicoherent sheaves, such that the definition works for any ringed space. Because manifolds are also examples of ringed spaces, this allows us to define quasicoherent sheaves on a manifold.

Proposition 2.6.12

Let A be a ring, then there is an adjunction

$$\mathrm{Mod}_A \underset{\Gamma}{\overset{\widetilde{\bullet}}{\rightleftarrows}} \mathrm{Mod}_{\mathcal{O}_{\operatorname{Spec} A}}$$

between $M \mapsto \widetilde{M}$ and global sections $\mathcal{F} \mapsto \Gamma(\operatorname{Spec} A, \mathcal{F})$.

Proof

This just amounts to showing that global sections

$$\hom_{\operatorname{Spec} A}(\widetilde{M}, \mathcal{F}) \xrightarrow{\Gamma} \hom_A(M, \Gamma\mathcal{F})$$

is bijective. This is clear because if $\varphi\colon M \to \Gamma\mathcal{F}$ is a module map then for any $f \in A$ by localization there is a unique φ_f making the following diagram commute

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & \Gamma(\operatorname{Spec} A, \mathcal{F}) \\ \downarrow & & \downarrow \\ M_f & \xrightarrow{\varphi_f} & \Gamma(\operatorname{Spec} A_f, \mathcal{F}) \end{array}$$

It is easy to see that the φ_f assemble into a $\mathcal{O}_{\operatorname{Spec} A}$-module morphism. Existence and uniqueness shows that Γ is a bijection. ■

In particular $M \mapsto \widetilde{M}$ commutes with colimits.

Proposition 2.6.13

Let X be a scheme. An $\mathcal{O}_X$-module is quasicoherent iff there exists an open cover $X = \bigcup_i U_i$ such that for all i, $\mathcal{F}|_{U_i}$ fits into an exact sequence

$$\mathcal{O}_{U_i}^{\oplus I} \to \mathcal{O}_{U_i}^{\oplus J} \to \mathcal{F}|_{U_i} \to 0.$$

Proof

If $\mathcal{F}$ is a quasicoherent sheaf take U_i to be an affine open cover. For an affine $\operatorname{Spec} A \subseteq X$ we have that

$$\mathcal{O}_{\operatorname{Spec} A}^{\oplus I} \cong (\widetilde{A})^{\oplus I} \cong \widetilde{A^{\oplus I}}.$$

Now if $\mathcal{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ then M fits into an exact sequence

$$A^{\oplus I} \to A^{\oplus J} \to M \to 0.$$

Because $\widetilde{\bullet}$ is a left-adjoint and thus right-exact, we have that the following sequence

$$\begin{array}{ccccccc} \widetilde{A^{\oplus I}} & \to & \widetilde{A^{\oplus J}} & \to & \widetilde{M} & \to & 0 \\ \cong & & \cong & & \cong & & \\ \mathcal{O}_{\operatorname{Spec} A}^{\oplus I} & \to & \mathcal{O}_{\operatorname{Spec} A}^{\oplus J} & \to & \mathcal{F}|_{\operatorname{Spec} A} & \to & 0 \end{array}$$

is exact, as desired.

Conversely if such an open cover exists, then we can refine it to be an affine cover. Because restricting $\mathcal{O}_X \mapsto \mathcal{O}_U$ is exact, we can just assume the cover to be affine. Thus we have for every $\operatorname{Spec} A \subseteq X$ in the cover, an exact sequence

$$\widetilde{A^{\oplus I}} \to \widetilde{A^{\oplus J}} \to \mathcal{F}|_{\operatorname{Spec} A} \to 0.$$

Let $M = \operatorname{coker}(A^{\oplus I} \to A^{\oplus J})$, so we have an exact sequence

$$A^{\oplus I} \to A^{\oplus J} \to M \to 0.$$

Applying $\widetilde{\bullet}$ then gives us the exact sequence

$$\widetilde{A^{\oplus I}} \to \widetilde{A^{\oplus J}} \to \widetilde{M} \to 0,$$

and the first arrow is the same as the previous one because $\widetilde{\bullet}$ and global sections are inverses from A-modules. Thus we must have that $\mathcal{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$, meaning $\mathcal{F}$ is quasicoherent as desired. ■

Thus for a general ringed space $(X, \mathcal{O}_X)$ we can take the condition in the above proposition as the definition of

quasicoherence.

In fact we can do a bit better. Let $(X, \mathcal{O}_X)$ be a ringed space with a ring of global sections $A = \mathcal{O}_X(X)$. Then for a module $M \in \mathrm{Mod}_A$ we can define its **associated presheaf** to be $\widetilde{M}_{\mathsf{pre}} \in \mathrm{Mod}_{\mathcal{O}_X}^{\mathsf{pre}}$ by

$$\widetilde{M}_{\mathsf{pre}}(U) = \mathcal{O}_X(U) \otimes_{\mathcal{O}_X(X)} M, \qquad U \subseteq X \text{ open}.$$

Its **associated sheaf** is then the sheafification $\widetilde{M} = \widetilde{M}_{\mathsf{pre}}^{\mathsf{sh}}$. These constructions are clearly functorial.

A first remark is that when $X = \operatorname{Spec} A$ then the associated presheaf of M is the associated sheaf because

$$\widetilde{M}(\mathrm{D}\, f) = M_f = A_f \otimes M = \mathcal{O}_{\operatorname{Spec} A}(\mathrm{D}\, f) \otimes_A M.$$

And in general the associated presheaf, and thus sheaf, of $A = \mathcal{O}_X(X)$ is X:

$$\widetilde{A}(U) = \mathcal{O}_X(U) \otimes_A A = \mathcal{O}_X(U).$$

Now an open subsheaf of an associated sheaf is itself an associated sheaf. More precisely if $\widetilde{M}$ is an associated sheaf of $\mathcal{O}_X$, then for $U \subseteq X$ open $\widetilde{M}|_U = \widetilde{\mathcal{O}_X(U) \otimes_{\mathcal{O}_X(X)} M}$. This is clearly true for the associated presheaf. Because restriction is left adjoint to inc_* for inclusion $U \hookrightarrow X$, both for presheaves and sheaves, we see that sheafification commutes with restriction. Indeed, sheafification is left adjoint to inclusion $\mathrm{Sh}(X) \hookrightarrow \mathrm{Psh}(X)$, and this commutes with inc_*, so sheafification and restriction does as well. Thus

$$\widetilde{\mathcal{O}_X(U) \otimes_{\mathcal{O}_X(X)} M} = \widetilde{\mathcal{O}_X(U) \otimes M_{\mathsf{pre}}^{\mathsf{sh}}} = \widetilde{M}_{\mathsf{pre}}|_U^{\mathsf{sh}} = \widetilde{M}_{\mathsf{pre}}^{\mathsf{sh}}|_U = \widetilde{M}|_U.$$

Proposition 2.6.14

Let $(X, \mathcal{O}_X)$ be a ringed space and $A = \mathcal{O}_X(X)$. Then there is an adjunction

$$\mathrm{Mod}_A \underset{\Gamma}{\overset{\widetilde{\bullet}}{\rightleftarrows}} \mathrm{Mod}_{\mathcal{O}_X}.$$

Proof

First we show this for presheaves: there is an adjunction

$$\mathrm{Mod}_A \underset{\Gamma}{\overset{\widetilde{\bullet}^{\mathsf{pre}}}{\rightleftarrows}} \mathrm{Mod}_{\mathcal{O}_X}^{\mathsf{pre}}.$$

In this case for an A-module M and an $\mathcal{O}_X$-module $\mathcal{F}$, global sections forms a bijection:

$$\hom_{\mathcal{O}_X}(\widetilde{M}, \mathcal{F}) \xrightarrow{\Gamma} \hom_A(M, \Gamma(X, \mathcal{F})).$$

Indeed recall that for an A-module B, $B \otimes_A \bullet$ is left-adjoint to the forgetful functor $\mathrm{Mod}_B \to \mathrm{Mod}_A$. Thus for any open $U \subseteq X$, there is a unique morphism $\widetilde{M}(U) \to \mathcal{F}(U)$ making the following diagram commute:

$$\begin{array}{ccc} M & \longrightarrow & \Gamma(X, \mathcal{F}) \\ \downarrow & & \downarrow \\ \mathcal{O}_X(U) \otimes M & \xrightarrow{\exists!} & \Gamma(U, \mathcal{F}) \end{array}$$

This demonstrates both uniqueness and existence (verifying this for $U \subseteq V$ instead of just $V = X$).

So we have a chain of adjunctions

$$\mathrm{Mod}_A \underset{\Gamma}{\overset{\widetilde{\bullet}^{\mathsf{pre}}}{\rightleftarrows}} \mathrm{Mod}_{\mathcal{O}_X}^{\mathsf{pre}} \underset{\mathrm{inc}}{\overset{\mathsf{sh}}{\rightleftarrows}} \mathrm{Mod}_{\mathcal{O}_X},$$

so that $\mathsf{sh} \circ \widetilde{\bullet}^{\mathsf{pre}} = \widetilde{\bullet}$ is left adjoint to $\Gamma \circ \mathrm{inc} = \Gamma$ as desired. ∎

Lemma 2.6.15

An $\mathcal{O}_X$-module $\mathcal{F}$ is associated iff there is an exact sequence

$$\mathcal{O}_X^{\oplus I} \to \mathcal{O}_X^{\oplus J} \to \mathcal{F} \to 0.$$

Proof

If $\mathcal{F} = \widetilde{M}$ is associated, then there are I, J making the following sequence exact:

$$A^{\oplus I} \to A^{\oplus J} \to M \to 0, \qquad A = \mathcal{O}_X(X).$$

Because $\widetilde{\bullet}$ is a left adjoint, it preserves direct sums and is right-exact. Furthermore because $\widetilde{A} = \mathcal{O}_X$, applying $\widetilde{\bullet}$ gives an exact sequence

$$\mathcal{O}_X^{\oplus I} \to \mathcal{O}_X^{\oplus J} \to \widetilde{M} \to 0.$$

Conversely if there is an exact sequence

$$\mathcal{O}_X^{\oplus I} \to \mathcal{O}_X^{\oplus J} \to \mathcal{F} \to 0,$$

then let M fit into the exact sequence

$$A^{\oplus I} \to A^{\oplus J} \to M \to 0.$$

As before applying $\widetilde{\bullet}$ gives the exact sequence

$$\mathcal{O}_X^{\oplus I} \to \mathcal{O}_X^{\oplus J} \to \widetilde{M} \to 0,$$

and thus $\widetilde{F} \cong \widetilde{M}$. ■

Say that an $\mathcal{O}_X$-module is **locally associated** if for every $p \in X$ there is a neighborhood $p \in U$ such that $\mathcal{F}|_U$ is an associated $\mathcal{O}_U$-module. The previous lemma then gives the following corollary immediately.

Corollary 2.6.16

An $\mathcal{O}_X$-module is quasicoherent iff it is locally associated.

Because open subsheaves of an associated sheaf are associated, if X is a scheme we see that it is locally associated iff every point lies in an *affine* neighborhood in which it is associated. Thus this corollary shows that quasicoherence, as defined in general, corresponds to quasicoherence defined over schemes. The proof of this is essentially the same as the one given proposition 2.6.13.

2.6.2 Finiteness of quasicoherent sheaves

For an A-module M, there are three finiteness conditions we care about:

(1) M is **finitely generated**: it has a finite number of generators. Equivalently there is a surjection $A^{\oplus n} \to M$ for a finite n.

(2) M is **finitely presented**: it has a finite number of generators and a finite number of relations. Equivalently there is a **finite presentation**: an exact sequence

$$A^{\oplus m} \to A^{\oplus n} \to M \to 0.$$

(3) M is **coherent**: it is finitely generated, and for any (not necessarily surjective) map $A^{\oplus n} \to M$, its kernel is finitely generated.

Note that being finitely presented is equivalent to there being a surjection $A^{\oplus n} \to M$ whose kernel is finitely generated. Indeed if $A^{\oplus m} \to A^{\oplus n} \to M \to 0$ is exact, then the kernel of $A^{\oplus n} \to M$ is the image of a map from $A^{\oplus m}$, which is finitely generated. Conversely if $A^{\oplus n} \to M$ has finitely generated kernel K, take a surjection $A^{\oplus m} \to K$; the composition $A^{\oplus m} \to A^{\oplus n} \to M \to 0$ is exact.

Thus coherence implies finitely presented. Clearly finitely presented implies finitely generated. We will show that if A is Noetherian, finitely generated implies coherence.

Proposition 2.6.17

If A is Noetherian, all three finiteness conditions above are equivalent.

Proof

Suppose M is a finitely generated A-module, and take a map $\alpha\colon A^{\oplus m} \to M$. Then $\ker\alpha$ is a submodule of $A^{\oplus m}$, which is Noetherian as the finite sum of Noetherian modules, and thus $\ker\alpha$ is Noetherian. In particular $\ker\alpha$ is finitely generated. ■

If A is Noetherian, then the category of finitely generated A-modules is an Abelian subcategory of Mod_A. Indeed, the zero module is finitely generated, and the sum of two finitely generated A-modules is finitely generated. We just need to show that the kernel and cokernel of a map between finitely generated A-modules is finitely generated. The cokernel is a quotient of its codomain and is thus finitely generated. The kernel is a submodule of the domain; since the domain is finitely generated, it is Noetherian so its submodules are finitely generated.

In fact even when A is not Noetherian, the category of coherent A-modules is an Abelian subcategory of Mod_A. We ignore the proof.

Proposition 2.6.18

The category of coherent A-modules is a full Abelian subcategory of Mod_A.

Note that the localization of finitely generated modules are finitely generated: if M is a finitely generated A-module, then $S^{-1}M$ is a finitely generated $S^{-1}A$-module. Conversely if $A = (f_1, \dots, f_n)$ and M_{f_i} is a finitely generated A_{f_i}-module, then M is a finitely generated A-module. Suppose $m_{ij} \in M$ generate M_{f_i}, then all the m_{ij} generate M. Indeed, for $m \in M$ there are $a_{ij} \in A$ and $k_i \geq 0$ such that for all i:

$$m = \frac{\sum_j a_{ij} m_{ij}}{f_i^{k_i}} \iff f_i^{N_i}\left(f_i^{k_i} m - \sum_j a_{ij} m_{ij}\right)$$

for some $N_i \geq 0$. We can assume that all the N_i and k_i are equal, and by updating a_{ij} to be $f_i^{N_i} a_{ij}$ we get that

$$f_i^N m = \sum_j a_{ij} m_{ij}.$$

Because $(f_i)_i$ is the unit ideal, so too is $(f_i^N)_i$ (because its radical is $(f_i)_i$ and $\sqrt{I} = (1)$ iff $I = (1)$). Thus there are $c_i \in A$ such that $\sum_i c_i f_i^N = 1$ and thus

$$m = \sum_i c_i f_i^N m = \sum_{i,j} c_i a_{ij} m_{ij}$$

as desired.

We summarize this as follows.

Proposition 2.6.19

Let M be an A-module.

(1) If M is a finitely generated A-module, then $S^{-1}M$ is a finitely-generated $S^{-1}A$-module.

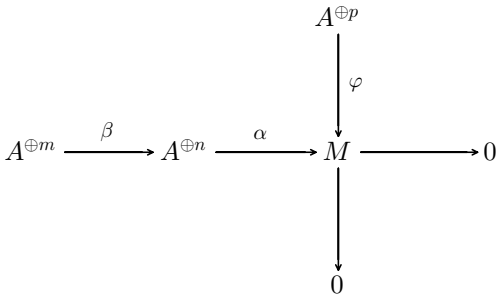
(2) If $A = (f_1, \dots, f_n)$ and M_{f_i} is a finitely-generated A_{f_i}-module for all i, then M is a finitely generated A-module.

Lemma 2.6.20

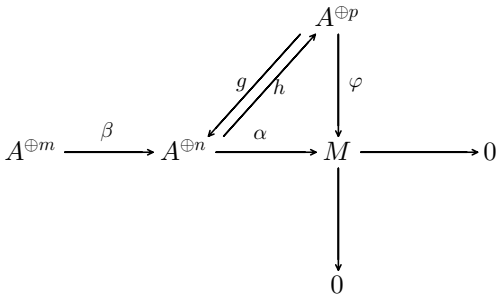
Suppose M is a finitely presented A-module. Then for any surjection $\varphi\colon A^{\oplus p} \to M$, its kernel is finitely generated.

Proof

Choose a finite presentation of M:



Because the rings are free, we can find lifts g, h making the following diagram commute



Explicitly, if $x_1, \dots, x_p$ generate $A^{\oplus p}$ then by surjectivity there are $y_i \in A^{\oplus n}$ such that $\alpha(y_i) = \varphi(x_i)$. So define $g(x_i) = y_i$.

Now consider the following sequence

$$A^{\oplus m} \oplus A^{\oplus k} \xrightarrow{\psi} A^{\oplus k} \xrightarrow{\varphi} M \to 0, \qquad \psi = (h \circ \beta) \oplus (\mathrm{id} - h \circ g).$$

We claim that this sequence is exact, and then we are finished. Firsly,

$$\varphi \circ h \circ \beta = \alpha \circ \beta = 0, \qquad \varphi - \varphi \circ h \circ g = \varphi - \alpha \circ g = \varphi - \varphi = 0$$

so that $\varphi \circ \psi = 0$. Now if $x \in \ker \varphi$ we want to find a $y \in A^{\oplus m}$ such that

$$x = \psi(y + x) = h(\beta(y)) + x - h(g(x)) \iff h(\beta(y)) = h(g(x)).$$

Now if $\varphi x = 0$ then $\alpha g x = 0$ so $gx \in \ker \alpha = \operatorname{im} \beta$, so there is a $y \in A^{\oplus m}$ such that $\beta y = gx$. Thus $h\beta y = hgx$ as desired. ∎

We already showed that finitely generated modules satisfy the hypothesis of the affine communication lemma.

We now show this for finite presentations and coherence.

Lemma 2.6.21

Let M be an A-module.

(1) If M is finitely presented, then $S^{-1}M$ is finitely presented as an $S^{-1}A$-module.

(2) If $(f_1, \dots, f_n) = A$ and M_{f_i} is a finitely presented A_{f_i}-module for all i, then M is a finitely presented A-module.

Proof

(1) is clear because localization is exact. For (2) we know then that M is finitely generated. Thus there is a surjection $\varphi\colon A^{\oplus n} \to M$, and its localizations $\varphi_{f_i}\colon A_{f_i}^{\oplus n} \to M_{f_i}$ are all surjective as well (as localization is exact). By the above lemma, this means that $\ker(\varphi_{f_i})$ is finitely generated. Because $\ker(\varphi_{f_i}) \cong (\ker\varphi)_{f_i}$, this means that $(\ker\varphi)_{f_i}$ is finitely generated for all i, and thus $\ker\varphi$ is. ■

For coherence we note that M is coherent iff it is finitely generated and all of its finitely generated submodules are finitely presented. Indeed if M is coherent and $N \subseteq M$ is a finitely generated submodule, let $A^{\oplus n} \to N$ be a surjection. Then the composition $A^{\oplus n} \to N \hookrightarrow M$ has a finitely generated kernel, so N is finitely presented. Conversely suppose all of M's finitely generated submodules are finitely presented. Let $\varphi\colon A^{\oplus n} \to M$ be any map, then consider its kernel K. Then $A^{\oplus n}/K$ is a submodule of M, thus finitely presented, and so by the above lemma the kernel of $A^{\oplus n} \to A^{\oplus n}/K$, which is K, is finitely generated.

In particular a finitely generated submodule of a coherent module is coherent.

Lemma 2.6.22

Let M be an A-module.

(1) If M is coherent, then $S^{-1}M$ is coherent as an $S^{-1}A$-module.

(2) If $(f_1, \dots, f_n) = A$ and M_{f_i} is a coherent A_{f_i}-module for all i, then M is a coherent A-module.

Proof

For (1) if $N' \subseteq S^{-1}M$ is a finitely generated submodule, then $N' = S^{-1}N$ for a finitely generated submodule $N \subseteq M$ (if x_i/s_i generate N' then take N generated by x_i). Then since M is coherent, N is finitely presented, and so by the previous lemma $N' = S^{-1}N$ is finitely presented as well. Thus $S^{-1}M$ is coherent.

For (2) let $N \subseteq M$ be finitely generated. Then $N_{f_i} \subseteq M_{f_i}$ is finitely generated, thus finitely presented. This is true for all i so by the previous lemma, N is finitely generated. Thus M is coherent. ■

Definition 2.6.23

A quasicoherent sheaf $\mathcal{F}$ is **finite type** (resp. **finitely presented**, **coherent**) if for every affine open $\operatorname{Spec} A \subseteq X$, $\Gamma(\operatorname{Spec} A, \mathcal{F})$ is a finitely generated (resp. finitely presented, coherent) A-module.

Note that coherent quasicoherent sheaves are finitely presented, which in turn are finitely generated. Because these three notions are equivalent over a Noetherian ring, these three notions are equivalent over a locally Noetherian scheme.

Because coherent modules are an Abelian subcategory of A-modules, coherent quasicoherent sheaves form an Abelian subcategory of QCoh_X. This can be seen because our construction of the kernel and cokernel of $\alpha\colon \mathcal{F} \to \mathcal{G}$ is the (co)kernel of $\beta\colon M \to N$, and M, N are coherent so the (co)kernel of β is as well.

Just like how quasicoherence can be generalized to arbitrary ringed spaces, so too can finite type, finitely presented, and coherence.

Definition 2.6.24

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{F}$ an $\mathcal{O}_X$-module.

(1) $\mathcal{F}$ is **finite type** if for every $x \in X$ there is an open $x \in U \subseteq X$ and a surjection

$$\mathcal{O}_U^{\oplus n} \to \mathcal{F}|_U \to 0$$

for some $n \geq 0$.

(2) $\mathcal{F}$ is **finitely presentable** if for every $x \in X$ there is an open $x \in U \subseteq X$ and $n, m \geq 0$ and an exact sequence

$$\mathcal{O}_U^{\oplus m} \to \mathcal{O}_U^{\oplus n} \to \mathcal{F}|_U.$$

(3) $\mathcal{F}$ is **coherent** if $\mathcal{F}$ is finite type and for all open $U \subseteq X$, the kernel of any map

$$\mathcal{O}_U^{\oplus n} \to \mathcal{F}|_U$$

is of finite type.

If $\mathcal{F}$ is quasicoherent these definitions agree with our previous definitions. Indeed the neighborhoods in (1) and (2) may be assumed to be affine, and because global sections and $\widetilde{\bullet}$ form an equivalence on Mod_A we get the desired result. And by taking U to be affine, we see that (3) implies that $\mathcal{F}$ is coherent (as in the previous definition). And because finite type can be taken locally, in (3) it is sufficient to consider only any open cover, so coherence implies (3).

2.6.3 Simple objects and length

For the remainder of this section $\mathcal{C}$ is an Abelian category.

Definition 2.6.25

An object $M \in \mathcal{C}$ is **simple** if its only subobjects are 0 and M itself. A **composition series** for an arbitrary object $M \in \mathcal{C}$ is a finite filtration

$$0 = X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_{n-1} \subsetneq X_n = M$$

such that the quotients X_{i+1}/X_i are all simple. The **length** of this composition series is n. An object with a composition series is said to have **finite length**.

Theorem 2.6.26 (Jordan-Hölder)

If M is of finite length, then all of its composition series have the same length. Furthermore, up to permutation, the quotients of any composition series of M are the same.

Proof

Suppose $0 = X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n = M$ and $0 = Y_0 \subsetneq Y_1 \subsetneq \cdots \subsetneq Y_p = M$ are two composition series. Define $Z_{ij} = X_i \cap Y_j \subseteq M$ (this is the pullback of $X_i \to M \leftarrow Y_j$). For reasons of abstract nonsense, the following are true:

(1) if $i \leq i'$ and $j \leq j'$, $Z_{i,j} \subseteq Z_{i',j'}$;

(2) $Z_{n,j} = Y_j$ and $Z_{i,p} = X_i$;

(3) $Z_{0,j} = Z_{i,0} = 0$;

(4) $Z_{i,j} \cap Z_{i',j'} = Z_{\min(i,i'),\min(j,j')}$.

The core of (1) and (4) is that $A \to X \times_M Y$ is monic if the composition $A \to X \times_M Y \to X$ is.

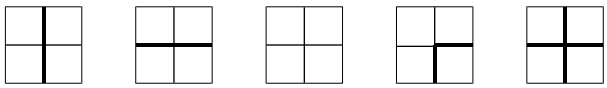
Now notice that the kernel of $Z_{ij} \to Z_{i+1,j}/Z_{i+1,j-1}$ is $Z_{ij} \cap Z_{i+1,j-1} = Z_{i,j-1}$. Thus we have that $Z_{ij}/Z_{i,j-1} \subseteq Z_{i+1,j}/Z_{i+1,j-1}$. Inductively then we have that $Z_{ij}/Z_{i,j-1}$ is either 0 or Y_j/Y_{j-1}. For $i = n$ this is clear as $Z_{nj} = Y_j$, and then inductively we have that $Z_{ij}/Z_{i,j-1} \subseteq Z_{i+1,j}/Z_{i+1,j-1} = Y_j/Y_{j-1}$. This is simple, so we have the desired result. Similarly $Z_{ij}/Z_{i-1,j}$ is either 0 or X_i/X_{i-1}.

Now define a matrix whose entries are Z_{ij}. Then between adjacent cells, draw a line if the quotient is nonzero. So for example if $Z_{ij}/Z_{i-1,j}$ is nonzero then draw a line between the $(i-1,j)$th and (i,j)th cell.

Take any 2×2 submatrix

$$\begin{matrix} Z_{ij} & Z_{i,j+1} \\ Z_{i+1,j} & Z_{i+1,j+1} \end{matrix} = \begin{matrix} A & B \\ C & D \end{matrix}.$$

We claim that the drawn lines are of one of the following five forms:



We know that $A = B \cap C$, so if $B = D$ then $A = C$. And if $C = D$ then $A = B$; this takes care of the first three cases. Otherwise suppose D/B and D/C are both nonzero. If $B = C$ then $A = B = C$ so that we are in the fourth case.

Otherwise we will see that $D/B \cong C/A$ and similarly $D/C \cong B/A$ so that we are in the last case. Indeed consider the map $C \to D/B$ which has kernel $C \cap B = A$, so we have an injection $C/A \hookrightarrow D/B$. Because D/B is simple, either C/A is zero or this is an isomorphism; but if $A = C$ then $C \subseteq B$ so we have a map $D/C \to D/B$. Because D/C is simple, this map is either the zero map or an injection (considering the kernel). If if it is an injection it is an isomorphism and thus $B = C$, a contradiction. But if it is the zero map this means that one of D/C or D/B is zero, contradicting that they are nonzero. Thus $C/A \cong D/B$ and similarly $B/A \cong D/C$, so we are in the fifth case.

Furthermore note that in all of the cases except the last, the bold lines are all isomorphic. In the the first case we have $D = B$ and $C = A$ so $B/A = D/C$. Similarly for the second. In the fourth case, $B = C = A$ so $D/B = D/C$. In the fifth case we saw that $D/B = C/A$ and $D/C = B/A$, so the straight lines are isomorphic (i.e. the two horizontal lines are isomorphic to one another, similar for vertical).

Note that on the rightmost and bottomost edge of the matrix, all of the horizontal lines are colored: $Z_{i,p}/Z_{i-1,p}$ and $Z_{n,j}/Z_{n,j-1}$ are isomorphic to the nonzero X_i/X_{i-1} and Y_j/Y_{j-1} respectively. Thus for any Y_j/Y_{j-1} we can follow the bold line all the way to the left and then make one turn downwards. This brings us to a X_i/X_{i-1}, and because the bold lines are isomorphic, we have that $Y_j/Y_{j-1} \cong X_i/X_{i-1}$. We can do this for X_i/X_{i-1} as well, and these processes are inverses. ■

Definition 2.6.27

Let $M \in \mathcal{C}$ be of finite length. Its **length** is the length of any of its composition series, and is denoted by $\ell(M)$. If M is not of finite length then define $\ell(M) = \infty$.

This is well-defined in lieu of Jordan-Hölder.

Note that subobjects of A/B are in correspondence with subobjects of A containing B. Indeed if $B \subseteq M \subseteq A$ then the kernel of $M \to A/B$ is $M \cap B = B$ so this gives an injection $M/B \hookrightarrow A/B$. Conversely if $N \subseteq A/B$ is a subobject, consider the pullback $M = N \times_{A/B} A$. Because $B \hookrightarrow A$, we have that $B \hookrightarrow M$ ($B \hookrightarrow A$ commutes with the zero map $B \to N$). These are mutual inverses.

Now if M has finite length, then so too do all of its subquotients. That is if we have a filtration $0 \subseteq M' \subseteq M'' \subseteq M$, then M''/M' has finite length. Let $M_\bullet$ be a composition series for M, then create a matrix as in the Jordan-Hölder theorem proof, whose cells are the intersections of $M_\bullet$ with $0 \subseteq M' \subseteq M'' \subseteq M$.

The horizontal lines in the matrix are all either zero or simple. This is because the kernel of $M_{i+1} \cap M' \hookrightarrow M_{i+1}/M_i$ is $M_i \cap M'$ so $(M_{i+1} \cap M')/(M_i \cap M')$ is a subobject of M_{i+1}/M_i. Meanwhile the vertical lines are subobjects of

M''/M' (or M/M'', M'): the kernel of $M_i \cap M'' \to M''/M'$ is $M_i \cap M'$ so $(M_i \cap M'')/(M_i \cap M')$ is a subobject of M''/M'.

The possible submatrices here are the same as in the proof above, except we have one additional option:



This is because if $D/B, D/C$ are nonzero and $B \neq C$, then we have an injection $C \to D/B$. As before this has kernel A so $C/A \subseteq D/B$. Here it may be that $A = C$ so $C \subseteq B$.

If we take a horizontal line M_i/M_{i-1} and follow the path which goes as left as possible and then down, we will either land on the line M''/M' or not. If it does land on M''/M' then keep track of the first vertical line in the path. This is a subobject of M''/M', and we claim that these form a composition series of M''/M'.

We show two things:

(1) B/A is a subobject of D/C,

(2) $\frac{D/C}{B/A}$ is either zero or simple.

For (1), the kernel of $B \to D/C$ is A so this is immediate. For (2), we split into non-trivial cases: in



we have $D = B$ and $C = A$ so

$$\frac{D/C}{B/A} = \frac{B/C}{B/C} = 0.$$

And in



we have $A = B = C$ so

$$\frac{D/C}{B/A} = \frac{D/B}{A/A} = D/B$$

which is simple as the horizontal lines are simple. And in



we have $A = C$ so

$$\frac{D/C}{B/A} = \frac{D/A}{B/A} = D/B$$

is simple. FInally in



we have $D/B \cong C/A$. Consider then the map $D \to (D/C)/(B/A)$ which is a surjection and B is a subobject of the kernel. So this quotients to a surjection $D/B \to (D/C)/(B/A)$, which is either an isomorphism or $(D/C)/(B/A) = 0$. This is because a simple object has no nontrivial quotients.

This means that by following these paths (left then down), the first vertical lines form a composition series of M''/M'.

Thus any filtration of a finite-length object M can be refined into a composition series. Indeed suppose $M' \subseteq M$, then we can find a composition series of M/M'. Because modules of M/M' correspond to modules $M' \subseteq M'' \subseteq M$, this forms a series $M' = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$ whose quotients are simple. So if we have a filtration $0 = M_0 \subsetneq \cdots \subsetneq M_n = M$, doing this for every $M_i \subsetneq M_{i+1}$ gives the desired composition series.

In this same way we see that if $M' \subseteq M$ is a subobject then

$$\ell(M) = \ell(M') + \ell(M/M').$$

This is because a composition series of M/M' corresponds to a filtration of $M' \subseteq M$ whose quotients are simple. Appending this to a composition series of M' gives a composition series of M. Note that this works without any assumptions on the objects: even if M is not of finite length, this identity holds.

Another way of writing this is that length is additive:

$$0 \to M' \to M \to M'' \to 0 \quad \implies \quad \ell(M) = \ell(M') + \ell(M'').$$

Thus the sum of two finite length objects is finite length, due to the exact sequence $0 \to M \to M \oplus N \to N \to 0$. And subobjects and quotients of finite length objects are also of finite length, in particular kernels and cokernels are. Thus the full subcategory of $\mathcal{C}$ consisting of objects of finite length is Abelian.

Now we focus on the case $\mathcal{C} = \mathrm{Mod}_A$ where A is a ring. Let us assume that A is not necessarily commutative, then simple modules are those which are isomorphic to $A/\mathfrak{m}$ for a maximal right ideal $\mathfrak{m} \leq A$. All such modules are simple, because there are no right ideals between $\mathfrak{m}$ and A. Note that A/I is simple iff I is a maximal right ideal, otherwise there is a $I \subsetneq J \subsetneq A$ and then $J/I \leq A/I$ is a non-trivial submodule.

And if M is a simple module, let $m \in M$ be nonzero. Now consider $f\colon A \to M$ given by $f(a) = am$. Then $\operatorname{im} f$ is a nonzero submodule of M, so it must be all of M. Thus $A/\ker f \cong M$ is simple, and thus $\ker f$ must be maximal as per the previous paragraph.

So in the case that A is commutative, the simple modules in Mod_A are fields which are surjected onto by A.

Lemma 2.6.28

Suppose M is a finite-length module, and $M_\bullet$ is a composition series for M. Now suppose that $M_i/M_{i-1} \cong A/\mathfrak{m}_i$ where $\mathfrak{m}_i \leq A$ are maximal ideals. Then M is annihlated by $\mathfrak{m}_1 \cdots \mathfrak{m}_n$, so it can be viewed as an $A/\mathfrak{m}_1 \cdots \mathfrak{m}_n$-module.

Proof

For each i define $b_i = a_0 \cdots a_i$. Then inductively we show that b_i annihlates M_i. For $M_0 = 0$ this is clear. Now we know that for $m \in M_i$, $a_i m \in M_i/M_{i-1}$ is zero, so $a_i m \in M_{i-1}$. Thus by induction $b_i m = b_{i-1} a_i m = 0$ as desired. ∎

Now suppose that in the context of the above lemma, the maximal ideals are $(\mathfrak{m}_1, \ldots, \mathfrak{m}_n)$. Removing repeated maximal ideals suppose we have $\mathfrak{n}_1, \ldots, \mathfrak{n}_s$ with multiplicities $\ell_1, \ldots, \ell_s$. Then by the chinese remainder theorem

$$A/\mathfrak{m}_1 \cdots \mathfrak{m}_n = \mathcal{A}/\mathfrak{n}_1^{\ell_1} \cdots \mathfrak{n}_s^{\ell_s} \cong A/\mathfrak{n}_1^{\ell_1} \times \cdots \times A/\mathfrak{n}_s^{\ell_s}.$$

For $1 \leq i \leq s$ let $e_i \in A$ be the image of $(0, \ldots, 1, \ldots, 0)$, i.e. the unique element where $e_i \equiv 1 \pmod{\mathfrak{n}_i^{\ell_i}}$ and $e_i \equiv 0 \pmod{\mathfrak{n}_j^{\ell_j}}$ for $j \neq i$.

Lemma 2.6.29

We have $M \cong e_1 M \times \cdots \times e_s M$, and $e_i M$ is a finite-length module whose simple quotients are $A/\mathfrak{n}_i$.

Proof

The obvious map $M \to e_1 M \times \cdots \times e_s M$, $m \mapsto (e_1 m, \ldots, e_s m)$, is injective. Indeed if $e_i m = 0$ for all i then $(e_1 + \cdots + e_n)m = 0$, and $e_1 + \cdots + e_n = 1$ in $A/\mathfrak{m}_1 \cdots \mathfrak{m}_n$. Since M is an $A/\mathfrak{m}_1 \cdots \mathfrak{m}_n$-module, this means $m = 0$.

This also shows that it is surjective: for $(e_1 m_1, \ldots, e_n m_n) \in e_1 M \times \cdots \times e_s M$, let $m = e_1 m_1 + \cdots + e_n m_n$. Because $e_i^2 = e_i$ and $e_i e_j = 0$ for $i \neq j$ in $A/\mathfrak{m}_1 \cdots \mathfrak{m}_n$, then we have $e_i m = e_i m_i$ as desired.

Because M is finite-length, so too are the $e_i M$ as submodules. Now consider $e_i M_j$, where $M_\bullet$ is the composition series of M. Then $e_i M_j$ forms a filtration of $e_i M$, though some components may be equal. Consider the map $e_i M_j \to M_j/M_{j-1}$ which has kernel $e_i M_j \cap M_{j-1} \supseteq e_i M_{j-1}$. Now suppose $M_j/M_{j-1} \cong A/\mathfrak{n}_k$.

If $i = k$ then $e_i - 1 \in \mathfrak{n}_k$ so that for all $m \in M_j$, $e_i m - m \in M_{j-1}$. So if $e_i m \in M_{j-1}$ then $m \in M_{j-1}$ and thus

$e_i m \in e_i M_{j-1}$; so $e_i M_j \cap M_{j-1} = e_i M_{j-1}$. Thus $e_i M_j / e_i M_{j-1}$ is a subobject of M_j/M_{j-1}, and so is either equal to zero or $A/\mathfrak{n}_i$.

Otherwise $e_i \in \mathfrak{n}_k$ so that $e_i m \in M_{j-1}$ for all $m \in M_j$, meaning $e_i M_j \subseteq M_{j-1}$. Thus $e_i M_j \cap M_{j-1} = e_i M_j$. And so the quotient is $e_i M_j / e_i M_{j-1} = 0$.

Thus the simple quotients of $e_i M$ are just $A/\mathfrak{n}_i$ (as the other quotients are zero). ■

Note that if M is a module with a composition series M_i whose quotients are all isomorphic to $A/\mathfrak{m}$, then $M_\mathfrak{p} \neq 0$ iff $\mathfrak{p} = \mathfrak{m}$. Indeed the composition series gives us a filtration

$$0 = (M_0)_\mathfrak{p} \subseteq \cdots \subseteq (M_n)_\mathfrak{p} = M_\mathfrak{p}.$$

Localization commutes with quotients, so the quotients here are

$$\frac{(M_i)_\mathfrak{p}}{(M_{i-1})_\mathfrak{p}} = \left(\frac{M_i}{M_{i-1}}\right)_\mathfrak{p} = (A/\mathfrak{m})_\mathfrak{p} = A_\mathfrak{p}/\mathfrak{m}A_\mathfrak{p}.$$

Now $\mathfrak{m}A_\mathfrak{p} \neq A_\mathfrak{p}$ iff $\mathfrak{m} \subseteq \mathfrak{p}$ iff $\mathfrak{p} = \mathfrak{m}$. Thus if $\mathfrak{p} \neq \mathfrak{m}$ then the quotients are zero, meaning $M_\mathfrak{p} = 0$. Thus

$$\operatorname{supp} M = \{\mathfrak{p} \in \operatorname{Spec} A \mid M_\mathfrak{p} \neq 0\} = \{\mathfrak{m}\}.$$

Now if $M = M_1 \times \cdots \times M_n$ where each M_i has simple quotients all of the form $A/\mathfrak{m}_i$, then

$$\operatorname{supp} M = \bigcup_i \operatorname{supp} M_i = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}.$$

The reason for the first equality is because localization commutes with finite products.

So by the previous lemma if M has finite length, then $\operatorname{supp} M$ consists only of finitely many maximal ideals. In other words, $\operatorname{supp} M$ is a finite subset of closed points of $\operatorname{Spec} A$.

Because QCoh_X is Abelian, there is a notion of simple objects and length. Our goal is to try and understand these objects.

2.6.4 Associated points

Recall that for an A-module M and $m \in M$,

$$\operatorname{supp} m = \{\mathfrak{p} \in \operatorname{Spec} A \mid m_\mathfrak{p} \neq 0\}.$$

Define the annihilator of m by $\operatorname{ann}_A(m) = \{a \in A \mid am = 0\}$. Then we have that

$$\operatorname{supp} m = \mathrm{V}(\operatorname{ann}_A(m)).$$

Indeed $m_\mathfrak{p} = 0$ iff there is a $a \notin \mathfrak{p}$ such that $am = 0$, i.e. iff $\operatorname{ann}_A(m)$ is not a subset of $\mathfrak{p}$. Thus $\mathfrak{p} \notin \operatorname{supp} m$ iff $\mathfrak{p} \notin \mathrm{V}(\operatorname{ann}_A(m))$.

Recall that we have

$$\operatorname{supp} M = \{\mathfrak{p} \in \operatorname{Spec} A \mid M_\mathfrak{p} \neq 0\} = \bigcup_{m \in M} \operatorname{supp} m.$$

If M is a principal module generated by $m \in M$ then

$$\operatorname{supp} M = \operatorname{supp} Am = \operatorname{supp} m = \mathrm{V}(\operatorname{ann} m).$$

Thus if M is finitely generated by $m_1, \ldots, m_n$ we have that

$$\operatorname{supp} M = \bigcup_{i=1}^n \operatorname{supp} m_i = \bigcup_{i=1}^n \mathrm{V}(\operatorname{ann} m_i)$$

so for a finitely-generated M, $\operatorname{supp} M \subseteq \operatorname{Spec} A$ is closed.

Recall that we extended these concepts to $\mathcal{O}_X$-modules. For an $\mathcal{O}_X$-module $\mathcal{F}$ and a global section $s \in \Gamma(X, \mathcal{F})$ we have

$$\operatorname{supp} s = \{p \in X \mid s_p \neq 0 \text{ in } \mathcal{F}_p\}, \qquad \operatorname{supp} \mathcal{F} = \{p \in X \mid \mathcal{F}_p \neq 0\} = \bigcup_{s \in \Gamma(X,\mathcal{F})} \operatorname{supp} s.$$

In particular $\operatorname{supp} \widetilde{M} = \operatorname{supp} M$.

Supports behave well in exact sequences: if $0 \to M' \to M \to M'' \to 0$ is exact then $\operatorname{supp} M = \operatorname{supp} M' \cup \operatorname{supp} M''$. We can extend this to $\mathcal{O}_X$-modules: if $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is exact then $\operatorname{supp} \mathcal{F} = \operatorname{supp} \mathcal{F}' \cup \operatorname{supp} \mathcal{F}''$. The proof boils down to the fact that if $0 \to A \to B \to C \to 0$ is exact, then $B = 0$ iff $A = C = 0$. Then because taking stalks is exact, $0 \to \mathcal{F}'_p \to \mathcal{F}_p \to \mathcal{F}''_p \to 0$ is exact. Then $p \notin \operatorname{supp} \mathcal{F}$ iff $\mathcal{F}_p = 0$ iff $\mathcal{F}'_p = \mathcal{F}''_p = 0$ as desired.

Proposition 2.6.30

If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is a short exact sequence of $\mathcal{O}_X$-modules, then

$$\operatorname{supp} \mathcal{F} = \operatorname{supp} \mathcal{F}' \cup \operatorname{supp} \mathcal{F}''.$$

Example 2.6.31

$\operatorname{supp} \mathcal{F}$ need not be closed in general. Indeed take $A = \mathbb{Z}$ and $M = \bigoplus_{p \text{ prime}} \mathbb{Z}/(p)$. Then for p prime we know that $(\mathbb{Z}/(p))_{(p)} = \mathbb{Z}/(p)$ because $\mathbb{Z}/(p)$ is a field and the localizing set $\mathbb{Z} - (p)$ maps to the nonzero elements of $\mathbb{Z}/(p)$. But $(\mathbb{Z}/(p))_{(0)} = 0$ because the localizing set $\mathbb{Z} - 0$ is mapped to all of $\mathbb{Z}/(p)$, including zero.

Thus

$$\operatorname{supp} M = \{(p) \mid p \in \mathbb{Z} \text{ prime}\},$$

which is not closed (because its associated ideal $\mathrm{I}(\operatorname{supp} M) = 0$ which has vanishing set all of $\operatorname{Spec} \mathbb{Z}$).

If M is finitely generated, and $x \in A$ has value zero at all the points in $\operatorname{supp} M$ (i.e. lies in all the prime ideals in $\operatorname{supp} M$), then there is a power x^n which annihilates M. Indeed if $m_1, \ldots, m_n$ generate M then $\operatorname{supp} M = \bigcup_i \mathrm{V}(\operatorname{ann} m_i)$. Thus x lies in all of the primes containing $\operatorname{ann} m_i$, thus $x \in \sqrt{\operatorname{ann} m_i}$. Then $x^{n_i} \in \operatorname{ann} m_i$, and by taking the maximum $n = n_i$ we have $x^n \in \operatorname{ann} m_i$ and so $x^n \in \operatorname{ann} M$.

Also if M is finitely generated an $x \in A$, then $\operatorname{supp}(M/xM) = \operatorname{supp} M \cap \mathrm{V}(x)$. Firstly $(xM)_{\mathfrak{p}} = x_{\mathfrak{p}} M_{\mathfrak{p}}$, because xM is the image of $M \xrightarrow{\times x} M$, localization commutes with images, and the localization of $\times x$ is $\times x_{\mathfrak{p}}$. Thus $\mathfrak{p} \notin \operatorname{supp}(M/xM)$ iff $x_{\mathfrak{p}} M_{\mathfrak{p}} = M_{\mathfrak{p}}$, i.e. iff $x_{\mathfrak{p}}$ acts surjectively on $M_{\mathfrak{p}}$. If $\mathfrak{p} \notin \operatorname{supp} M$ then $M_{\mathfrak{p}} = 0$ so this is clear, and if $\mathfrak{p} \notin \mathrm{V}(x)$ then $x \notin \mathfrak{p}$ so $x_{\mathfrak{p}}$ is a unit and so $\operatorname{supp}(M/xM) \subseteq \operatorname{supp} M \cap \mathrm{V}(x)$. Now conversely suppose $\mathfrak{p} \notin \operatorname{supp}(M/xM)$ and $\mathfrak{p} \in \mathrm{V}(x)$, so $x \in \mathfrak{p}$; we want to show that $\mathfrak{p} \notin \operatorname{supp} M$, i.e. that $M_{\mathfrak{p}} = 0$.

So in our case we have a finitely generated $A_{\mathfrak{p}}$-module $M_{\mathfrak{p}}$ and a non-unit $x_{\mathfrak{p}}$ which acts surjectively on $M_{\mathfrak{p}}$. By Nakayama's lemma this means $M_{\mathfrak{p}} = 0$. Indeed we can rephrase our situation as having a local ring $(A, \mathfrak{m})$ $(= (A_{\mathfrak{p}}, \mathfrak{p}A_{\mathfrak{p}}))$ and a finitely-generated module M $(= M_{\mathfrak{p}})$ such that a non-unit $x \in A$ $(= x_{\mathfrak{p}})$ acts surjectively on M. Because $x \in \mathfrak{m}$ this means that $\mathfrak{m}M = 0$. As M is finitely generated, this means that M must be zero by Nakayama.

Definition 2.6.32

Let M be an A-module. Its **associated prime ideals** (or **associated points**, if we are taking a geometric viewpoint) are those *prime* ideals of A which are of the form $\operatorname{ann}_A(m)$ for $m \in M$. The set of associated points of M is denoted $\operatorname{ass} M \subseteq \operatorname{Spec} A$.

The associated points/prime ideals of A are its associated points viewed as a module ovr itself.

If A is a domain then $\operatorname{ann}_A(a) = 0$ for all $a \in A$, which is prime, so $\operatorname{ass}(A) = \{0\}$.

Lemma 2.6.33

Let $M' \subseteq M$ be A-modules, then

(1) $\mathfrak{p} \in \operatorname{ass} M$ iff there is an injection $A/\mathfrak{p} \hookrightarrow M$,

(2) $\operatorname{ass} M' \subseteq \operatorname{ass} M$,

(3) $\operatorname{ass} M \subseteq \operatorname{supp} M$.

Proof

If there is an injection $\varphi\colon A/\mathfrak{p} \hookrightarrow M$ then

$$a \in \operatorname{ann}_A(\varphi 1) \iff a\varphi(1) = 0 \iff \varphi(a) = 0 \iff a \in \mathfrak{p}$$

so that $\operatorname{ann}_A(\varphi 1) = \mathfrak{p}$, so $\mathfrak{p}$ is an associated point of M. Conversely if $\mathfrak{p} = \operatorname{ann}_A(m)$ is associated then $A \to M$ given by $a \mapsto am$ has kernel $\mathfrak{p}$, giving rise to the desired injection.

(2) follows from (1) as if $\mathfrak{p} \in \operatorname{ass} M'$ then the injection $A/\mathfrak{p} \hookrightarrow M'$ composes with inclusion to give an injection $A/\mathfrak{p} \hookrightarrow M$.

For (3) if $\mathfrak{p} \in \operatorname{ass} M$ then there is an injection $A/\mathfrak{p} \hookrightarrow M$. Because localization is exact and thus preserves monomorphisms, this gives rise to an injection $(A/\mathfrak{p})_\mathfrak{p} \hookrightarrow M_\mathfrak{p}$. And $(A/\mathfrak{p})_\mathfrak{p} = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p} = \kappa(\mathfrak{p})$ is nonzero, so $M_\mathfrak{p}$ is nonzero, meaning $\mathfrak{p} \in \operatorname{supp} M$ as desired. ■

A simpler proof of (3) is just that $\operatorname{ann}_A(m)$ is independent of if we view $m \in M'$ or $m \in M$.

Lemma 2.6.34

Suppose A is Noetherian, and $m \in M$ is nonzero. Then there is a multiple $n = xm$ such that for any nonzero $yn \neq 0$, we have $\operatorname{ann} yn = \operatorname{ann} n$.

Proof

In general $\operatorname{ann} n \subseteq \operatorname{ann} yn$. Define

$$\Sigma = \{\operatorname{ann} xm \mid xm \neq 0\}$$

which is a collection of ideals of A, and as A is Noetherian has a maximal element $\operatorname{ann} xm$. Then for any y, $\operatorname{ann} yxm \subseteq \operatorname{ann} xm$ and thus there is equality. ■

In this case $\operatorname{ann} n$ is prime. Indeed if $ab \in \operatorname{ann} n$ then $abn = 0$. So either $bn = 0$, so $b \in \operatorname{ann} n$, or $a \in \operatorname{ann} bn = \operatorname{ann} n$. Thus we have proven the following.

Proposition 2.6.35

If M is a nonzero module over a Noetherian ring A, then $\operatorname{ass}_A M$ is nonempty. Furthermore, for any nonzero $m \in M$, there is an associated prime $\mathfrak{p}$ containing $\operatorname{ann} m$ and $\mathfrak{p} = \operatorname{ann} xm$ for some $x \in A$.

Proposition 2.6.36

Suppose M is a nonzero module over a Noetherian ring A. Then the natural map

$$M \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{ass} M} M_\mathfrak{p}$$

is an injection.

Proof

Let K denote the kernel, and suppose it is nonzero. Then by the above proposition it has an associated prime $\mathfrak{p}$ which by monotonicity is an associated prime of M. It is the annihilator of $m \in K \subseteq M$, but because m is in the kernel of the map, $m_\mathfrak{p} = 0$. Thus there is an $a \notin \mathfrak{p}$ such that $am = 0$, contradicting the fact that $\operatorname{ann}_A(m) = \mathfrak{p}$. ■

In the above injection we may restrict the $\mathfrak{p}$ to be those associated primes which are maximal amongst the associated primes.

Proposition 2.6.37

Let A be Noetherian and $f \in A$. Then f is a zerodivisor on M iff f vanishes at an associated point of M. In other words $\bigcup \operatorname{ass} M$ is the set of zerodivisors of M.

Proof

If f vanishes at $\mathfrak{p} \in \operatorname{ass} M$ then $f \in \mathfrak{p} = \operatorname{ann} m$ so $fm = 0$, meaning f is a zerodivisor.

Now suppose that f vanishes at no associated point, and consider the commutative square

$$\begin{array}{ccc} M & \hookrightarrow & \prod_{\mathfrak{p}\in \operatorname{ass} M} M_{\mathfrak{p}} \\ \downarrow{\scriptstyle \times f} & & \downarrow{\scriptstyle \times f} \\ M & \hookrightarrow & \prod_{\mathfrak{p}\in \operatorname{ass} M} M_{\mathfrak{p}} \end{array}$$

Note that multiplication $M_{\mathfrak{p}} \xrightarrow{\times f} M_{\mathfrak{p}}$ is injective. By assumption $f \notin \mathfrak{p}$, so it is a unit of $A_{\mathfrak{p}}$. Thus the right vertical arrow is injective, and thus so too must be the left. ■

A quicker proof of the converse in the above proof is that if $M \xrightarrow{\times f} M$ is not injective then it has a nonzero kernel K. Then it has an associated point $\mathfrak{p} = \operatorname{ann}_A(m)$ for $m \in K$. This is also an associated point of M, and $fm = 0$ implies that $f \in \mathfrak{p}$, a contradiction.

Note that $\operatorname{ass}(A/\mathfrak{p}) = \{\mathfrak{p}\}$ for $\mathfrak{p} \leq A$ prime. This is because for any nonzero $a \in A/\mathfrak{p}$, $x \in \operatorname{ann}(a)$ iff $xa \in \mathfrak{p}$ iff $x \in \mathfrak{p}$ (because $a \notin \mathfrak{p}$), so $\operatorname{ann}(a) = \mathfrak{p}$.

Proposition 2.6.38

Suppose

$$0 \to M' \to M \to M'' \to 0$$

is a short exact sequence of A-modules. Then

$$\operatorname{ass} M' \subseteq \operatorname{ass} M \subseteq \operatorname{ass} M' \cup \operatorname{ass} M''.$$

Proof

The first inclusion was already proven. Now suppose $\mathfrak{p} = \operatorname{ann} m \in \operatorname{ass} M$. We want to find a submodule of M' or M'' isomorphic to $A/\mathfrak{p}$. Consider the sub-short exact sequence

$$0 \to Am \cap M' \to Am \to Am/(Am \cap M') \to 0.$$

It is sufficient to prove the proposition for this short exact sequence, as $\mathfrak{p} \in \operatorname{ass} Am$, and $\operatorname{ass}(Am \cap M') \subseteq \operatorname{ass} M'$, $\operatorname{ass}(Am/(Am \cap M')) \subseteq \operatorname{ass}(M'')$. Thus it is sufficient to prove the proposition in the case

$$0 \to I/\mathfrak{p} \to A/\mathfrak{p} \to A/I \to 0.$$

So we want to show that $\mathfrak{p}$ is an associated point of either $I/\mathfrak{p}$ or A/I. If $I/\mathfrak{p} \neq 0$ then choose a nonzero element $a \in I/\mathfrak{p}$, and as it is an element of $A/\mathfrak{p}$ we have $\operatorname{ann}(a) = \mathfrak{p}$ as desired. Otherwise $\mathfrak{p} = I$ in which case $A/I = A/\mathfrak{p}$ so clearly $\mathfrak{p}$ is an associated point. ■

Example 2.6.39

The inclusions above may be strict, in particular $\operatorname{ass} M$ may be a proper subset of $\operatorname{ass} M' \cup \operatorname{ass} M''$. Recall that for an integral domain A, $\operatorname{ass}(A) = \{0\}$. Consider then the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2 \to 0.$$

We have that $\operatorname{ass}(\mathbb{Z}/2) = \{(2)\}$ which is not contained in $\operatorname{ass}(\mathbb{Z})$.

In a similar vein we have the following result.

Proposition 2.6.40

Suppose A is Noetherian and we have a short exact sequence of A-modules

$$0 \to M' \to M \to M'' \to 0.$$

Suppose $\mathfrak{p} \in \operatorname{ass} M''$ but $\mathfrak{p} \notin \operatorname{supp} M'$. Then $\mathfrak{p} \in \operatorname{ass} M$.

Proof

Let $m'' \in M''$ with $\mathfrak{p} = \operatorname{ann}(m'')$. Then choose a lift $m \in M$ of m'', and consider the inclusion of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \ker(Am \to Am'') & \longrightarrow & Am & \longrightarrow & Am'' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' & \longrightarrow & 0 \end{array}$$

Note that indeed $\ker(Am \to Am'') \subseteq M'$ because if $am = 0$ in M'', then $am \in M'$ as $M' = \ker(M \to M'')$. Thus we have

$$\operatorname{supp} \ker(Am \to Am'') \subseteq \operatorname{supp} M', \qquad \operatorname{ass}(Am) \subseteq \operatorname{ass} M, \qquad \operatorname{ass}(Am'') \subseteq \operatorname{ass}(M''),$$

and so we can consider the top row. That is, we can consider a short exact sequence of the form

$$0 \to \mathfrak{p}/I \to A/I \to A/\mathfrak{p} \to 0,$$

and the hypothesis is $\mathfrak{p} \notin \operatorname{supp}(\mathfrak{p}/I)$.

Define $B = A/I$ so that $\mathfrak{q} = \mathfrak{p}/I$ is a prime ideal of B and our short exact sequence is

$$0 \to \mathfrak{q} \to B \to B/\mathfrak{q} \to 0.$$

And because localizing at $\mathfrak{p}$ is the same as localizing at $\mathfrak{q}$ (for B-modules), this means that $\mathfrak{q} \notin \operatorname{supp} \mathfrak{q}$.

Because A is Noetherian, $\mathfrak{q}$ is finitely generated, and so $\operatorname{supp} \mathfrak{q}$ is closed. Note that $\mathfrak{p} \notin \mathrm{V}(I)$ implies $\bigcap \mathrm{V}(I) = \sqrt{I}$ is not a subset of $\mathfrak{p}$, as this would imply $\mathfrak{p} \in \mathrm{V}(\mathfrak{p}) \subseteq \mathrm{V}(I)$. Thus there is a $b \in \bigcap \operatorname{supp} \mathfrak{q}$ but $b \notin \mathfrak{q}$.

From the former we must have that b^n annihilates $\mathfrak{q}$ for some $n \geq 0$. This is because if $\mathfrak{q} = (x_1, \ldots, x_n)$ then $\operatorname{supp} \mathfrak{q} = \mathrm{V}(\operatorname{ann}(x_1) \cdots \operatorname{ann}(x_n))$ and so $b^n \in \operatorname{ann}(x_1) \cdots \operatorname{ann}(x_n) \subseteq \operatorname{ann}(\mathfrak{q})$ for some n. But from the latter $b^n \notin \mathfrak{q}$.

Thus $\mathfrak{q} \subseteq \operatorname{ann}(b^n)$ (because b^n annihilating $\mathfrak{q}$ is the same as $\mathfrak{q}$ annihilating b^n), but because $b^n \notin \mathfrak{q}$ we have $\operatorname{ann}(b^n) \subseteq \mathfrak{q}$. This is since if $xb^n = 0 \in \mathfrak{q}$ then we must have $x \in \mathfrak{q}$. Thus $\mathfrak{q} = \operatorname{ann}_B(b^n)$ is an associated prime of B, viewed as a B-module. Viewed as an A-module, $\mathfrak{p} = \operatorname{ann}_A(b^n)$ because $ab^n = 0$ iff $a + I \in \operatorname{ann}_B(b^n) = \mathfrak{q}$ iff $a \in \mathfrak{p}$. ∎

Lemma 2.6.41

Suppose M is a finitely generated module over a Noetherian ring A. Then M has a finite filtration

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

where the quotients are all of the form $M_{i+1}/M_i \cong A/\mathfrak{p}_i$ for $\mathfrak{p}_i \leq A$ a prime.

Proof

From the hypotheses on M we conclude it is Noetherian itself. Furthermore by proposition 2.6.35 there is a prime $\mathfrak{p}_0 \leq A$ such that $A/\mathfrak{p}_0$ embeds into M, so define $M_1 = A/\mathfrak{p}_0$. Now suppose that inductively we've defined $M_0, \ldots, M_n$. If $M_n \neq M$, then M/M_n is a Noetherian A-module so we can embed $A/\mathfrak{p}_n \hookrightarrow M/M_n$. This corresponds to a submodule $M_n \subsetneq M_{n+1} \subseteq M$ with quotient $M_{n+1}/M_n \cong A/\mathfrak{p}_n$.

Because M is Noetherian and this chain is increasing, it must terminate at some point, i.e. $M_n = M$. ■

Lemma 2.6.42

Suppose M is an A-module with a filtration as in the previous lemma. Then every associated prime of M is one of the $\mathfrak{p}_i$s. Similarly

$$\operatorname{supp} M = \bigcup_i \mathrm{V}(\mathfrak{p}_i).$$

Proof

We inductively show that $\operatorname{ass} M_i \subseteq \{\mathfrak{p}_0, \ldots, \mathfrak{p}_{i-1}\}$. For $M_0 = 0$ this is clear, and then for M_{i+1} we have a short exact sequence

$$0 \to M_i \to M_{i+1} \to M_{i+1}/M_i = A/\mathfrak{p}_i \to 0.$$

By proposition 2.6.38 we have

$$\operatorname{ass} M_{i+1} \subseteq \operatorname{ass} M_i \cup \operatorname{ass} A/\mathfrak{p}_i \subseteq \{\mathfrak{p}_0, \ldots, \mathfrak{p}_{i-1}\} \cup \{\mathfrak{p}_i\}$$

as desired.

For the second claim we follow essentially the same proof: inductively we show $\operatorname{supp} M_i = \mathrm{V}(\mathfrak{p}_0) \cup \cdots \cup \mathrm{V}(\mathfrak{p}_{i-1})$ using proposition 2.6.30. ■

Corollary 2.6.43

A finitely generated module over a Noetherian ring has finitely many associated points.

Lemma 2.6.44

Suppose M is finitely generated over a Noetherian A. Then suppose $\mathfrak{p} \leq A$ is a prime ideal corresponding to an irreducible component of $\operatorname{supp} M \subseteq \operatorname{Spec} A$. Then $\mathfrak{p}$ is an associated point of M.

Note:

$\operatorname{supp} M$ is closed in $\operatorname{Spec} A$, so its closed irreducible subsets are of the form $\mathrm{V}(\mathfrak{p})$ for a prime ideal $\mathfrak{p} \leq A$.

Proof

Let M_i be the filtration from the previous lemma with $M_{i+1}/M_i \cong A/\mathfrak{p}_i$. So by the previous lemma $\operatorname{supp} M = \bigcup_i \mathrm{V}(\mathfrak{p}_i)$, and in general $\operatorname{supp} M_n = \bigcup_{i<n} \mathrm{V}(\mathfrak{p}_i)$. The irreducible components of $\operatorname{supp} M$ are then of the form $\mathrm{V}(\mathfrak{p}_i)$ where $\mathfrak{p}_i$ is minimal amongst the $\mathfrak{p}_j$s (as it is then minimal in $\operatorname{supp} M$).

So if $\mathfrak{p}_i$ is minimal amongst the $\mathfrak{p}_j$s, suppose further that $\mathfrak{p}_i$ is its first occurance. Thus we have a short exact sequence

$$0 \to M_i \to M_{i+1} \to M_{i+1}/M_i = A/\mathfrak{p}_i \to 0.$$

Then we have $\mathfrak{p}_i \in \operatorname{ass}(A/\mathfrak{p}_i)$ and $\mathfrak{p}_i \notin \operatorname{supp} M_i$ as $\operatorname{supp} M_i = \mathrm{V}(\mathfrak{p}_0) \cup \cdots \cup \mathrm{V}(\mathfrak{p}_{i-1})$ and none of the $\mathfrak{p}_0, \ldots, \mathfrak{p}_{i-1}$ are contained in $\mathfrak{p}_i$ (it is minimal). So by proposition 2.6.40 $\mathfrak{p}_i \in \operatorname{ass} M_{i+1} \subseteq \operatorname{ass} M$ as desired. ■

Recall that by proposition 2.6.37, $f \in A$ is a zero divisor of M iff it vanishes at some associated point. Because $\operatorname{supp} A = \operatorname{Spec} A$, by the previous lemma we see that for Noetherian A, $f \in A$ is a zerodivisor iff it lies in some minimal prime ideal of A.

One direction of this holds true without assuming Noetherian-ness. If $f \in A$ lies in a minimal prime, it is a zero divisor. Indeed suppose $f \in \mathfrak{p}$ is minimal, then by minimality $\mathfrak{p}A_\mathfrak{p}$ is the unique prime ideal of $A_\mathfrak{p}$, and thus $f \in A_\mathfrak{p}$ is nilpotent. Thus there is an $x \in A - \mathfrak{p}$ such that $xf^n = 0$, i.e. f is a zero-divisor.

Proposition 2.6.45

If M is finitely generated over a Noetherian A, then $\operatorname{supp} M = \overline{\operatorname{ass} M}$.

Proof

As $\operatorname{ass} M \subseteq \operatorname{supp} M$ and $\operatorname{supp} M$ is closed, we have one direction of inclusion. Now if $\mathfrak{p} \in \operatorname{supp} M$, then it lies in some irreducible component of $\operatorname{supp} M$, say $\mathfrak{p} \in \mathrm{V}(\mathfrak{q}) = \overline{\mathfrak{q}}$. Then $\mathfrak{q}$ is an associated point of M by the above lemma, so $\mathfrak{q} \in \operatorname{ass} M$ and $\mathfrak{p} \in \overline{\mathfrak{q}} \subseteq \overline{\operatorname{ass} M}$. ■

Recall that radicals commute with localization, so $\operatorname{nil}(A)_\mathfrak{p} \cong \operatorname{nil}(A_\mathfrak{p})$. Thus $\operatorname{nil}(A)_\mathfrak{p} \neq 0$ iff $A_\mathfrak{p}$ is nonreduced. In other words $\operatorname{supp} \operatorname{nil}(A)$ is the locus of points $\mathfrak{p}$ where $A_\mathfrak{p}$ is nonreduced.

Now if A is Noetherian, $\mathfrak{p} \in \operatorname{ass} A$ iff there is a $f \in A$ such that $\operatorname{supp} f = \mathrm{V}(\mathfrak{p})$. For one direction recall that $\operatorname{supp} f = \mathrm{V}(\operatorname{ann} f)$. So if $\mathfrak{p} = \operatorname{ann} f$ is associated this is immediate. Conversely if $\operatorname{supp} f = \mathrm{V}(\mathfrak{p})$ then $\mathfrak{p}$ corresponds to an irreducible component of $\operatorname{supp} Af = \operatorname{supp} f$, which is finitely generated. Thus $\mathfrak{p} \in \operatorname{ass} Af \subseteq \operatorname{ass} A$ as desired.

Recall that if A is Noetherian and M finitely generated, there is a filtration

$$0 = M_0 \subsetneq \cdots \subsetneq M_n = M, \qquad \frac{M_{i+1}}{M_i} = \frac{A}{\mathfrak{p}_i},$$

and $\operatorname{ass} M$ are among the $\mathfrak{p}_i$. But it is not necessarily the case that each $\mathfrak{p}_i$ is associated. But what is true is that each $\mathfrak{p}_i$ *contains* an associated prime.

Indeed, $\operatorname{supp}(A/\mathfrak{p}_i) = \mathrm{V}(\mathfrak{p}_i) \subseteq \operatorname{supp} M$ (recall $\operatorname{supp} M$ is the union of these vanishing sets). As $\mathrm{V}(\mathfrak{p}_i) = \overline{\mathfrak{p}_i}$ is irreducible, it is contained in an irreducible component of $\operatorname{supp} M$. So $\mathrm{V}(\mathfrak{p}_i) \subseteq \mathrm{V}(\mathfrak{p})$ which is irreducible, and thus $\mathfrak{p}$ is an associated point of M, and $\mathfrak{p}_i \leq \mathfrak{p}$.

Proposition 2.6.46

If M is an A-module and $S \subseteq A$ multiplicative then

$$\operatorname{supp}_{S^{-1}A} S^{-1}M = \operatorname{supp}_A M \cap \operatorname{Spec} S^{-1}A$$

where we view $\operatorname{Spec} S^{-1}A$ as a subset of $\operatorname{Spec} A$.

Proof

For a prime ideal $S^{-1}\mathfrak{p} \leq S^{-1}A$ note that $\mathfrak{p} \cap S = \emptyset$ meaning $S \subseteq A - \mathfrak{p}$. Localizing at $S^{-1}\mathfrak{p}$ means localizing by $S^{-1}A - S^{-1}\mathfrak{p}$, and because we can focus on non-units, this is equivalent to localizing by $(S^{-1}A - S^{-1}\mathfrak{p}) \cap A$. Now $a \in S^{-1}A \cap A$ is in $S^{-1}\mathfrak{p}$ iff $a \in \mathfrak{p}$: if $a = p/s$ for $p \in \mathfrak{p}, s \in S$, then $tsa = tp$ for $t \in S$, so $tsa \in \mathfrak{p}$ and $ts \notin \mathfrak{p}$ so $a \in \mathfrak{p}$. Thus $(S^{-1}A - S^{-1}\mathfrak{p}) \cap A = A - \mathfrak{p}$.

Thus localizing by S and then at $S^{-1}\mathfrak{p}$ is the same as localizing at $\mathfrak{p}$. And so

$$S^{-1}\mathfrak{p} \in \operatorname{supp}_{S^{-1}A} S^{-1}M \iff (S^{-1}M)_{S^{-1}\mathfrak{p}} = M_{\mathfrak{p}} \neq 0 \iff \mathfrak{p} \in \operatorname{supp}_A M$$

giving the desired result. ∎

Proposition 2.6.47

Let M be a finitely generated module over a Noetherian ring A. Then for a multiplicative $S \subseteq A$,

$$\operatorname{ass}_{S^{-1}A} S^{-1}M = \operatorname{ass}_A M \cap \operatorname{Spec} S^{-1}A.$$

Proof

First suppose $\mathfrak{p} \in \operatorname{Spec} S^{-1}A \cap \operatorname{ass}_A M$, then there is an injection $A/\mathfrak{p} \hookrightarrow M$. Localization preserves injections and so we have an injection $S^{-1}(A/\mathfrak{p}) = S^{-1}A/S^{-1}\mathfrak{p} \hookrightarrow S^{-1}M$. Because $S^{-1}\mathfrak{p}$ is prime, we have $\mathfrak{p} \in \operatorname{ass}_{S^{-1}A} S^{-1}M$.

Now suppose $\mathfrak{q} = S^{-1}\mathfrak{p} \in \operatorname{ass}_{S^{-1}A} S^{-1}M$ so $\mathfrak{q} = \operatorname{ann}_{S^{-1}A}(m/s)$ for $m \in M$ and $s \in S$. Because $s \in S$ is a unit, we have $\mathfrak{q} = \operatorname{ann}_{S^{-1}A}(m)$. Thus $\operatorname{supp}_{S^{-1}A} m = \mathrm{V}_{S^{-1}A}(\mathfrak{q})$, and by the previous proposition

$$\operatorname{supp}_{S^{-1}A} m = \operatorname{supp}_A m \cap \operatorname{Spec} S^{-1}A, \qquad \mathrm{V}_{S^{-1}A}(\mathfrak{q}) = \mathrm{V}(\mathfrak{p}) \cap \operatorname{Spec} S^{-1}A.$$

Thus $\mathfrak{p} \in \operatorname{supp}_A m$, and it is minimally so: if $\mathfrak{p}' \in \operatorname{supp}_A m$ and $\mathfrak{p}' \leq \mathfrak{p}$ then $S^{-1}\mathfrak{p}' \leq \mathfrak{q}$ and $S^{-1}\mathfrak{p}' \in \operatorname{supp}_{S^{-1}A} m$. Thus $\operatorname{ann}_{S^{-1}A}(m) = \mathfrak{q} \leq S^{-1}\mathfrak{p}'$ so that $S^{-1}\mathfrak{p}' = \mathfrak{q}$ and thus $\mathfrak{p}' = \mathfrak{p}$. So $\mathfrak{p}$ is corresponds to an irreducible component of $\operatorname{supp}_A m$ and thus is an associated prime of Am, and thus of M. ∎

Definition 2.6.48

The associated points of M which are *not* the generic points of irreducible components of $\operatorname{supp} M$ are called **embedded points** or **embedded primes**.

If A is reduced, then A has no embedded primes. Suppose that $\mathfrak{p} = \operatorname{ann} a$ is an embedded prime, so $\mathfrak{p}$ is not minimal. Let $\mathfrak{q}$ be a minimal prime, so there is a $b \in \mathfrak{p} - \mathfrak{q}$, then $ab = 0$ and thus $a \in \mathfrak{q}$. So a lies in all minimal primes, and since every prime contains a minimal prime, a lies in all primes. Thus a is nilpotent, but A is reduced so $a = 0$ a contradiction.

It is not true that nonreduced rings have embedded primes. For instance $k[x]/(x^2)$ has no embedded primes: its only prime is $(x) = \operatorname{ann} x$, and is thus minimal.

But if $\mathfrak{p}$ is an embedded prime of A, then $A_{\mathfrak{p}}$ is nonreduced. Indeed, suppose $\mathfrak{p} = \operatorname{ann}(a)$, then $\mathfrak{p}$ is not minimal so there are primes $\mathfrak{q} < \mathfrak{p}$. These form the non-maximal prime ideals of $A_{\mathfrak{p}}$. And as before $a \in \mathfrak{q}$ for all of them, so a is in the intersection of all the prime ideals of $A_{\mathfrak{p}}$.

Definition 2.6.49

Let $\mathcal{F}$ be a coherent sheaf on a locally Noetherian scheme X. Then $\mathcal{F}$'s **associated points** are the points $p \in X$ which lie in an affine subscheme $p \in \operatorname{Spec} A \subseteq X$ such that p is an associated point of $\Gamma(\operatorname{Spec} A, \mathcal{F}) \in \operatorname{Mod}_A$.

Recall that for a Noetherian A, $\mathfrak{p} \leq A$ is an associated point of M iff $\operatorname{V}(\mathfrak{p})$ is an irreducible component of $\operatorname{supp} m$ for some $m \in M$. Indeed if $\mathfrak{p} = \operatorname{ann} m$ then $\operatorname{V}(\operatorname{ann} m) = \operatorname{supp} m$. Conversely if $\operatorname{V}(\mathfrak{p})$ is an irreducible component of $\operatorname{supp} m$, then it is an irreducible component of $\operatorname{supp} Am$ which is finitely generated and thus $\mathfrak{p}$ is an associated prime of $Am \subseteq M$.

So for A Noetherian and M an A-module, a prime $\mathfrak{p} \leq A$, the following are equivalent:

(1) $\mathfrak{p}$ is associated,

(2) $\operatorname{V}(\mathfrak{p}) = \operatorname{supp} m$ for some $m \in M$,

(3) $\operatorname{V}(\mathfrak{p})$ is an irreducible component of $\operatorname{supp} m$ for some $m \in M$.

Thus if X is a locally Noetherian scheme and $\mathcal{F}$ a coherent sheaf, then for $p \in X$ the following are equivalent:

(1) p is associated,

(2) $\overline{p} = \operatorname{supp} s$ for some local section $s \in \Gamma(U, \mathcal{F})$,

(3) $\overline{p}$ is an irreducible component of $\operatorname{supp} s$ for some local section $s \in \Gamma(U, \mathcal{F})$.

Indeed by the previous equivalence clearly (1) implies (2) implies (3). Now if $\overline{p}$ is an irreducible component of $\operatorname{supp} s$ for $s \in \Gamma(U, \mathcal{F})$, let $\operatorname{Spec} A \subseteq U$ be an open affine subset. Then $\operatorname{supp} s \cap \operatorname{Spec} A = \operatorname{supp} s|_{\operatorname{Spec} A}$, and $\overline{p} \cap \operatorname{Spec} A$ remains an irreducible component of $\operatorname{supp} s \cap \operatorname{Spec} A = \operatorname{supp} s|_{\operatorname{Spec} A}$. So by the previous equivalence (3) implies (1).

If X is locally Noetherian, $\mathcal{F}$ coherent, and $U \subseteq X$ is open, then

$$\Gamma(U, \mathcal{F}) \to \prod_{p \in \operatorname{ass} \mathcal{F} \cap U} \mathcal{F}_p$$

is injective. Indeed, this is true for any affine subscheme $\operatorname{Spec} A \subseteq U$ by proposition 2.6.36. So any section in the kernel of this map is in the kernel of all the maps for affine subschemes, which cover U, and thus the section must be zero.

Definition 2.6.50

An open subscheme $U \subseteq X$ is **scheme-theoretically dense** if a section of $V \subseteq X$ is zero when it restricts to zero on $U \cap V$.

Note that if $U \subseteq X$ is scheme-theoretically dense, then it is topologically dense. This is because a scheme-theoretically dense subset intersects all open subsets of X. If $U \cap V = \emptyset$ then $\mathcal{O}_X(V)$ is the zero ring, and thus V cannot contain any affine subschemes, so $V = \emptyset$.

If X is locally Noetherian, $U \subseteq X$ is scheme-theoretically dense iff it contains all of X's associated points. Suppose that U contains all of X's associated points, and let $\operatorname{Spec} A \subseteq X$ be affine open. There is an injection

$$A \hookrightarrow \prod_{p \in \operatorname{ass} A} A_p,$$

and because U contains all of A's associated points, so too does $U \cap \operatorname{Spec} A$, and thus in the injection above f maps to zero, so $f = 0$.

Conversely suppose U doesn't contain an associated point p. Then $p \in \operatorname{Spec} A \subseteq X$ corresponds to $p = \operatorname{ann} f$ for nonzero $f \in A$. Now let $q \in U \cap \operatorname{Spec} A$, then p is not contained in q as if it were then $p \leq q$ so that $q \in \overline{p}$ but $p \notin U$ is open. Because p is not contained in q, there is an $x \in p - q$, and so x becomes a unit in A_q. Because $x \in \operatorname{ann} f$, this means that f is equal to 0 in A_q (as $xf = 0$ and x is a unit), so $f \in q$ for all $q \in U \cap \operatorname{Spec} A$. Thus $f|_{U \cap \operatorname{Spec} A} = 0$ but $f|_{\operatorname{Spec} A} \neq 0$.

Note that the collection of scheme-theoretically dense open subsets of a scheme X is cofiltered: if U, V are scheme-theoretically dense then so too is $U \cap V$.

Definition 2.6.51

Let X be a scheme. We define the **total fraction ring** to be the filtered colimit over the scheme-theoretically dense subsets of X:

$$\operatorname{colim}_{U \subseteq X \text{ dense}} \mathcal{O}_X(U).$$

An **rational function** is an element of the total fraction ring, equivalently a section over a scheme-theoretically dense open subset of X.

Note that if $f \in \mathcal{O}_X(U)$ and $g \in \mathcal{O}_X(V)$ are rational functions, then $f = g$ (as rational functions) iff $f|_{U\cap V} = g|_{U\cap V}$. Clearly this is sufficient, but it is also necessary: if $f|_W = g|_W$ for $W \subseteq U \cap V$, then $f - g = 0$ on W which is scheme-theoretically dense, so $f - g = 0$ on $U \cap V$.

Thus we see that any rational function f has a **domain of definition**: a maximally open dense $U \subseteq X$ for which f is a section. This is because if we let (U_i, f_i) be all the rational functions equal to f, then by the previous paragraph, the f_i agree on intersections. Thus we can glue the f_i to get a section of $U = \bigcup_i U_i$. We say that a rational function f is **regular** at a point $p \in X$ if p lies in f's domain of definition.

Note that if X is integral, then all of its non-empty open subsets are scheme-theoretically dense. Indeed, let η be its generic point so $\mathcal{O}_{X,\eta} = K(X)$. Then by proposition 2.5.21, all of the canonical maps $\mathcal{O}_X(U) \to K(X)$, $f \mapsto f_\eta$, are injective. So $f = 0$ iff $f_\eta = 0$, thus if $f \in \mathcal{O}_X(U)$ is zero in $U \cap V$, it is zero in U. In particular the total fraction ring of an integral scheme is its fraction field.

3 Morphisms of Schemes

3.1 Types of morphisms

3.1.1 Open embeddings, finite morphisms

Recall that a morphism of locally ringed spaces $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a topological map $\varphi\colon X \to Y$ equipped with a morphism of sheaves $\varphi^\sharp\colon \mathcal{O}_X \to \varphi_*\mathcal{O}_Y$ such that the associated map of stalks $\varphi_p\colon \mathcal{O}_{X,\varphi p} \to \mathcal{O}_{Y,p}$ maps the maximal ideal $\mathfrak{m}_{X,\varphi p}$ into $\mathfrak{m}_{Y,p}$ for all $p \in Y$. A morphism of schemes is then a morphism of locally ringed spaces. It is clear that an isomorphism of ringed spaces is an isomorphism of locally ringed spaces.

Furthermore recall the mutual right adjunction

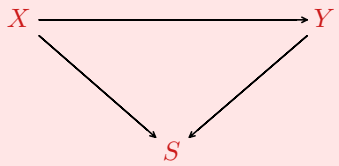
$$\hom_{\mathrm{LocRSp}}(X, \operatorname{Spec} A) \cong \hom_{\mathrm{Ring}}(A, \Gamma(X, \mathcal{O}_X))$$

which was proven in theorem 2.3.16.

The affine scheme $\operatorname{Spec}\mathbb{Z}$ is terminal in the category of affine schemes (because $\mathbb{Z}$ is the initial ring), and it is thus terminal in the category of schemes by taking an affine cover.

Definition 3.1.1

Let S be a scheme. The category of **S-schemes** is the over category Sch_S. That is, it is the category whose objects are schemes X with a morphism $X \to S$ (called the **structure morphism**), and morphisms are commutative triangles



For a ring A, we may define an A-scheme to be a scheme whose structure sheaf is valued in the category of A-algebras. A $\operatorname{Spec} A$-scheme determines an A-algebra, as the structure morphism gives a global section $A \to \Gamma(X, \mathcal{O}_X)$ and all sections are $\Gamma(X, \mathcal{O}_X)$-algebras. The converse follows by theorem 2.3.16. So an A-scheme is the same as a $\operatorname{Spec} A$-scheme.

Definition 3.1.2

A class of morphisms of schemes is called **reasonable** if it contains all isomorphisms and satisfies the following properties:

(1) The class is closed under composition,

(2) The class is preserved by base change (pullback): if $X \to Y$ is in the class, then for any $Y' \to Y$, the induced map $X \times_Y Y' \to Y'$ is also in the class. (This assumes the pullback exists, and we will see that pullbacks of schemes always exist.)

(3) The class is **local on target**. That is to say, (a) if $\pi\colon X \to Y$ is in the class, then for any open $V \subseteq Y$, the restriction $\pi^{-1}V \to V$ is in the class. And (b) if there is an open cover $Y = \bigcup_i V_i$ such that $\pi^{-1}V_i \to V_i$ is in the class for all i then π is in the class.

Note that if a property has (1) and (2) then it is closed under products. That is to say if $X \to Y$ and $X' \to Y'$ are morphisms of S-schemes with the property then $X \times_S X' \to Y \times_S Y'$ has the property. Indeed we note that $X \times_S X' \to X' \times_S Y$ has the property as it is the morphism

$$X \times_S X' \cong X \times_Y (Y \times_S X') \to Y \times_S X' \cong X' \times_S Y.$$

Similarly $X' \times_S Y \to X' \times_S Y'$ has the property, and thus $X \times_S X' \to Y \times_S Y'$ does.

Example 3.1.3

Clearly the class of isomorphisms is reasonable.

Definition 3.1.4

A morphism $X \to Y$ is an **open embedding** (or **open immersion**) if it factors as

$$(X, \mathcal{O}_X) \xrightarrow{\sim} (U, \mathcal{O}_Y|_U) \hookrightarrow (Y, \mathcal{O}_Y)$$

for $U \subseteq Y$ open. That is X is isomorphic to an open subscheme of Y.

Open embeddings clearly satisfy (1) and (3) of being reasonable. Now if $\iota\colon U \to Z$ is an open embedding and $\rho\colon Y \to Z$ is an arbitrary morphism, then $U \times_Z Y = (\rho^{-1}U, \mathcal{O}_Y|_{\rho^{-1}U})$. This is easy enough to check, and this is clearly an open subscheme of Y. Thus the class of open embeddings is reasonable.

If $X \subseteq Y$ is an open subscheme and Y is locally Noetherian, then so too is X (clearly, as an affine open subscheme of X is an affine open subscheme of Y). And if Y is Noetherian so too is X as Y is a Noetherian space and thus X is quasicompact.

Definition 3.1.5

A property of morphisms is **local on the source** when (a) if $\pi\colon X \to Y$ has the property and $U \subseteq X$ is an open subscheme, then $\pi|_U\colon U \to Y$ has the property. And (b) if $X = \bigcup_i U_i$ is an open cover where $\pi|_{U_i}\colon U_i \to Y$ has the property then so does π.

Definition 3.1.6

Let $\pi\colon X \to Y$ be a morphism of topological spaces. It is **quasicompact** if for every quasicompact open

$U \subseteq Y$, $\pi^{-1}U \subseteq X$ is quasicompact. And it is **quasiseparated** if for every quasiseparated open $U \subseteq Y$, $\pi^{-1}U \subseteq X$ is quasiseparated.

A morphism of schemes has one of these properties if the underlying topological map has it.

For a morphism of schemes it is sufficient to check quasicompactness/quasiseparatedness on affine open subsets. This is because any open subset of a scheme can be covered with an affine open cover, and if it is quasicompact then this cover can be taken to be finite, and its preimage is a finite union of quasicompact spaces and thus quasicompact. And similarly for quasiseparatedness: a space with an open cover of quasiseparated subspaces is quasiseparated.

The composition of quasicompact or quasiseparated morphisms is quasicompact or quasiseparated respectively.

Example 3.1.7

Clearly if X is a Noetherian space then any map $X \to Y$ is quasicompact. In particular if X is a Noetherian scheme (this is not true for X quasicompact!). And if X is a quasiseparated space then any map $X \to Y$ is quasiseparated (as all open subsets of X are quasiseparated).

Proposition 3.1.8

Let $\pi\colon X \to Y$ be a morphism of schemes.

(1) π is quasicompact iff there is an open affine cover $Y = \bigcup_i U_i$ such that $\pi^{-1}U_i$ is quasicompact.

(2) π is quasiseparated iff there is an open affine cover $Y = \bigcup_i U_i$ such that $\pi^{-1}U_i$ is quasiseparated.

Proof

Both points are proven with affine communication. ■

Definition 3.1.9

A morphism of schemes $\pi\colon X \to Y$ is **affine** if $\pi^{-1}U$ is affine for all $U \subseteq Y$ affine.

Because quasicompactness and quasiseparatedness can be checked on affine open subsets of Y, affine morphisms are quasicompact and quasiseparated.

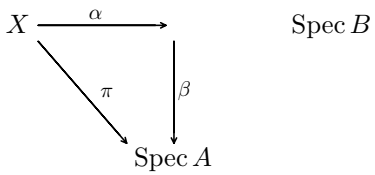
Proposition 3.1.10

A morphism $\pi\colon X \to Y$ is affine iff there is an open affine cover $Y = \bigcup_i U_i$ such that $\pi^{-1}U_i$ are all affine.

Proof

We check that the property of an affine $U \subseteq Y$ having $\pi^{-1}U$ being affine satisfies the conditions of the Affine communication lemma. Clearly for $\operatorname{Spec} A \subseteq Y$ if $\pi^{-1} \operatorname{Spec} A = \operatorname{Spec} B$ then $\pi^{-1} \operatorname{Spec} A_f = \operatorname{Spec} B_{\pi^\sharp f}$. And if $(f_1, \ldots, f_n) = A$ and $\pi^{-1} \operatorname{Spec} A_{f_i} = \operatorname{Spec} B_i$. Now we may suppose that our map is $\pi\colon X \to \operatorname{Spec} A$, i.e. the codomain is $\operatorname{Spec} A$, and we want to show X is affine. So define $B = \Gamma(X, \mathcal{O}_X)$ so that $X \to \operatorname{Spec} A$ factors through $X \to \operatorname{Spec} B$ (arising from the identity $B \to \Gamma(X, \mathcal{O}_X)$).

So we have the commutative triangle



And we want to show that α is an isomorphism. Now $\operatorname{Spec} A = \bigcup_i \mathrm{D}(f_i)$ and $\beta^{-1}\mathrm{D}(f_i) = \mathrm{D}(\beta^\sharp f_i) = \operatorname{Spec} B_{\beta^\sharp f_i}$. And $\pi^{-1}\mathrm{D}(f_i) = \operatorname{Spec} B_i$. Furthermore we have that $X = \pi^{-1}\operatorname{Spec} B$ is quasicompact and quasiseparated by the above proposition.

By The Qcqs lemma there is a natural isomorphism $B_{\beta^\sharp f_i} = \Gamma(X, \mathcal{O}_X)_{\beta^\sharp f_i} \cong \Gamma(X_{\beta^\sharp f_i}, \mathcal{O}_X) = A_i$. And so α induces an isomorphism $\operatorname{Spec} B_i \to \operatorname{Spec} B_{\beta^\sharp f_i}$. Because the B_i cover $\operatorname{Spec} B$ we have that α is an isomorphism $X \to \operatorname{Spec} B$ as desired. ■

Corollary 3.1.11

Let $Z \subseteq \operatorname{Spec} A$ be closed such that locally Z is cut out by a single equation. That is, there is an open affine cover $\operatorname{Spec} A = \bigcup_i \operatorname{Spec} A_i$ such that $Z \cap \mathrm{V}_{A_i}(f_i)$ for $f_i \in A_i$. Then $Y = \operatorname{Spec} A - Z$ is affine.

Proof

Under the inclusion $\iota: Y \to \operatorname{Spec} A$, the preimage of each $\operatorname{Spec} A_i \subseteq \operatorname{Spec} A$ is $\mathrm{D}_{A_i}(f_i) = \operatorname{Spec} A_{i,f_i}$ which is affine. By the above proposition ι is an affine morphism, and so $Y = \iota^{-1}\operatorname{Spec} A$ is affine. ■

Recall that a morphism of rings $A \to B$ is **finite** if it endows B with an A-algebra structure such that B is finitely generated as an *A-module*. And it is of **finite type** if B is a finitely generated *A-algebra*.

Definition 3.1.12

A morphism of schemes $\pi: X \to Y$ is **finite** if it is affine and for $\operatorname{Spec} B \subseteq Y$, $\pi^{-1}\operatorname{Spec} B = \operatorname{Spec} A$ and $\pi: \operatorname{Spec} A \to \operatorname{Spec} B$ makes A into a finite B-algebra.

It is **locally of finite type** if for every affine open $\operatorname{Spec} A \subseteq X$ and $\operatorname{Spec} B \subseteq Y$ such that $\pi \operatorname{Spec} A \subseteq \operatorname{Spec} B$, A is a finite type B-algebra. It is **of finite type** if it is locally of finite type and quasicompact (so the preimage of affine open in Y can be covered by finitely many affine opens in X).

Proposition 3.1.13

$\pi: X \to Y$ is locally of finite type iff $Y = \bigcup_i \operatorname{Spec} B_i$ and $\pi^{-1}\operatorname{Spec} B_i = \bigcup_j \operatorname{Spec} A_{ij}$ where A_{ij} is a finite type B_i-algebra. And it is of finite type if there are finitely many A_{ij}.

Proof

See homework 2. ■

Proposition 3.1.14

$\pi: X \to Y$ is finite iff $Y = \bigcup_i \operatorname{Spec} B_i$ and $\pi^{-1}\operatorname{Spec} B_i = \operatorname{Spec} A_i$ making A_i into a finite B_i-algebra.

Proof

By affine communication, if $\pi^{-1}\operatorname{Spec} B = \operatorname{Spec} A$ then $\pi^{-1}\operatorname{Spec} B_f = \operatorname{Spec} A_{\pi^\sharp f}$ and $A_{\pi^\sharp f}$ is a finite B_f-algebra. Conversely if $B = (f_1, \dots, f_n)$ and $A_{\pi^\sharp f_i}$ are finite B_{f_i}-algebras, then A is a finite B-algebra. ■

3.1.2 Chevalley's theorem

TODO: This

3.1.3 Closed embeddings

Definition 3.1.15

A morphism of schemes $\pi\colon X \to Y$ is a **closed embedding** if it is affine, and for $\operatorname{Spec} B \subseteq Y$ and $\pi^{-1}\operatorname{Spec} B = \operatorname{Spec} A$, the induced map $B \to A$ is surjective. A **closed subscheme** of a scheme Y is a subset $X \subseteq Y$ which is a scheme and for which the inclusion is a closed embedding.

Clearly closed embeddings are finite and closed embeddings are closed under composition. Furthermore by affine locality we may show easily that the property of being a closed embedding is affine-local on the target.

Remark 3.1.16

If $\pi\colon X \to Y$ is a closed embedding, then π is a topological closed embedding $\pi^\sharp\colon \mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is surjective. Indeed, if $Y = \bigcup_i \operatorname{Spec} B_i$ is an affine open cover and $\pi^{-1}\operatorname{Spec} B_i = \operatorname{Spec} A_i$ then the restriction $\pi\colon \operatorname{Spec} A_i \to \operatorname{Spec} B_i$ is a closed embedding (corresponding to $\operatorname{Spec} B/I \cong \mathrm{V}(I) \hookrightarrow \operatorname{Spec} B$). Because being a topological closed embedding is local on target, π is.

Thus the stalks of $\pi_*\mathcal{O}_X$ are $(\pi_*\mathcal{O}_X)_{\pi p} \cong \mathcal{O}_{X,p}$ for $p \in X$ and otherwise zero when the point is outside of π's image. Then on $\operatorname{Spec} B \subseteq Y$ we have that $\mathcal{O}_Y|_{\operatorname{Spec} B} \to (\pi_*\mathcal{O}_X)|_{\operatorname{Spec} B}$ is $\operatorname{Spec} B \to \operatorname{Spec} A$, whose stalks are surjective. Thus $\pi^\sharp$'s stalks are surjective, and thus $\pi^\sharp$ is.

The converse is also true: if π is a topological closed embedding and $\pi^\sharp$ is an epimorphism of sheaves, then π is a closed embedding.

Definition 3.1.17

Let Y be a scheme (or a ringed space), then a sub-$\mathcal{O}_Y$-module of $\mathcal{O}_Y$ is called an **ideal sheaf**. This is equivalent to its sections being ideals of $\mathcal{O}_Y$'s sections.

Lemma 3.1.18

If $\pi\colon X \to Y$ is affine then $\pi_*\mathcal{O}_X$ is a quasicoherent sheaf on Y.

Proof

By lemma 2.6.7, it is sufficient to show that for $\operatorname{Spec} A \subseteq Y$ and $f \in A$, we have $\Gamma(\operatorname{Spec} A, \pi_*\mathcal{O}_X)_f \cong \Gamma(\operatorname{Spec} A_f, \pi_*\mathcal{O}_X)$. By assumption $\pi^{-1}\operatorname{Spec} A = \operatorname{Spec} B$ for some B and so $\pi^{-1}\operatorname{Spec} A_f = \operatorname{Spec} B_{\pi^\sharp f}$. Thus we have the desired isomorphism. ■

In particular if π is a closed embedding then $\pi_*\mathcal{O}_X$ is a quasicoherent sheaf on Y. Because quasicoherent sheaves form an Abelian category, the kernel of $\mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is quasicoherent. Call it $\mathcal{I}_{X/Y}$, and it fits into an exact sequence of quasicoherent sheaves on Y

$$0 \to \mathcal{I}_{X/Y} \to \mathcal{O}_Y \to \pi_*\mathcal{O}_X \to 0,$$

called the **closed subscheme exact sequence** for $\pi\colon X \hookrightarrow Y$. From such an exact sequence we recover X and π. Taking local sections on an affine open $\operatorname{Spec} B \subseteq Y$ is exact on quasicoherent sheaves, because quasicoherent sheaves on $\operatorname{Spec} B$ are in equivalence with B-modules (via global sections and $M \mapsto \widetilde{M}$). So if $I = \Gamma(\operatorname{Spec} B, \mathcal{I}_{X/Y})$ then we have an exact sequence

$$0 \to I \to B \to \Gamma(\pi^{-1} \operatorname{Spec} B, \mathcal{O}_X) \to 0,$$

so that $\Gamma(\pi^{-1} \operatorname{Spec} B, \mathcal{O}_X) = B/I$, and we have that $\pi^{-1} \operatorname{Spec} B = \operatorname{Spec}(B/I) \subseteq \operatorname{Spec} B$ (viewing X as a subspace of Y).

Conversely if $\mathcal{I}$ is a quasicoherent sheaf of ideals, so it fits into an exact sequence where $\mathcal{F}$ is quasicoherent on Y

$$0 \to \mathcal{I} \to \mathcal{O}_Y \to \mathcal{F} \to 0.$$

For $\operatorname{Spec} B \subseteq Y$ affine open let $\mathcal{I}(B) = \Gamma(\operatorname{Spec} B, \mathcal{I})$. Now notice that for $\operatorname{Spec} A, \operatorname{Spec} B \subseteq Y$ affine open, $\operatorname{Spec} A/\mathcal{I}(A) \subseteq \operatorname{Spec} A$ and $\operatorname{Spec} B/\mathcal{I}(B) \subseteq \operatorname{Spec} B$ restrict to the same closed subscheme of the intersection $\operatorname{Spec} A \cap \operatorname{Spec} B$. We may cover the intersection $\operatorname{Spec} A \cap \operatorname{Spec} B$ with open affine subschemes which are simultaneously distinguished in both $\operatorname{Spec} A$ and $\operatorname{Spec} B$.

Now because $\mathcal{I}$ is quasicoherent then $\mathcal{I}|_{\operatorname{Spec} A} = \widetilde{\mathcal{I}(A)}$ and so $\mathcal{I}(\operatorname{Spec} A_f) = \mathcal{I}(A)_f$. Thus if $\operatorname{Spec} A_f = \operatorname{Spec} B_g \subseteq \operatorname{Spec} A \cap \operatorname{Spec} B$ we have $\mathcal{I}(A)_f = \mathcal{I}(B)_g$ and $A_f = B_g$. Thus $\operatorname{Spec} A_f/\mathcal{I}(A)_f = \operatorname{Spec} B_g/\mathcal{I}(B_g)$, and because $\operatorname{Spec} A_f/\mathcal{I}(A)_f = \operatorname{Spec} A/\mathcal{I}(A) \cap \operatorname{Spec} A_f \subseteq \operatorname{Spec} A$, we see that we can glue $\operatorname{Spec} A/\mathcal{I}(A)$ together to get a closed subscheme X of Y.

So we have proven the following.

Proposition 3.1.19

$\pi\colon X \hookrightarrow Y$ is a closed embedding iff π is a closed topological embedding, $\pi^\sharp$ is surjective, and $\ker(\pi^\sharp)$ is a quasicoherent sheaf of ideals.

Conversely if $\mathcal{I}$ is a quasicoherent sheaf of ideals, then it defines a closed subscheme of Y.

Remark 3.1.20

In fact we need not require that $\ker(\pi^\sharp)$ be quasicoherent, this can be proven from the first two assumptions (see homework 2).

Definition 3.1.21

Let $Z_i \subseteq X$ be (arbitrarily many) closed subschemes of X. These correspond to quasicoherent sheaves of ideals $\mathcal{I}_i \leq \mathcal{O}_X$. Their sum $\sum_i \mathcal{I}_i$ is a quasicoherent sheaf of ideals, and thus corresponds to a closed subscheme of X which we call the **scheme-theoretic intersection** of Z_i.

If there are only finitely many Z_i, then $\bigcap_i \mathcal{I}_i$ is a quasicoherent sheaf of ideals and thus corresponds to a closed subscheme of X called the **scheme-theoretic union** of Z_i.

Example 3.1.22

Note that for any affine scheme $X = \operatorname{Spec} A$, its closed subschemes are of the form $\operatorname{Spec}(A/I) = \mathrm{V}_A(I)$. Then the scheme theoretic intersection and union of closed subschemes are the usual affine intersection and union:

$$\bigcap_i \mathrm{V}_A(I_i) = \mathrm{V}\left(\sum_i I_i\right), \qquad \mathrm{V}_A(I) \cup \mathrm{V}_A(J) = \mathrm{V}_A(IJ) = \mathrm{V}_A(I \cap J).$$

In order for these definitions to be valid we must show that the sum of quasicoherent sheaves of ideals is quasicoherent, and similarly for the intersection. This is clear as if $\mathcal{I}_i|_{\operatorname{Spec} A} = \widetilde{I}_i$ then $(\sum_i \mathcal{I}_i)|_{\operatorname{Spec} A} = \widetilde{\sum_i I_i}$. And

similarly the intersection of finitely many quasicoherent sheaves of ideals is quasicoherent (because localization by $f \in A$ commutes with finite intersections of ideals).

Furthermore note that the underlying set of $\bigcap_i Z_i$ is the intersection of Z_i's underlying sets, and similar for union. This is true affine-locally and thus everywhere.

We see that closed embeddings are monic. Indeed suppose $\pi\colon Z \hookrightarrow X$ is a closed embedding and $\varphi_i\colon Y \to Z$ are morphisms with $\pi \circ \varphi_1 = \pi \circ \varphi_2$. Then as topological maps $\varphi = \varphi_1 = \varphi_2$ (because π is an injective topological map), and

$$(\pi \circ \varphi_i)^\sharp = \mathcal{O}_X \xrightarrow{\pi^\sharp} \pi_*\mathcal{O}_Z \xrightarrow{\pi_*\varphi_i^\sharp} \pi_*\varphi_*\mathcal{O}_Y.$$

And because $\pi^\sharp$ is surjective, $\pi_*\varphi_1^\sharp = \pi_*\varphi_2^\sharp$. Because π is a closed topological embedding we have that for $p \in Z$, the stalk of $\pi_*\varphi_i^\sharp$ is just the stalk of $\varphi_i^\sharp$ at p. Thus $\varphi_1^\sharp = \varphi_2^\sharp$ as they agree at all stalks of Z.

So we have proven (the case for open embeddings is easy).

Proposition 3.1.23

A closed or open embedding of schemes is monic.

Let Y be a scheme and $s \in \Gamma(Y, \mathcal{O}_Y)$ a global section. This defines a presheaf of ideals, $(s) \leq \mathcal{O}_Y$ defined on $U \subseteq Y$ by $\Gamma(U, (s)) = (s|_U) \leq \Gamma(U, \mathcal{O}_Y)$ (it is the principal ideal generated by s's restriction). This is quasicoherent because for $a \in A$ and $f \in A$ we have $(a_f) = (a)_f \leq A_f$ and so $(s)|_{\operatorname{Spec} A} = \widetilde{(s|_{\operatorname{Spec} A})}$, and thus it is a quasicoherent sheaf.

Definition 3.1.24

For a global section $s \in \Gamma(Y, \mathcal{O}_Y)$, the closed subscheme of Y whose associated quasicoherent sheaf of ideals is (s) is called the **vanishing scheme of** s, denoted $\mathrm{V}(s)$. For a subset $S \subseteq \Gamma(Y, \mathcal{O}_Y)$ we define its vanishing scheme by

$$\mathrm{V}(S) = \bigcap_{s \in S} \mathrm{V}(s).$$

Affine-locally $\mathrm{V}(s)$ has the form $\mathrm{V}(s)|_{\operatorname{Spec} B} = \mathrm{V}_B(s|_{\operatorname{Spec} B})$. We may extend the above discussion to any ideal $I \leq \Gamma(Y, \mathcal{O}_Y)$ and define $\mathcal{I}$ similarly. Then it is easy to see that $\mathrm{V}(S) = \mathrm{V}((S))$ where $(S) \leq \Gamma(Y, \mathcal{O}_Y)$ is the ideal generated by S.

3.1.4 The scheme-theoretic image

Definition 3.1.25

Let $\iota\colon Z \hookrightarrow Y$ be a closed subscheme, giving an exact sequence $0 \to \mathcal{I}_{Z/Y} \to \mathcal{O}_Y \to \iota_*\mathcal{O}_Z \to 0$. For a morphism $\pi\colon X \to Y$, we say that its **image lies in** Z if $\mathcal{I}_{Z/Y} \to \mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is zero. That is (locally) if a function of Y vanishes on Z, then it pulls back to the zero function on X.

If π's image lies in Z_i, then it also lies in their intersection (because $\mathcal{I}_i \to \mathcal{O}_Y \to \iota_*\mathcal{O}_Z$ are zero, so too must $\sum_i \mathcal{I}_i \to \mathcal{O}_Y \to \iota_*\mathcal{O}_Z$). So we define π's **scheme-theoretic image** to be the smallest closed subscheme Z in which π's image lies (take the intersection of all such Z).

When $Y = \operatorname{Spec} B$ is affine then for $\pi\colon X \to Y$, its image is just $Z = \operatorname{Spec} B/I$ where $I \leq B$ is the kernel of $\pi^\sharp\colon B \to \Gamma(X, \mathcal{O}_X)$. This is because $\mathcal{I} \to \mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is zero iff it is zero on the global sections, iff $\mathcal{I}(B)$ is contained in the kernel.

It would be nice to extend this affine-locally. So for $\operatorname{Spec} B \subseteq Y$ open affine let $\mathcal{I}(B) \leq B$ be the kernel of $B \to \Gamma(\operatorname{Spec} B, \pi_*\mathcal{O}_X)$. There is a natural map $\varphi\colon \mathcal{I}(B)_g \to \mathcal{I}(B_g)$ for $g \in B$, induced by the map $\mathcal{I}(B) \to \mathcal{I}(B_g)$. If this map is an isomorphism, then $\mathcal{I}$ forms a quasicoherent sheaf of ideals on Y. So we want to determine when this map is an isomorphism.

φ is clearly injective: if $r \in B$ and r/g^n is zero on $\pi^{-1} \operatorname{Spec} B_g$ then because g is invertible in $\operatorname{Spec} B_g$, we have that r must pull back to zero. For surjectivity we need to determine when the following holds: given r/g^n on

$\operatorname{Spec} B_g$ such that it pulls back to zero on $\pi^{-1}\operatorname{Spec} B_g$, is there some m such that rg^m pulls back to zero on $\pi^{-1}\operatorname{Spec} B$.

If $\pi^{-1}\operatorname{Spec} B$ is reduced, then $\pi^\sharp(r)$ vanishes on $\pi^{-1}\operatorname{D} g$, and g vanishes on $\operatorname{V}(g)$, so that $\pi^\sharp(rg)$ vanishes on $\pi^{-1}\operatorname{Spec} B = \pi^{-1}\operatorname{D}(g) \cup \pi^{-1}\operatorname{V}(g)$ (by considering that its value at each point is zero).

And if $\pi^{-1}\operatorname{Spec} B = \operatorname{Spec} A$ is affine, then let $r' = \pi^\sharp(r)$ and $g' = \pi^\sharp(g)$ in A. Then if $r' = 0$ on $\operatorname{D}(g') = \pi^{-1}\operatorname{D} g$, then there is an m such that $r'(g')^m = 0$ in A, so rg^m pulls back to zero on $\operatorname{Spec} B$. And in general if $\pi^{-1}\operatorname{Spec} B$ is quasicompact, then we can cover it with finitely many affine open subsets such that $rg^{m_i} = 0$ when pulling back, and take m to be the maximum of m_i.

Theorem 3.1.26

Let $\pi\colon X \to Y$ be a morphism of schemes such that either X is reduced or π is quasicompact. Then the scheme-theoretic image of X may be computed affine-locally: on $\operatorname{Spec} A \subseteq Y$ it is $\operatorname{Spec} A/I$ where I is the kernel of $\pi^\sharp\colon A \to \Gamma(\operatorname{Spec} A, \pi_*\mathcal{O}_X)$.

Corollary 3.1.27

The closure set-theoretic image of a morphism is contained in the underlying set of the scheme-theoretic image. They are equal under the conditions of the above theorem.

Proof

Let $\iota\colon Z \hookrightarrow Y$ be a closed then $0 \to \mathcal{I}_{Z/Y} \to \mathcal{O}_Y \to \iota_*\mathcal{O}_Z \to 0$ is exact and thus for $p \notin Z$ we have that $(\iota_*\mathcal{O}_Z)_p = 0$ so that $\mathcal{I}_{Z/Y,p} \cong \mathcal{O}_{Y,p}$. If the image of $\pi\colon X \to Y$ is contained in Z then $\mathcal{I}_{Z/Y} \to \mathcal{O}_Y \to \pi_*\mathcal{O}_X$ is zero. Taking stalks this means that $\mathcal{O}_{Y,p} \to (\pi_*\mathcal{O}_X)_p$ is zero. Because these are rings this means $(\pi_*\mathcal{O}_X)_p$ is zero. If p is in the image of π then there would have to be an open $p \in U \subseteq Y$ such that $\pi^{-1}U \neq \emptyset$ and thus the colimit is not zero, a contradiction. So the set-theoretic image of π is contained in the underlying set of the scheme theoretic image, and the scheme theoretic image is closed.

Now let Z^{set} be the set-theoretic image of π, and let $U = Y - Z^{\mathsf{set}}$. Then $\pi^{-1}U = \emptyset$, and by the above theorem the scheme-theoretic image may be computed locally. So on U we have that the scheme-theoretic image is empty. ∎

Let $X^{\mathsf{set}} \subseteq Y$ be a closed subset. Consider the following three closed subschemes of Y.

(1) The smallest closed subscheme of Y whose underlying set contains X^{set}.

(2) For an affine open $\operatorname{Spec} B \subseteq Y$ consider the radical ideal $\mathcal{I}(B) = \mathcal{I}(X^{\mathsf{set}} \cap \operatorname{Spec} B)$, and glue together the $\operatorname{Spec}(B/\mathcal{I}(B))$s to form a closed subscheme. It is easy to see that this is well-defined, as the ideals I form a quasicoherent sheaf.

(3) Let W be the scheme

$$W = \coprod_{p \in X^{\mathsf{set}}} \operatorname{Spec} \kappa(p)$$

where $\kappa(p) = \mathcal{O}_{Y,p}/\mathfrak{m}_{Y,p}$ is the residue field at p. There is a canonical map $\rho\colon W \to Y$ which sends the unique point of $\kappa(p) \in W$ to $p \in Y$. The coordinates of this are the morphisms $\operatorname{Spec} \kappa(p) \to Y$ whose sheaf morphism corresponds to mapping a section to its value at p (or zero if it lies in an open set disjoint from p). Then consider the closed subscheme which is the scheme-theoretic image of ρ.

We claim all of these closed subschemes are equal.

(1) and (2) form the same closed subscheme, because if $X^{\mathsf{set}} \subseteq Z$ then in $\operatorname{Spec} B$ we must have that $X^{\mathsf{set}} \cap \operatorname{Spec} B \subseteq Z \cap \operatorname{Spec} B$, and $X^{\mathsf{set}} \cap \operatorname{Spec} B = \operatorname{V}_B(\mathcal{I}(B))$ so that $\mathcal{I}_{Z/Y}(B) \subseteq \mathcal{I}(B)$. Thus $\mathcal{I}$ is the largest such quasicoherent sheaf and thus defines the smallest such closed subscheme. Notice that this shows that the underlying set of this scheme is X^{set}.

For (3) notice that the stalk of $\rho_*\mathcal{O}_W$ at $p \in Y$ is $\kappa(p)$ when $p \in X^{\mathsf{set}}$ and zero otherwise. Thus $\mathcal{I}_{Z/Y} \to \mathcal{O}_Y \to \rho_*\mathcal{O}_W$ is zero iff for every $p \in X^{\mathsf{set}}$ the map $\mathcal{I}_{Z/Y,p} \to \mathcal{O}_{Y,p} \to \kappa(p)$ is zero, iff $\mathcal{I}_{Z/Y,p} \subseteq \mathfrak{m}_{Y,p}$ in $\mathcal{O}_{Y,p}$. Because $\mathcal{I}_{Z/Y,p}$ is an ideal of $\mathcal{O}_{Y,p}$ this is iff it is proper. Now $\mathcal{I}_{Z/Y,p}$ is only not proper when $p \in Z$, because it is all of

$\mathcal{O}_{Y,p}$ only when $(\iota_*\mathcal{O}_Z)_p = 0$ which is iff $p \notin Z$. And so the image of ρ is contained in Z iff $X^{\mathsf{set}} \subseteq Z$. That is, its image is the smallest closed subscheme containing X^{set}.

By (2) it is clear that this closed subscheme is reduced, as $B/\mathcal{I}(B)$ is. This motivates the following definition.

Definition 3.1.28

Let $X \subseteq Y$ be a closed sub*set* of the scheme Y. Then the closed subscheme constructed above (characterized by any of (1), (2), or (3)) is called the **reduced subscheme structure** on X and is denoted X^{red}.

When $X = Y$ then Y^{red} is called the **reduction** of Y.

Note that on the affine open $\operatorname{Spec} B \subseteq Y$, Y^{red} corresponds to $\operatorname{Spec} B/\operatorname{nil}(B)$.

3.2 Fiber products

We will show that the category of schemes has fiber products (pullbacks). We do this in steps.

Lemma 3.2.1

Suppose $\alpha\colon X \to Z$ and $\beta\colon Y \to Z$ are morphisms of affine schemes. Then their fiber product $X \times_Z Y$ exists (in the category of schemes) and is affine.

Proof

Let $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, $Z = \operatorname{Spec} C$, then their fiber product is a representation of

$$\operatorname{Sch}(\bullet, \operatorname{Spec} A) \times_{\operatorname{Sch}(\bullet, \operatorname{Spec} C)} \operatorname{Sch}(\bullet, \operatorname{Spec} B),$$

and following the spec-global sections functor we see that

$$\begin{aligned}\operatorname{Sch}(\bullet, \operatorname{Spec} A) \times_{\operatorname{Sch}(\bullet, \operatorname{Spec} C)} \operatorname{Sch}(\bullet, \operatorname{Spec} B) &\cong \operatorname{Ring}(A, \Gamma\bullet) \times_{\operatorname{Ring}(C, \Gamma\bullet)} \operatorname{Ring}(B, \Gamma\bullet) \\ &\cong \operatorname{Ring}(A \otimes_C B, \Gamma\bullet) \\ &\cong \operatorname{Sch}(\bullet, \operatorname{Spec}(A \otimes_C B)).\end{aligned}$$

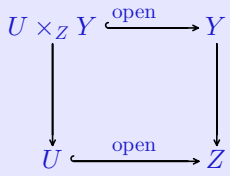
So that the fiber product is $\operatorname{Spec}(A \otimes_C B)$ which is affine. ∎

Note:

It follows that $\mathbb{A}^1_k \times_k \mathbb{A}^1_k \cong \mathbb{A}^2_k$, because $k[x] \otimes_k k[y] \cong k[x, y]$ via the obvious map $p(x) \otimes q(y) \mapsto p(x)q(y)$. But as sets it is not true that $\left|\mathbb{A}^2_k\right| \cong \left|\mathbb{A}^1_k\right| \times \left|\mathbb{A}^1_k\right|$, as there are prime ideals of $\mathbb{A}^2_k$ not of the form $(x-a, y-b)$ (i.e. which are not maximal) or $(x-a)$ or $(y-b)$. So the underlying set of a fiber product of schemes is not the fiber product of underlying sets.

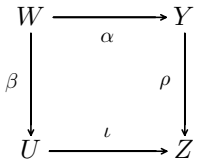
Lemma 3.2.2

Suppose $\iota: U \hookrightarrow Z$ is an open embedding and $\rho: Y \to Z$ is a morphism. Then $U \times_Z Y$ is $(\rho^{-1}U, \mathcal{O}_Y|_{\rho^{-1}U})$. In particular the fiber product of an open embedding is an open embedding:



Proof

Suppose we have the following commutative square



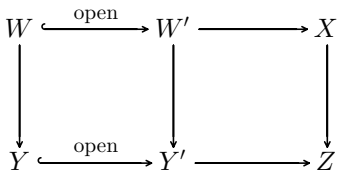
Then as topological maps we must have that $\alpha(W) \subseteq \rho^{-1}U$. And so restricting α's codomain to the open $\rho^{-1}U$ gives us the desired map $\overline{\alpha}: W \to \rho^{-1}U$ making the diagram commute. ■

Lemma 3.2.3

Suppose $\alpha: X \to Z$ is a morphism of affine schemes, and $\beta: Y \to Z$ factors as $Y \overset{\iota}{\hookrightarrow} Y' \to Z$ where ι is an open embedding and Y' affine. Then $X \times_Z Y$ exists.

Proof

Consider the diagram below:



The right square has a pullback (because the schemes are affine), and the left square has a pullback (because open embeddings have pullbacks). Thus the rectangle as a whole is a pullback. ■

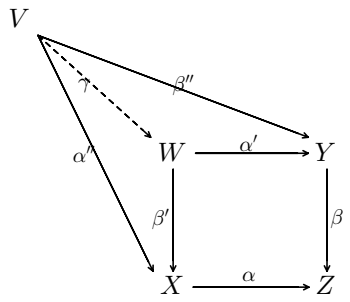
Lemma 3.2.4

Suppose X, Z are affine and Y is arbitrary. Then $X \times_Z Y$ exists.

Proof

First assume that Y has a cover by two affine open subsets Y_1, Y_2. Let $Y_{12} = Y_1 \cap Y_2$, so that by lemma 3.2.1 $W_i = X \times_Z Y_i$ exists and so does $W_{12} = X \times_Z Y_{12}$ by lemma 3.2.3. Furthermore by the proof of lemma 3.2.3 the canonical map $X \times_Z Y_{12} \hookrightarrow X \times_Z Y_i$ is an open embedding (it is the map $W \hookrightarrow W'$ in the proof). And so we can glue W_1 and W_2 along W_{12}, call this scheme W.

There are obvious maps $W \xrightarrow{\alpha'} Y$ and $W \xrightarrow{\beta'} X$. We want to show that $W = X \times_Z Y$. So suppose that we have $\alpha'' : V \to X$ and $\beta'' : V \to Y$ which commute with α, β. That is, we must fill in the below diagram with a γ:



Define $V_i = (\beta'')^{-1} Y_i$ and $V_{12} = (\beta'')^{-1} Y_{12} = V_1 \cap V_2$. Then for each i there is a unique map $V_i \to W_i$ fitting into the pullback ($W_i = Y_i \times_Z X$). Thus a unique $\gamma_i : V_i \to W$ fitting into the diagram above (with V_i for V). And similarly there is a unique map $\gamma_{12} : V_{12} \to W$ fitting into the diagram. By uniqueness, γ_1 and γ_2 must agree on V_{12} with γ_{12}, and thus there is a unique gluing $\gamma : V \to W$.

In the general case, cover Y with an open cover of affine open subsets Y_i. As before $Y_{ij} = Y_i \cap Y_j$ has a pullback, so let

$$W_i = Y_i \times_Z X, \qquad W_{ij} = Y_{ij} \times_Z X.$$

Then we may glue the W_i along W_{ij} together to form a scheme W.

Now given $\alpha'' : V \to X$ and $\beta'' : V \to Y$ as before, define $V_i = (\beta'')^{-1} Y_i$ and $V_{ij} = (\beta'')^{-1} Y_{ij} = V_i \cap V_j$. As before for each i there is a unique $V_i \to W_i$ fitting into the diagram, giving us a unique $\gamma_i : V_i \to W$. And there are unique $\gamma_{ij} : V_{ij} \to W$, so γ_i and γ_j must agree on $V_i \cap V_j$ with γ_{ij}. Thus they uniquely glue together to give γ. ■

Lemma 3.2.5

If Z is affine and X, Y are arbitrary then $X \times_Z Y$ exists.

Proof

Consider the above lemma and its proof. In it we know that $W_i = X \times_Z Y_i$ exists because Y_i is affine (this is what the lemma proves). And $W_{ij} = X \times Y_{ij}$ exists by lemma 3.2.2. Thus the proof of the lemma works again (now that we know the lemma is true) for general X as well. ■

Lemma 3.2.6

Suppose X, Y are abitrary and Z is an open subscheme of an affine scheme Z'. Then $X \times_Z Y$ exists.

Proof

In general if $Z \to Z'$ is monic then $X \times_Z Y$ and $X \times_{Z'} Y$ satisfy the same universal property, so one exists iff the other does. So this lemma follows from the previous one. ■

Theorem 3.2.7

For arbitrary schemes X, Y, Z, the fiber product $X \times_Z Y$ exists.

Proof

Cover Z with affine open subschemes Z_i, and let $X_i = \alpha^{-1}Z_i$ and $Y_i = \beta^{-1}Z_i$. Further define $Z_{ij} = Z_i \cap Z_j$ and $X_{ij} = \alpha^{-1}Z_{ij} = X_i \cap X_j$ and similar for Y. Then by lemma 3.2.5 $W_i X_i \times_{Z_i} Y_i$ exists for all i, and by lemma 3.2.6 $W_{ij} = X_{ij} \times_{Z_{ij}} Y_{ij}$ exists for all ij.

Note that as before $X_i \times_{Z_i} Y_i = X_i \times_Z Y_i$, and furthermore we can see that $X_i \times_Z Y_i = X \times_Z Y_i$. This is because they both satisfy the same universal property: any V with $\alpha'' : V \to X$ and $\beta'' : V \to Y_i$ must have that $\alpha\alpha'' = \beta\beta''$ which means that $\alpha\alpha''$ lies in Z_i and thus α'' lies in X_i. So we may identify $W_i = X \times_Z Y_i$. And similarly $W_{ij} = X \times_Z Y_{ij}$, and then we may glue the W_is along the W_{ij}s, and for the same reason as in lemma 3.2.4 this gives us the fiber product $X \times_Z Y$. ■

3.2.1 Representable functors

In this section we will give a more sophisticated perspective on fiber products and provide an alternative proof as to their existence. We assume only lemma 3.2.1 and lemma 3.2.2.

Recall that a (contravariant) functor is **representable** if it is naturally isomorphic to $\hom(\bullet, X)$. And for a diagram $D : I \to \mathcal{C}$, its limit exists iff the limit of $\mathsf{y} \circ D : I \to \mathrm{Psh}(\mathcal{C})$ is representable (here y is the Yoneda embedding, $\mathsf{y}(X) = \hom(\bullet, X)$). Note that $\mathsf{y} \circ D$'s limit exists because $\mathrm{Psh}(\mathcal{C})$ is complete.

Writing $h_X = \hom(\bullet, X)$, we see that $X \times_Z Y$ exists iff $h_X \times_{h_Z} h_Y$ is representable. So we will give an alternative perspective on the above proof in light of this.

Note that if U_i is an open cover of Y, a morphism $Y \to X$ is determined by its restrictions $U_i \to X$. And given morphisms $U_i \to X$ which agree on $U_i \cap U_j$, they in turn glue to give $Y \to X$. Thus we have an equalizer diagram

$$h_X(Y) \to \prod h_X(U_i) \rightrightarrows \prod h_X(U_i \cap U_j),$$

which is to say that h_X forms a sheaf on every scheme Y. In general if F is a contravariant functor $\mathrm{Sch}^{\mathsf{op}} \to \mathrm{Set}$ which forms a sheaf on every scheme Y (i.e. $F|_{\mathrm{open}(Y)}$ is a sheaf), then F is called a **Zariski sheaf**. So if F is representable it is a Zariski sheaf.

Lemma 3.2.8

For schemes X, Y, Z, $h_X \times_{h_Z} h_Y$ is a Zariski sheaf.

Proof

Let $\alpha : X \to Z$ and $\beta : Y \to Z$ so that for any scheme W

$$h_X(W) \times_{h_Z(W)} h_Y(W) = \{(f : W \to X, g : W \to Y) \mid \alpha f = \beta g\}.$$

Now suppose U_i cover W and we are given $f_i : U_i \to X$ and $g_i : U_i \to Y$ such that $\alpha f_i = \beta g_i$ and $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ and similar for g_is. Then the f_i and g_i glue uniquely together to give $f : W \to X$ and $g : W \to Y$, and because locally we have $\alpha f = \beta g$ we must have this globally. ■

Definition 3.2.9

A subfunctor $h' \hookrightarrow h$ is an **open subfunctor** if for all schemes X and maps $h_X \to h$, the fibered product $h_X \times_h h'$ is representable, say by a scheme U, and $h_U \to h_X$ (induced from $h_X \times_h h' \to h_X$) corresponds to an open embedding of schemes $U \to X$.

Lemma 3.2.10

A morphism $h_W \to h_Z$ is an open subfunctor iff $W \to Z$ is an open embedding.

Proof

Suppose that $h_W \to h_Z$ is an open subfunctor, then taking $X = Z$ we have that $h_Z \times_{h_Z} h_W \cong h_W$ (which is already representable) and then by definition $W \to Z$ is an open embedding.

Conversely if $W \to Z$ is an open embedding then for a scheme X by lemma 3.2.2 we know that $X \times_Z W$ exists and is isomorphic to h_U where $U \hookrightarrow X$ is open. Thus $h_X \times_{h_Z} h_W$ is representable by h_U and $U \hookrightarrow X$ is an open embedding. ■

Lemma 3.2.11

If $h'' \hookrightarrow h'$ and $h' \hookrightarrow h$ are open subfunctors, then $h'' \hookrightarrow h$ is an open subfunctor.

Proof

Let X be a scheme, and so

$$h_X \times_h h'' \cong h_X \times_h (h' \times_{h'} h'') \cong (h_X \times_h h') \times_{h'} h''.$$

By assumption there is an open subscheme $U \hookrightarrow X$ so that $h_X \times_h h' \cong h_U$, so

$$h_X \times_h h'' \cong h_U \times_{h'} h'',$$

and again there is an open subscheme $V \hookrightarrow U$ such that this is isomorphic to h_V, and $V \hookrightarrow U \hookrightarrow X$ is an open embedding as desired. ■

Lemma 3.2.12

(1) If $h' \hookrightarrow h$ and $h'' \hookrightarrow h$ are open subfunctors, their pullback $h' \times_h h''$ is an open subfunctor of h', h'', h. We call the pullback their **intersection**.

(2) If $U, V \hookrightarrow X$ are open subschemes of X, then the intersection of $h_U, h_V \hookrightarrow h_X$ is $h_{U \cap V}$.

Proof

For (1), let X be a scheme so that

$$h_X \times_{h'} (h' \times_h h'') \cong h_X \times_h h'' \cong h_U$$

for some open subscheme $U \hookrightarrow X$. Thus $h' \times_h h''$ is an open subfunctor of h', and by symmetry for h''. As these are open subfunctors of h, by the previous lemma the intersection is an open subfunctor of h.

For (2) we need to show that $h_U \times_{h_X} h_V \cong h_{U \cap V}$. This follows directly from lemma 3.2.2. ■

Lemma 3.2.13

Let $\alpha: X \to Z$ and $\beta: Y \to Z$ be morphisms of schemes. Let $U \subseteq X, V \subseteq Y, W \subseteq Z$ be open subschemes so that $\alpha(U), \beta(V) \subseteq W$. Show that the natural map $h_U \times_{h_W} h_V \hookrightarrow h_X \times_{h_Z} h_Y$ is an open embedding.

Proof

We want to show that for any $h_T: h_X \times_{h_Z} h_Y$, the pullback $h_T \times_{h_X \times_{h_Z} h_Y} h_V \times_{h_W} h_U$ is represented by an open subscheme of T. The map $h_T \to h_X \times_{h_Z} h_Y$ is determined by morphisms $\varphi: T \to X$ and $\psi: T \to Y$. Then we claim that

$$h_T \times_{h_X \times_{h_Z} h_Y} h_V \times_{h_W} h_U \cong h_{\varphi^{-1}U \cap \psi^{-1}V}.$$

Indeed a pair of morphisms $h \to h_T$ and $h: h_V \times_{h_W} h_U$ gives morphisms from h to h_T, h_V, h_U which commute with the base morphisms, and thus a morphism $h \to h_U \times_{h_X} h_T$ and $h \to h_V \times_{h_Y} h_T$. Because $U \subseteq X$ and $V \subseteq Y$ are open, we know that these fiber products are $h_{\varphi^{-1}U}$ and $h_{\psi^{-1}V}$, so we have morphisms $h \to h_{\varphi^{-1}U}$ and $h \to h_{\psi^{-1}V}$. These commute with the base morphisms giving us a morphism $h \to h_{\varphi^{-1}U} \times_{h_T} h_{\psi^{-1}V} = h_{\varphi^{-1}U \cap \psi^{-1}V}$. All of the steps here gave unique morphisms, so we have shown the desired result. ■

Definition 3.2.14

Let h_i be a collection of open subfunctors of h. Then for a scheme X and a morphism $h_X \to h$, we have that $h_X \times_h h_i = h_{U_i}$ for some open subscheme $U_i \hookrightarrow X$. Then h_i **covers** h if the U_i cover X for all X.

Lemma 3.2.15

Suppose we have morphisms of schemes $X \to Z$ and $Y \to Z$. Let $\{Z_i\}_i$ be an affine cover of Z, and $\{X_{ij}\}_j$ be an affine cover of Z_i's preimage in X, and $\{Y_{ij}\}_j$ an affine cover of Z_i's preimage in Y. Then $\left\{h_{X_{ij}} \times_{h_{Z_i}} h_{Y_{ik}}\right\}_{ijk}$ is an open cover of $h_X \times_{h_Z} h_Y$.

Proof

By lemma 3.2.13 these are open subfunctors of $h_X \times_{h_Z} h_Y$. By the same lemma for $h_T \to h_X \times_{h_Z} h_Y$ corresponding to $T \xrightarrow{\varphi} X$ and $T \xrightarrow{\psi} Y$, we have that

$$h_T \times_{h_X \times_{h_Z} h_Y} h_{X_{ij}} \times_{h_{Z_i}} h_{Y_{ik}} \cong h_{\varphi^{-1}X_{ij} \cap \psi^{-1}Y_{ik}}.$$

So we claim that $\varphi^{-1}X_{ij} \cap \psi^{-1}Y_{ik}$ cover T. Indeed for $t \in T$ we have that $\alpha\varphi(t) = \beta\psi(t) \in Z_i$ for some i. Then $\varphi(t) \in X_{ij}$ for some j and $\psi(t) \in Y_{ik}$ for some k, and thus $t \in \varphi^{-1}X_{ij} \cap \psi^{-1}X_{ik}$ as desired. ■

Lemma 3.2.16

Let h be a Zariski sheaf which has an open cover by representable functors. Then h is representable.

Proof

Let $h_{U_i} \hookrightarrow h$ be an open cover. Then $h_{U_i} \times_h h_{U_j}$ is representable by some open $U_{ij} \hookrightarrow U_i$. It follows then that $U_{ij} \cong U_{ji}$ and so we can glue the U_i along the U_{ij}s to get a scheme X.
Now we construct a morphism $h_X \to h$. By Yoneda this is equivalent to giving an element of $h(X)$. Because h is a Zariski sheaf, we can do this locally. The U_i cover X, so we can give compatible sections of $h(U_i)$, which is the same as $h_{U_i} \to h$. Taking the given open embeddings $h_{U_i} \hookrightarrow h$ we see that we may glue them together (they agree on overlaps) to give us $h_X \to h$.

Now let Y be another scheme, we want to show that $h_X(Y) \to h(Y)$ is an isomorphism. Given a morphism $h_Y \to h$ we have that $h_Y \times_h h_{U_i} \cong h_{V_i}$ cover Y. Then we have morphisms $h_{V_i} \to h_{U_i} \hookrightarrow h_X$, which glue together to give $h_Y \to h_X$. And conversely if $h_Y \to h_X$ is a morphism, then this gives us a morphism $h_Y \to h_X \to h$. These two procedures are natural inverses, so we see that $\hom(\bullet, h_X) \cong \hom(\bullet, h)$ and thus $h \cong h_X$ as desired. ■

Theorem 3.2.17

The fiber products of schemes exists.

Proof

We know that $h_X \times_{h_Z} h_Y$ is a Zariski sheaf by lemma 3.2.8. Furthermore by lemma 3.2.15 we can cover this by representable functors. Thus by the previous lemma $h_X \times_{h_Z} h_Y$ is representable, and so $X \times_Z Y$ exists. ■

3.2.2 Computing fiber products

There are a few ways of computing fiber products (base change). The first is if we want to base change relative to an open embedding, then we know what the fiber product looks like by lemma 3.2.2.

Secondly we know that for rings $A \otimes_B B[x_1, \ldots, x_n] \cong A[x_1, \ldots, x_n]$. We form morphisms in both directions by mapping $a \otimes p(\overline{x}) \mapsto ap(\overline{x})$, and conversely mapping x_i to $1 \otimes x_i$ and using the universal property of polynomial algebras. In particular $A[\overline{x}] \cong A \otimes_{\mathbb{Z}} \mathbb{Z}[\overline{x}]$.

Thus we have that

$$\operatorname{Spec} A \times_{\operatorname{Spec} B} \mathbb{A}^n_B \cong \mathbb{A}^n_A.$$

In particular

$$\mathbb{A}^n_A \cong \operatorname{Spec} A \times_{\operatorname{Spec} \mathbb{Z}} \mathbb{A}^n_{\mathbb{Z}}.$$

In general then we define the following.

Definition 3.2.18

Let X be a scheme, then we define $\mathbb{A}^n_X$ by $X \times_{\mathbb{Z}} \mathbb{A}^n_{\mathbb{Z}}$.